Geography and Environment Discussion Paper Series

Green Foreign Direct Investment is flowing far beyond renewables: a taxonomy-guided LLM analysis

Juan Alvarez-Vilanova,[1] Riccardo Crescenzi,[1] Lee Mager[2]

[1] *Department of Geography and Environment, London School of Economics, Houghton Street, London WC2A 2AE, United Kingdom*

[2] *LSE Law School, London School of Economics, Houghton Street, London WC2A 2AE, United Kingdom*

Corresponding author: *r.crescenzi@lse.ac.uk*

March 2026

Abstract

Foreign direct investment (FDI) finances and diffuses the capital, skills and know-how needed for the low-carbon transition, yet 'green FDI' remains systematically under-measured. Sector proxies – typically renewable energy and waste remediation – miss green activities embedded within other industries, yielding a partial and geographically-biased picture. Here we combine Large Language Models with the EU Taxonomy for Sustainable Activities to classify investment projects against sector-specific green criteria. Applying this taxonomy-guided framework to 109,084 inward greenfield FDI projects into the European Union and the United Kingdom (2013–2024), we identify 15.7% of FDI value as green – around twice the share captured by traditional measures. We show that sector metrics miss large volumes in manufacturing and services, and that beyond-energy green FDI is more strongly linked to extra-European investors, implying distinct geopolitical dependencies. Blinded human coding and robustness tests confirm high accuracy and reproducibility. Together, these results enable scalable monitoring of investment alignment with climate objectives.

Keywords: green FDI, EU Taxonomy, large language models, greenfield investment, decarbonisation, green transition

JEL Classification: F21, Q55, Q56, C88

1. Introduction

To avert the worst effects of climate change and meet the goals of the 2015 Paris Agreement, economies must rapidly shift technologies, infrastructures and production systems towards low-carbon alternatives (Intergovernmental Panel on Climate Change (IPCC), 2022). Foreign Direct Investment (FDI) can accelerate this transition by mobilising capital for costly or novel technologies and processes (UNCTAD, 2023), especially as public budgets in many nations face competing demands. Beyond financing the green transition, FDI enables the cross-border diffusion of green technologies, skills and knowledge (Rogelj et al., 2018), as multinational firms mobilise expertise and capabilities in host economies. Yet the extent to which cross-border investment is truly supporting decarbonisation remains difficult to observe.

The core obstacle is measurement. Most 'green FDI' statistics rely on broad sectoral classifications, labelling entire industries – such as renewable energy or waste management – as 'green' while overlooking potentially green activities embedded within all other sectors. This has skewed the debate towards renewable energy, fuelling intense competition among countries and sub-national jurisdictions to attract these projects as visible, high-profile anchors for their decarbonisation and wider sustainability strategies. Yet renewable energy projects are geographically constrained by natural resource availability – such as solar irradiance or wind conditions – limiting their reach to specific areas within specific countries. Moreover, as policymakers increasingly seek to harness green FDI for broader industrial and regional development goals (European Commission: Directorate-General for Regional and Urban Policy, 2024), concerns have emerged that renewable energy projects are capital-intensive, sensitive to geopolitical and regulatory uncertainty, and may offer limited long-term local economic benefits in terms of job creation and technology transfer (Marin and Vona, 2019). Conversely, potentially transformative green investments can occur inside manufacturing and services – through process change, enabling inputs, low-carbon materials and environmental technologies – but they are typically invisible to sector-based approaches.

Recent reports have begun to highlight the growth of FDI in activities such as electric vehicles, battery manufacturing, and environmental technologies, suggesting that green investment is increasingly flowing beyond traditional renewable energy sectors (OECD, 2024*b*; UNCTAD, 2023). However, without a project-level classification framework that is both systematic and technically grounded, the scale, sectoral composition and geography of green FDI cannot be assessed, compared or monitored. This paper addresses this gap by developing and applying a bottom-up approach to identifying green FDI across all sectors of the economy. By combining large language models (LLMs) with a technical taxonomy defining sector-specific qualifying activities through technical screening criteria rather than self-declared labels, this paper develops a scalable, bottom-up framework for mapping green FDI across all sectors of the economy.

Developing such a framework is not merely a measurement exercise. The absence of a universally accepted system for classifying green investment has long hampered both scholarship and policy, opening the door to greenwashing (OECD, 2020) and limiting our ability to engage with longstanding debates over whether FDI acts as an environmental "haven" or promotes environmental upgrading and technology transfer (Cole et al., 2006; Eskeland and Harrison, 2003; Marin and Vona, 2019). While related efforts have emerged in adjacent domains – such as environmental patent classification (Amendolagine et al., 2023; Hascic and Migotto, 2015), green job taxonomies (Dierdorff et al., 2009), and green revenue tracking (London Stock Exchange Group (LSEG), 2024) – none provide a comprehensive framework for systematically identifying green FDI.

Against this backdrop, the establishment of a comprehensive and technically rigorous taxonomy – such as the EU Taxonomy of Sustainable Activities (EUTSA) – marks a major advance. Developed over four years by a technical expert group, the taxonomy goes beyond labelling entire sectors as green; it identifies specific activities that contribute to environmental objectives, whether because they are inherently sustainable, because they represent lower-emission alternatives to conventional practices, or because they enable green outcomes elsewhere in the economy (European Commission, 2020, 2021; Lucarelli et al., 2023). Detailed screening criteria accompany each activity. For example, the taxonomy distinguishes low-emission steel production methods, such as Electric Arc Furnaces (EAF), from conventional high-emission processes (Griffin and Hammond, 2021; International Energy Agency (IEA), 2020; World Steel Association, 2021).

However, applying such standards to individual investment projects has long been infeasible at scale. We overcome this constraint by combining the EUTSA with recent advances in Large Language Models (LLMs), which can follow structured, domain-specific instructions with high accuracy and reproducibility (Bubeck et al., 2023; Qin et al., 2023), including in green-finance applications (Colesanti Senni et al., 2024; Zou et al., 2025). Recent work shows that GPT-based classification of structured qualitative attributes is robust across domains and generally indistinguishable from human annotation (Asirvatham et al., 2026; Besley et al., 2026). We develop a taxonomy-guided classification pipeline that evaluates project-level FDI descriptions against EUTSA criteria. We apply the approach to inward greenfield projects into the EU27+UK over 2013–2024 using Orbis-Crossborder, leveraging the regulatory homogeneity and policy salience of the European setting while designing the workflow to be transferable to other jurisdictions with emerging taxonomies. Although the empirical focus is European, the workflow is designed to be transferable to other jurisdictions adopting (or designing their own) green taxonomies, and we provide a full replication package enabling researchers with access to the underlying data to apply the framework to other geographies.

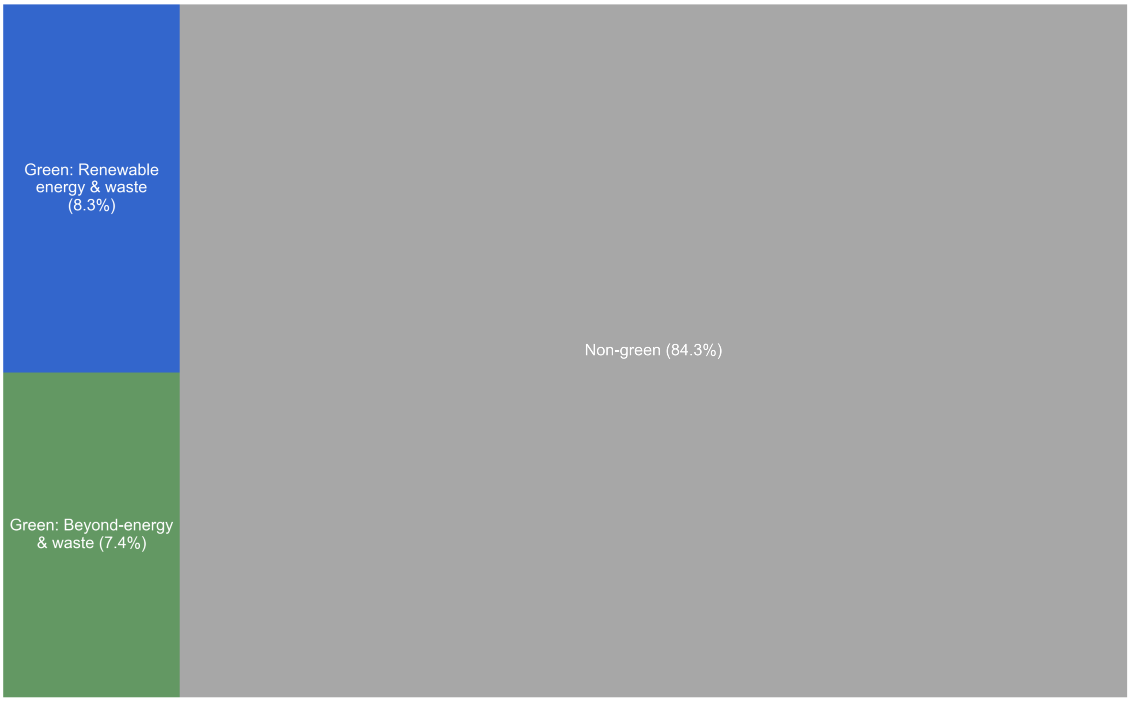
The analysis yields three contributions. First, it provides a scalable and technically anchored method to identify green FDI at the project level, distinguishing qualifying investments from green-language "noise". Second, it documents that a substantial and growing share of green FDI occurs outside renewable energy and waste, concentrated in manufacturing and knowledge-intensive activities central to industrial decarbonisation. Third, it reveals pronounced spatial and origin–destination heterogeneity, implying that the geography of the green transition depends not only on natural endowments but also on industrial capabilities and innovation ecosystems. Together, these results enable a more complete evidence base for place-sensitive green industrial policy and for monitoring the international allocation of capital underpinning decarbonisation.

2. Results

2.1 A large (and growing) flow of green FDI beyond renewable energy

Using our EUTSA-guided LLM classification (Methods), we uncover that nearly one in six dollars (15.7%[1]) of inward FDI into the EU27+UK over 2013–2024 can be classified as 'green', in the sense of contributing to one of the 6 environmental objectives that the EUTSA considers in its definition of 'green'.[2] About half of this green FDI lies outside of the Renewable Energy & Waste (REW) sectors – equivalent to 7.4% of total FDI capital value (Figure 1). This indicates that a substantial level of greening activity is occurring beyond the sectors many FDI statistics have traditionally focused on to capture green FDI.

Figure 1: Share of 'green' inward FDI into EU27+UK (FDI capital value, 2013–2024).



Source: Own elaboration with data from Orbis-Crossborder / Bureau van Dijk (2025).

[1] Given that some FDI projects in our data (around 11%) have no textual descriptions, these shares are calculated using the total value of FDI projects that do have a description.

[2] Climate change mitigation; climate change adaptation; protection of biodiversity and ecosystems; transition to a circular economy; pollution prevention and control; protection of water resources.

Figure 2 illustrates two examples of green FDI beyond REW (henceforth referred to as beyond-energy green FDI): a 2021 investment in a green ammonia plant in Norrbotten, Sweden, and an e-mobility R&D lab in Delft, the Netherlands. Such projects would be overlooked by sector-based classifications, but the application of the EUTSA-based approach captures them as green investment projects due to their involvement of activities that contribute toward EUTSA environmental objectives.

Figure 2: Two examples of green FDI projects identified beyond the energy/waste sector.

Example A: E-mobility R&D, Netherlands

Sector: Scientific R&D (NACE M72)
Investment value: $10mn
EUTSA objective: Climate change mitigation

*"In 2020, it was announced that a Switzerland-based electrical equipment manufacturer had opened an **e-mobility research and development centre** in Delft, Netherlands."*

Example B: Green ammonia plant, Sweden

Sector: Manufacturing – chemicals (NACE C20)
Investment value: $1.1bn
EUTSA objective: Climate change mitigation

*"In 2021, it was announced that a Spain-based fertilisers manufacturer is to open a **green ammonia and fertilisers manufacturing plant** in Norrbotten, Sweden, to serve the global market."*

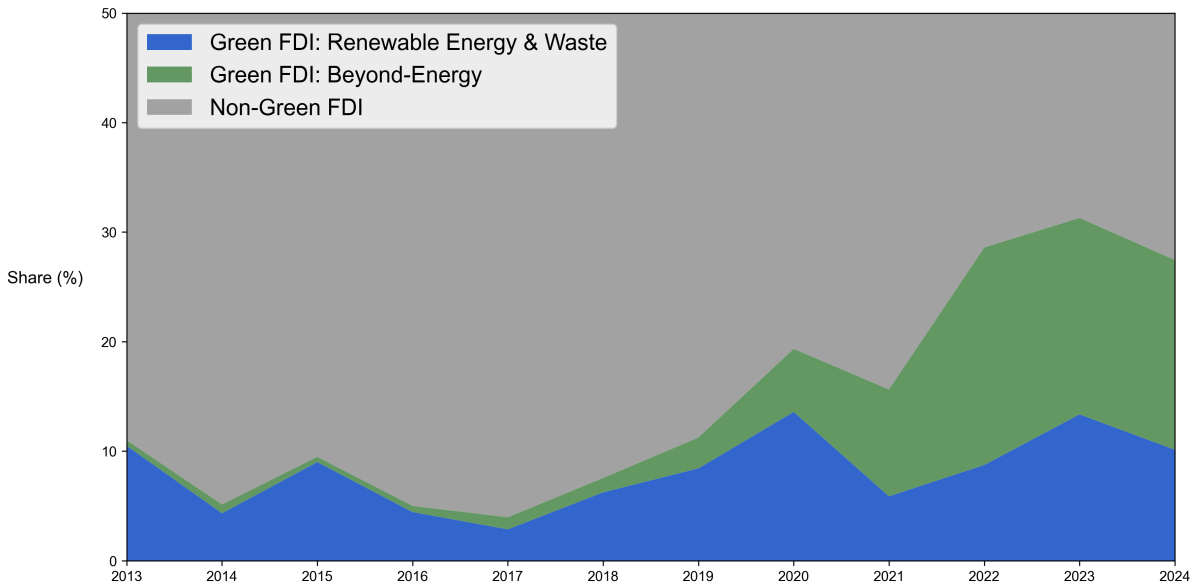
Source: Own elaboration using data from Orbis-Crossborder / Bureau van Dijk (2025).

It should be noted that, to assess the sensitivity of our findings, we also implement an alternative classification using a more 'lenient' LLM approach which provides more generic EU Taxonomy text (see Methods for more detail). This approach, as expected, identifies a larger share of FDI as green (21.3%, with 12.9% beyond energy). However, as our objective is to provide a conservative and robust identification of green FDI, we rely on the stricter classification as our main approach throughout this analysis. Results from the more lenient classification are presented in the Appendix as an 'upper bound', for comparison.

Looking at the time profile of green FDI, in Figure 3 below, it is possible to notice the stagnation in share of inward FDI into renewable energy, which has hovered at around 10% since 2013. This is in stark contrast with the steady growth of green FDI beyond renewable energy. By 2022–2024, more than a quarter of total inward FDI into the EU27+UK was green, with beyond-energy green accounting for the majority. While REW green FDI already represented a notable share 12 years ago, green FDI in other sectors was virtually absent until only five years ago. This mirrors recent evidence from UNCTAD (2023), BCG (2022) and (OECD, 2024*a*), which highlight growing inflows of green investment beyond energy and into broader infrastructure and manufacturing, such as the electric vehicle (EV) industry.[3]

[3]While sources such as (UNCTAD, 2023), BCG (2022) and (OECD, 2024*a*) usefully document growing FDI flows into sectors beyond renewable energy, these analyses do not apply a systematic, economy-wide classification framework. The criteria by which individual investments are labelled as green are not always transparent, and

Figure 3: Evolution in the share of 'green' inward FDI into EU27+UK (FDI capital value, 2013–2024).



Source: Own elaboration with data from Orbis-Crossborder / Bureau van Dijk (2025).

The share of FDI in REW sectors that we identify broadly aligns with the findings of previous research focusing on these sectors. For example, (Knutsson and Ibarlucea Flores, 2022) quantify the share of renewable energy FDI for OECD countries using data from the fDi Markets database for 2012–2021. Our estimates for REW green FDI broadly align with previous research. (Knutsson and Ibarlucea Flores, 2022), using fDi Markets data, report renewable energy shares of 21% in 2020 and 6% in 2021, consistent with our series for the same period. The World Bank (2023) similarly finds that global renewable energy FDI amounted to \$2.9 trillion (2013–2021), just over 20% of total FDI, though this figure includes geographies and investment types beyond our scope.[4]

For beyond-energy green FDI, no direct benchmark exists. The closest effort is UNCTAD's 2010 World Investment Report, which identified \$344bn of "low-carbon" investments in 2003–2009 (about 4.3% of total FDI).[5] This figure is broadly in line with our estimates for the earlier

coverage is typically limited to a selected set of sectors rather than spanning the full breadth of economic activity. None applies a technically grounded framework – such as the EU Taxonomy – that defines qualifying activities through explicit screening criteria at the activity level.

[4]According to (UNCTAD, 2024), global inward FDI for the 2013–2021 period (inclusive) totalled USD 14.3 trillion. Figures are available here: https://unctadstat.unctad.org/datacentre/dataviewer/US.FdiFlowsStock

[5]In the 2010 UNCTAD report, the authors evaluate 1,725 FDI projects occurring between 2003 and 2009 in three pre-selected "business areas" which they consider likeliest to contain "low-carbon FDI projects": alternative/renewable energy, recycling activities and environmental technology manufacturing. While this constitutes a notable effort to ascertain green FDI projects beyond energy in a bottom-up way, it does not cover FDIs in all sectors of the economy, as "*it is not feasible to scrutinize each individual FDI case to separate out those which are definitively low carbon from a total numbering some 22,000 in 2009 alone*" (UNCTAD, 2010 p. 111). As such, this constitutes a lower bound estimate of low-carbon FDI in the economy, since "*. . . there are also some low-carbon foreign investments in other industries and activities.*" (UNCTAD, 2010 p. 111). Moreover, in the absence of a comprehensive

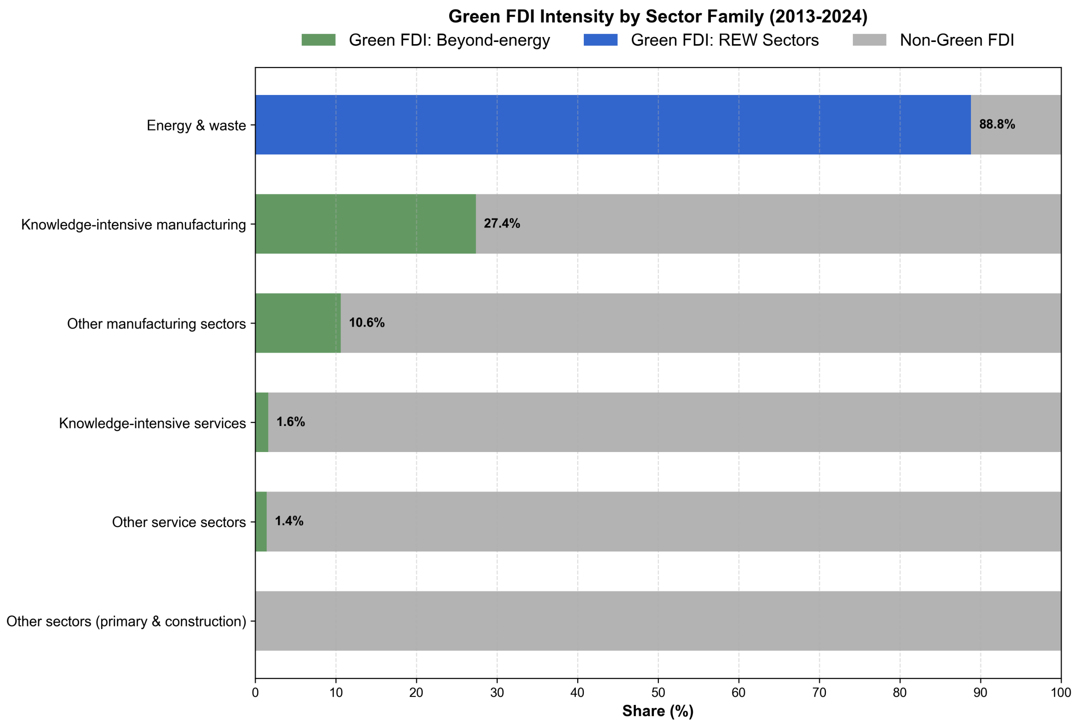
years of our series,[6] though methodological and geographic differences make strict comparison difficult.

2.2 Sectoral and spatial trends of green FDI in EU27+UK

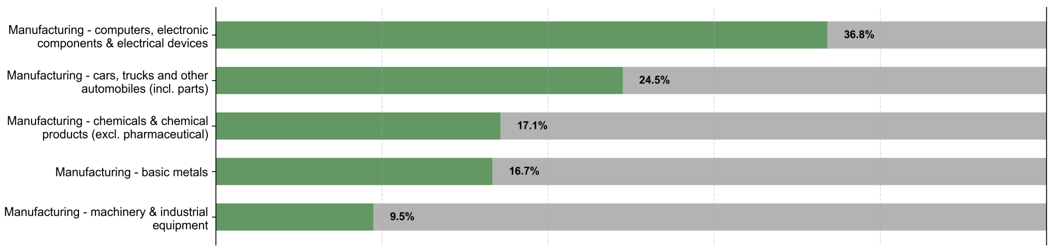
Beyond-energy green FDI is concentrated into manufacturing sectors, and in particular knowledge-intensive manufacturing.

Figure 4: Share of inward FDI into EU27+UK that is ‘green’ (FDI capital value, 2013–2024), by sector group (top) and top 5 manufacturing sectors by green FDI share (bottom).

(a) Share by broad sector group



(b) Top 5 manufacturing sectors by green FDI share



Source: Own elaboration with data from Orbis-Crossborder / Bureau van Dijk (2025).

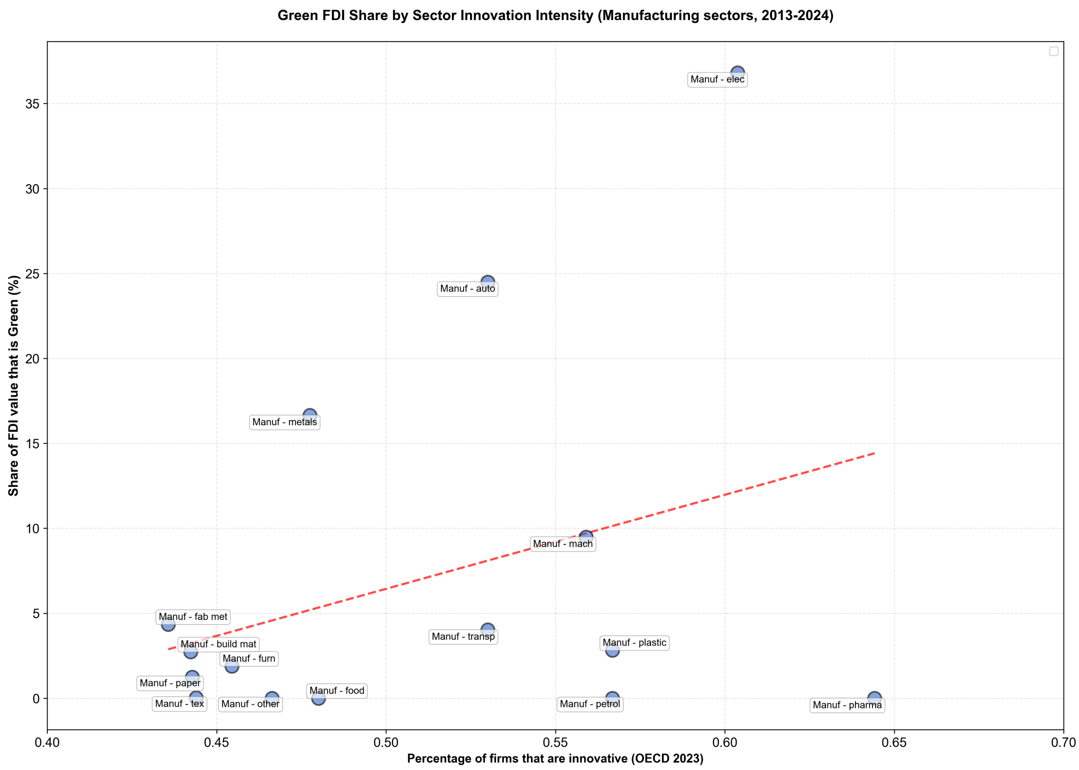
framework for defining green activities (such as the EUTSA), this exercise was based on relatively ad hoc search for evidence of low-carbon processes/technologies within each FDI project, which the authors note as an additional limitation and can have implications for the replicability and/or accuracy of such a classification.

[6] According to (UNCTAD, 2024), global inward FDI for the 2003–2009 period (inclusive) totalled USD 8.1 trillion. Figures are available here: `https://unctadstat.unctad.org/datacentre/dataviewer/US.FdiFlowsStock`

Figure 4 shows that 37% of FDI into manufacturing sectors between 2013–2024 was green, with especially high shares in knowledge-intensive sectors like "Computers, electronic & electrical devices" and "Cars, trucks and automobiles," where more than one-third of projects qualified. By contrast, in service sectors only ∼2% of inward FDI over the same period is classified as green.[7]

These sectoral patterns, in particular the higher share for knowledge-intensive sectors, suggest that being green beyond the energy sector is fundamentally linked to innovation capacity: the ability to develop, adopt, and implement new technologies and processes that enable low-carbon production, sustainable materials, or environmental solutions. This interpretation is supported by examining the relationship between green FDI and innovation across sectors, particularly manufacturing sectors. Figure 5 plots the share of green FDI beyond-energy in manufacturing sectors against the share of highly innovative firms in each sector (as classified by (OECD, 2024*a*)), revealing a positive correlation: sectors with higher concentrations of innovative firms attract substantially larger shares of green FDI.

Figure 5: Share of inward FDI into EU27+UK that is 'green' (FDI capital value, 2013–2024) in each manufacturing sector vs. share of innovative firms in sector (2023).

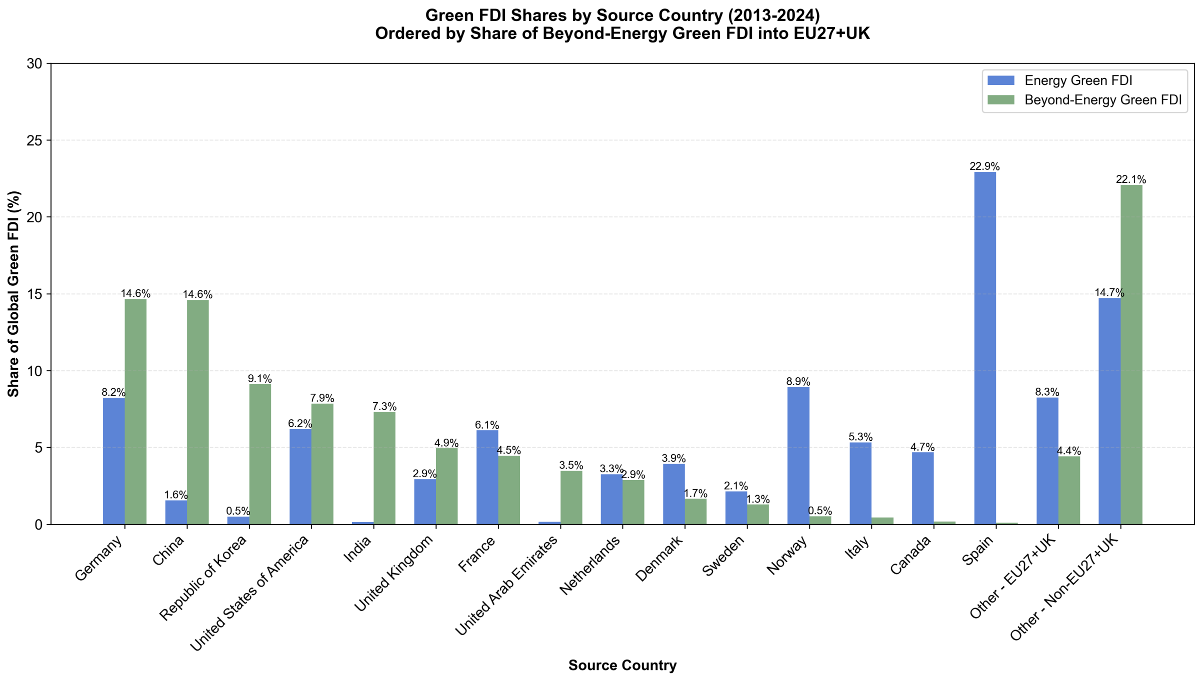


Source: Own elaboration with data from Orbis-Crossborder / Bureau van Dijk (2025); OECD (2024).

[7] Our more lenient classification approach (see Methods and Appendix) does identify a significantly larger share of services FDI as green, suggesting that our methodology is capable of identifying green service-sector FDI, but that meeting the Taxonomy's substantive contribution criteria in these sectors is inherently more challenging than in manufacturing. We believe this stricter approach is methodologically sound – requiring clear evidence of green contributions – though it likely represents a lower bound.

The origins of green FDI reveal distinct geographical patterns depending on the type of investment (Figure 6). Green FDI in REW sectors is predominantly intra-European, with the top source countries including Germany, France, Spain, and other EU27+UK nations, alongside Norway. By contrast, beyond-energy green FDI is more likely to originate from outside Europe. Among the top five source countries for non-REW green FDI are China, South Korea, the United States, and India – major global players in green manufacturing technologies such as electric vehicles, batteries, and advanced materials. This pattern suggests that while Europe's green energy transition is largely self-financed through regional capital, the broader industrial greening of the continent relies more heavily on extra-European investors, particularly from Asia and North America, who bring critical technologies and manufacturing capabilities.

Figure 6: Share of inward FDI into EU27+UK that is 'green' (FDI capital value, 2013–2024) by country of origin, energy vs. beyond-energy green FDI, top 15 source countries.

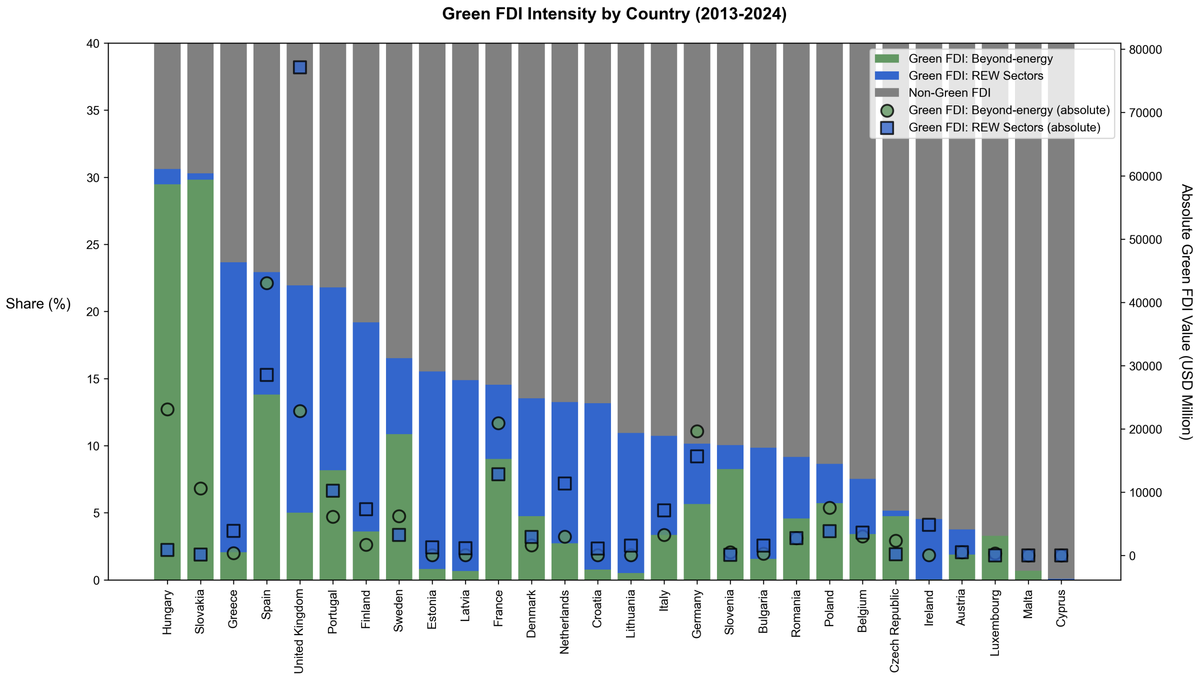


Source: Own elaboration with data from Orbis-Crossborder / Bureau van Dijk (2025).

Turning from the origins to the destinations of green FDI, country-level patterns vary sharply (Figure 7). Southern European countries such as Greece, Croatia and Portugal received large inflows of REW FDI projects, particularly solar. Similar REW-dominated patterns are visible in the Baltic states and, to a lesser extent, the UK, reflecting offshore wind investment. By contrast, Hungary and Slovakia stand out in terms of intensity of green FDI beyond the renewable energy sector, primarily stemming from large investments these countries have seen in 'green' automotive (EVs) projects. A few countries, including Spain, Sweden and France, attracted substantial shares of both REW and non-REW green FDI. At the other end of the spectrum, Austria, Ireland and the Czech Republic show some of the lowest green FDI shares in the region.

Moving beyond country-level aggregates is essential for understanding the true geography of

Figure 7: Value and share of inward FDI into EU27+UK that is 'green' (FDI capital value, 2013–2024), by country.



Source: Own elaboration with data from Orbis-Crossborder / Bureau van Dijk (2025).

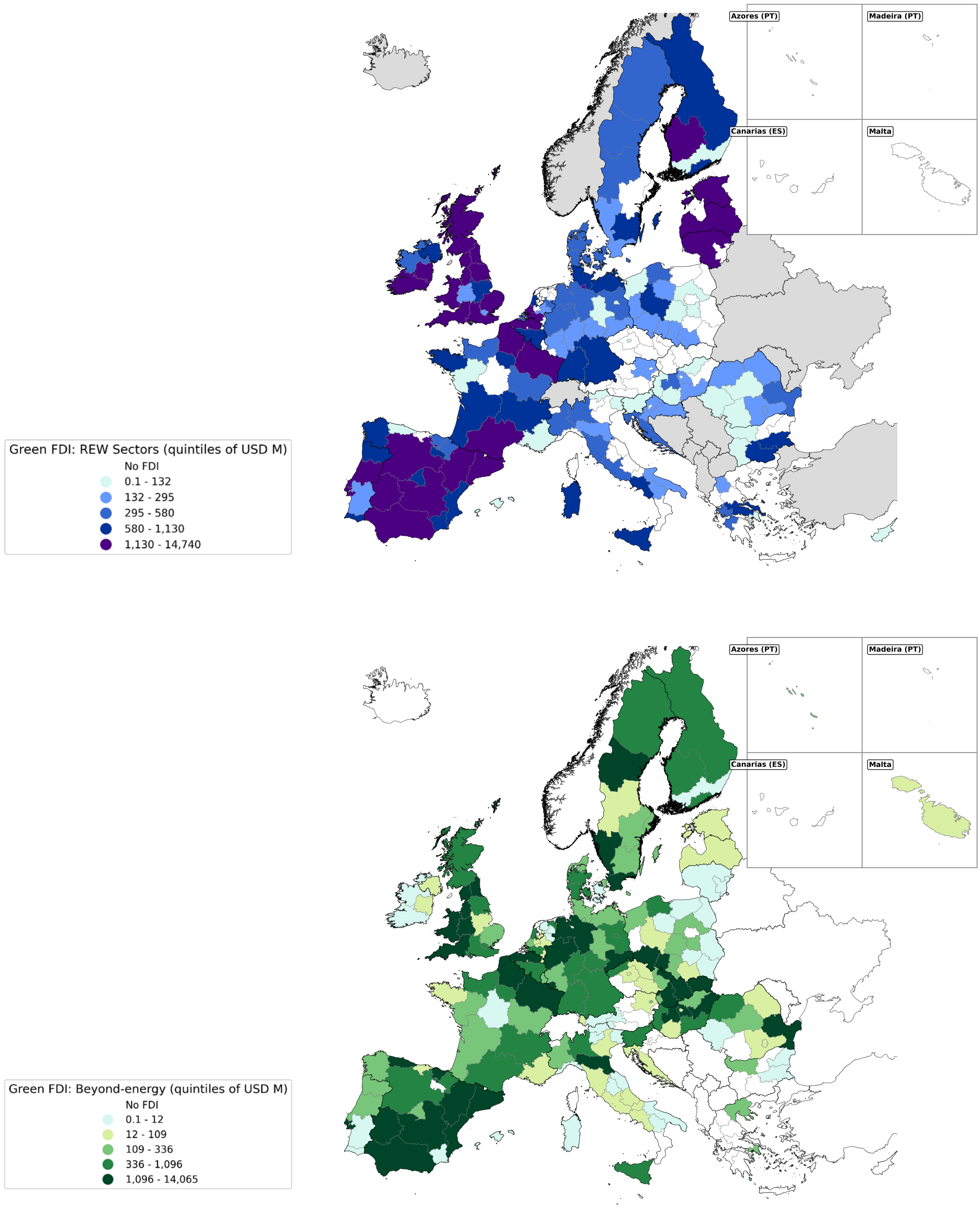
green FDI. Within any given country, some sub-national regions may be blessed with abundant wind or solar resources while others are not. The critical question is whether regions lacking these natural advantages are nevertheless harnessing FDI to support their green transitions – perhaps through different types of green investment more closely aligned with their existing industrial capabilities. Shifting the analysis to the sub-national[8] scale reveals precisely this pattern (Figure 8). REW investments cluster predictably in regions with favourable natural resources: sun-rich areas of Spain, Portugal, southern France, Greece and Italy, and windy areas around the UK, Ireland and the Baltic Sea. By contrast, beyond-energy green FDI shows a more diverse geography, unconstrained by geo-physical factors. Eastern Europe's industrial regions – notably in Hungary, Slovakia and Poland – have attracted large inflows into green manufacturing, particularly EVs and batteries. In Western Europe, advanced metropolitan regions such as Catalonia, Flanders, North-West England and North Rhine–Westphalia received a broader mix, combining green manufacturing with R&D, professional services and IT activities.

Our analysis reveals regions that are attracting substantial volumes of green FDI, yet have remained largely undetected in previous green investment statistics. These investments contribute meaningfully to environmental objectives and bring essential capital, skills, and knowledge to support local transitions toward a greener, low-carbon economy (Rogelj et al., 2018; UNCTAD, 2023).

Figure 9 plots the share of green FDI into European regions, distinguishing between REW

[8]NUTS level 2 or 3, depending on subnational administrative geography (see SI for more detail).

Figure 8: The regional distribution of green FDI (USD value) in EU27+UK: Renewable Energy & Waste FDI (top panel) and beyond-energy green FDI (bottom panel).



Source: Own elaboration using data from Orbis-Crossborder / Bureau van Dijk (2025).

Sector green FDI on the horizontal axis and green FDI beyond renewables on the vertical axis. The size of each circle represents total regional FDI (both green and non-green). Moreover, the colour of each circle denotes the region's potential for renewable energy generation, with regions in the top quintile in this measure shown in red and those in the lowest in blue, based on the 2019 ENSPRESO dataset of solar and offshore/onshore wind potential capacity for European regions, built by the Joint Research Centre (further detail on how this measure was computed is provided in the Supplementary Information).

The top-right quadrant identifies regions that are leaders in both types of green FDI, ranking within the top two quintiles for both REW green FDI and beyond-energy green FDI. These regions represent sub-national economies where investment in renewable energy complements (and potentially enables) broader green activities across other sectors. They are the *Green Leaders*, making up just under 10% of the EU27+UK population. Such regions are potentially leveraging the availability of green energy to generate green products and services in other sectors of the economy.

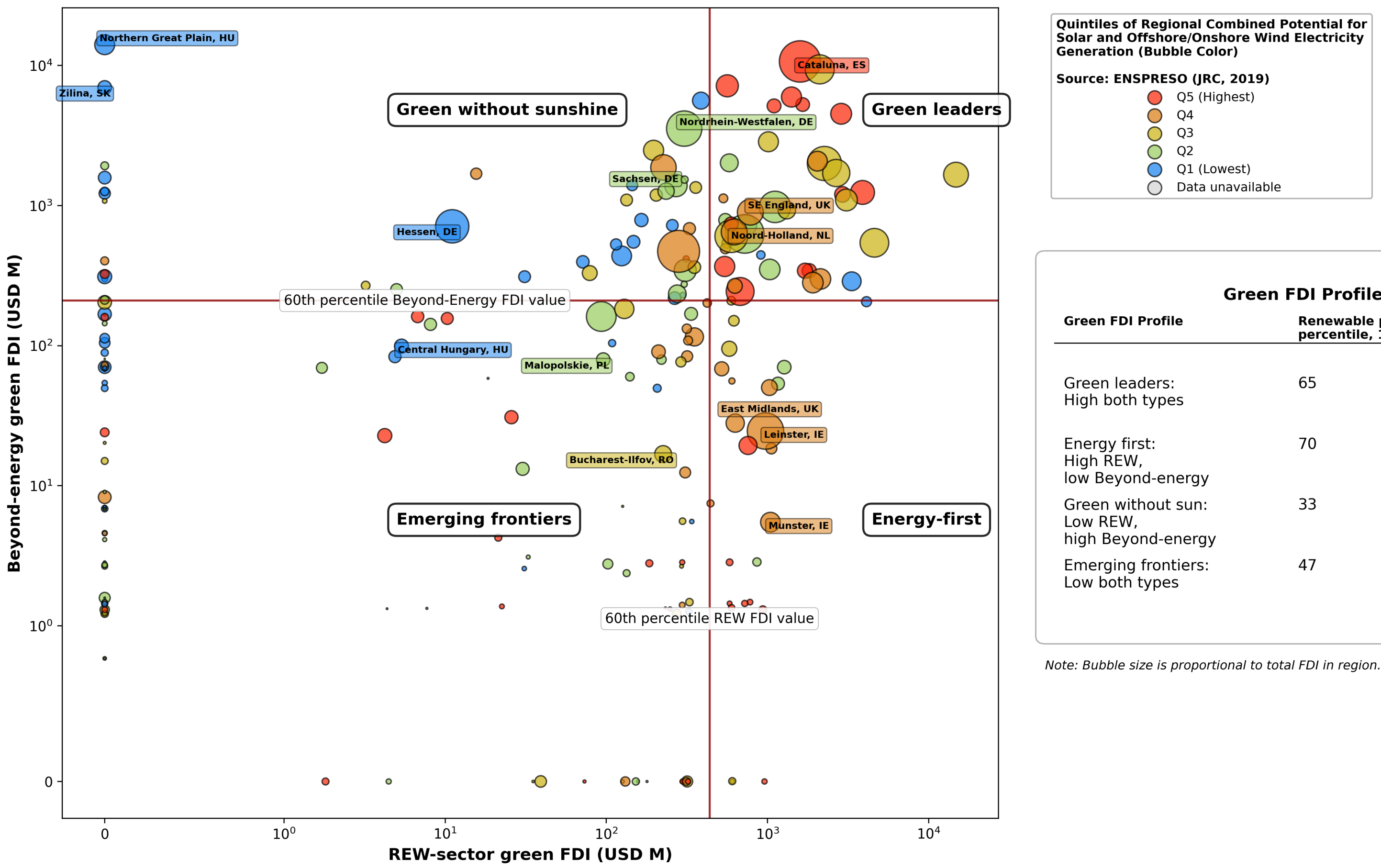
The bottom-right quadrant highlights regions that are strong performers in attracting REW Sector green FDI but record comparatively low levels of beyond-energy green FDI. These *Energy-First* regions are often characterised by abundant natural resources – wind or solar – that make them natural magnets for renewable energy investment. However, some regions are sparsely populated or lack an industrial base, which structurally limits their capacity to attract green FDI beyond energy generation; these areas will continue to supply clean power to wider industrial ecosystems. By contrast, regions with stronger industrial capabilities could, with targeted policy efforts, attract complementary green investments in manufacturing and other sectors and capture broader local gains from the green transition.

The top-left quadrant showcases regions that are leaders in attracting green FDI beyond the REW Sectors, despite not being major renewable energy hubs. These regions demonstrate that economic greening can occur independently of local renewable energy generation potential. Often lacking abundant natural assets such as sun or wind, they nevertheless play a pivotal role in the green transition by hosting FDI in green manufacturing and other sectors where they hold traditional strengths. These are the *Green Without Sunshine* regions – proof that industrial capabilities and innovation ecosystems can drive green transformation even in resource-constrained contexts. *Green Without Sunshine* regions account for 28.7% of the EU27+UK population. By attracting green FDI despite varying levels of renewable energy potential, they reveal a broader land of opportunity for greening through private investment – previously overlooked due to limitations in conventional FDI tracking. This wider landscape brings 125 million EU citizens into the fold of the green transition.

Finally, the bottom-left quadrant includes regions with low levels of both REW Sector and beyond-energy green FDI. However, this quadrant is far from homogeneous. It comprises two distinct types of regions. First, those with limited capacity to attract REW Sector FDI but which may still leverage existing inflows of other green FDI, particularly in niche manufacturing or service sectors. Second, regions with emerging potential in both dimensions, where a combination of modest renewable energy resources and nascent green industrial activity could form the basis for future growth. These areas represent the *Emerging Frontiers* of the green transition – regions where targeted policy support and strategic investment promotion could unlock latent potential and catalyse broader greening (with or without favourable natural conditions in terms of renewable energy).

Figure 9: Green FDI by region in EU27+UK (2013–2024): Share of REW Sector green FDI vs. Share of Beyond-Energy Green FDI (FDI USD value), symmetric log scale.

Green FDI by EU27+UK region (2013-2024): REW-sector green FDI vs beyond-energy green FDI, USD M (Symmetric Log Scale)



Green FDI Profile Statistics

Green FDI Profile	Renewable potential percentile, 100 highest	Num. regions & population
Green leaders: High both types	65	40 regions, 197.7M pop (34.6% of EU27+UK)
Energy first: High REW, low Beyond-energy	70	24 regions, 55.5M pop (9.7% of EU27+UK)
Green without sun: Low REW, high Beyond-energy	33	47 regions, 124.8M pop (21.8% of EU27+UK)
Emerging frontiers: Low both types	47	123 regions, 193.3M pop (33.8% of EU27+UK)

Note: Bubble size is proportional to total FDI in region.

Source: Own elaboration using data from Orbis-Crossborder / Bureau van Dijk (2025).

3. Discussion

This study develops a novel taxonomy-guided LLM pipeline to identify green FDI at the project level, addressing a core measurement constraint in research and policy on the green transition. By operationalising the EU Taxonomy for Sustainable Activities (EUTSA) on large-scale investment descriptions, we show that project-level classification can be conducted with high accuracy and reproducibility. This approach moves beyond the prevailing focus on REW sectors as the sole proxy for green FDI, revealing a broader spectrum of green activities across manufacturing and services.

Our findings show that nearly one in six dollars (15.7%) of inward FDI into the EU27+UK between 2013 and 2024 contributes to decarbonisation objectives, with approximately half of this green FDI in sectors outside renewable energy and waste (REW). The sectoral composition uncovered here is therefore inconsistent with the prevailing tendency to equate green FDI with renewables alone and highlights the growing role of international capital in transforming a wider array of economic activities (Ghodsi et al., 2025). In particular, the expansion of green FDI into electric vehicles, battery manufacturing, and other energy-intensive industries suggests that FDI is increasingly enabling low-carbon transformation from within sectors traditionally associated with high emissions (Hertwich and Wood, 2018).

A long-standing debate in the literature concerns whether FDI promotes environmental upgrading in host economies or instead relocates pollution to jurisdictions with weaker standards (Cole et al., 2006; Eskeland and Harrison, 2003; Marin and Vona, 2019). Engaging rigorously with this question has been hampered by the absence of systematic, project-level data on which investments are genuinely green. By providing a scalable method to identify green FDI across all sectors, this paper offers a measurement foundation that future research can use to test these hypotheses more directly – for instance, by linking classified green investments to firm-level productivity, innovation or emissions outcomes in host economies.

The positive correlation between sectoral innovation intensity and green FDI share underscores that greening beyond renewable energy is closely tied to innovation capacity – the ability to develop, adopt, and implement new low-carbon technologies and processes. This reflects the centrality of industrial and technological transformation in the early stages of the green transition. The finding that innovation-intensive sectors attract disproportionately greater quantities of green FDI suggests that the green transition in manufacturing is not simply about capital reallocation, but about deep technological change requiring significant R&D capabilities and organisational innovation.

The macro-regional origins of green FDI further illuminate the nature of this transition. While renewable energy investments flow predominantly from within Europe – reflecting regional expertise and policy alignment – green FDI beyond the energy sector relies heavily on extra-European sources, particularly from China, South Korea, the United States, and India. This pattern reveals an important dimension of the global green economy: Europe's broader industrial greening is critically dependent on international technology transfer and manufacturing know-how from leading economies in green technologies. The dominance of Asian investors in non-REW green FDI, especially in electric vehicles and battery manufacturing, highlights how global value chains are reshaping the geography of the green transition (Pavlínek, 2023; UNCTAD, 2023). This dependency also raises questions about technological sovereignty and the strategic implications of relying on external actors for critical green industrial capabilities.

The origins of green FDI tell only part of the story; equally important is where these investments land and why. Recent work has shown that multinational firms systematically avoid host countries with elevated physical climate risks (Li and Gallagher, 2022), suggesting that climate considerations already influence where FDI lands. Our findings extend this picture: renewable energy investments tend to reflect endowments and planning constraints, whereas beyond-energy green FDI is concentrated in activities where location advantages are shaped by industrial capabilities, supply-chain positioning and innovation ecosystems. The destination patterns we document suggest that regions lacking wind or solar advantages may nonetheless participate meaningfully in the green transition through capability-based pathways, including process upgrading and enabling inputs. For example, areas with strong cultural or environmental heritage may resist disruptive energy projects but can still attract green FDI into low-impact sectors such as R&D or professional services. Likewise, industrial regions can pursue transformation from within by targeting low-carbon variants of traditional sectors, rather than abandoning them altogether. This perspective aligns with recent work emphasising the importance of regional upgrading through the evolution of existing sectors toward higher value-added and greener activities (Crescenzi and Harman, 2023). This research highlights how regions can leverage internationalisation, FDI and Global Value Chains to foster green upgrading, rather than relying solely on sectoral shifts or internal natural resources (Crescenzi and Harman, 2026).

The EU27+UK provides an ideal testbed for this first application of our methodology. The European Union offers a unique combination of regulatory homogeneity, a single market for capital and investment, and a stable policy commitment to sustainability through the European Green Deal. The development of the EUTSA specifically for the EU context further strengthens the reliability and relevance of our classification framework. In contrast, other jurisdictions – such as the United States – have experienced more frequent shifts in their sustainability policy commitments,[9] which may complicate longitudinal analysis. Nonetheless, the growing international interest in sustainability taxonomies suggests that this approach can be scaled globally, with appropriate adaptation to local regulatory and institutional contexts.

The LLM-based classification framework developed here is both scalable and adaptable. It can be applied to other jurisdictions with sustainability taxonomies or ESG frameworks, enabling comparative analyses and supporting global efforts to monitor and promote green investment. This approach offers a powerful tool for real-time tracking of green investment

[9]At the time of writing, President Trump had just withdrawn the United States from the Paris Climate Agreement for the second time: https://www.bbc.co.uk/news/articles/cp80ln97py5o

flows, with potential applications in evaluating corporate sustainability strategies, informing development planning, and guiding investment promotion in emerging economies. Its ability to detect nuanced patterns across sectors and geographies makes it particularly valuable for researchers and policymakers seeking to align investment with climate and sustainability goals.

Several limitations should be acknowledged. Project descriptions in greenfield FDI databases are typically derived from press releases and news reports, and can be brief and lacking in technical detail. Given the granularity of the EU Taxonomy's screening criteria, such descriptions may omit precisely the details needed to confirm or exclude classification, potentially leading to misclassification through omission rather than error. Future work could address this by supplementing investment descriptions with additional firm- or project-level documentation where available. Moreover, taxonomy criteria evolve over time and differ across jurisdictions, requiring periodic updating and careful calibration when extending the approach beyond Europe. Future research can strengthen the empirical and policy relevance of this framework by linking classified green investments to firm-level outcomes (innovation, productivity, employment), integrating richer technical documentation where available, and adapting the pipeline to other sustainability taxonomies and data sources to support comparative monitoring. Finally, the rapid pace of LLM development means that any implementation reflects a snapshot in time. Newer model generations – including those with extended reasoning capabilities – operate on different architectures and prompting techniques[10] than 'standard' token prediction LLMs. Systematically evaluating and adapting the framework to such models including continual human review represents a natural direction for future work.

4. Methods

4.1 Data: Orbis-Crossborder FDI and the EUTSA

Our analysis combines two sources: project-level inward FDI data and a comprehensive classification of green activities.

The FDI data come from Orbis-Crossborder (Bureau van Dijk/Moody's Analytics), which records announced FDI projects worldwide. We focus exclusively on greenfield FDI in the EU27+UK between 2013 and 2024. While Orbis-Crossborder also captures mergers and acquisitions (M&As), we exclude these from our analysis because the literature suggests that greenfield FDI has greater potential for technology spillovers and knowledge transfer to host economies compared to M&As (Javorcik, 2004; Wang and Wong, 2009). Given that our conceptual motivation for studying green FDI emphasises its role both as a source of finance and as a conduit for transferring green capabilities and knowledge, this distinction matters:

[10]See OpenAI's guide on 'prompting reasoning models': https://developers.openai.com/api/docs/guides/reasoning-best-practices/

greenfield investments are more likely to introduce new technologies, processes, and practices that can diffuse to local firms and workers.

Within the universe of inward FDI into the EU27+UK, we retain project 'types' listed as 'new' investments, 'expansions', 'co-locations' or 're-locations'. We exclude 'rumoured' projects that lack confirmation. We also exclude sectors that broadly constitute the public sector, as these fall outside the scope of private FDI flows. After applying these filters and removing records with insufficient information, our sample comprises 109,084 greenfield projects, of which approximately 89% contain short descriptive text. These descriptions typically consist of 100–800 characters detailing the investor, activity, technology, and location, and form the basis for our text-based classification. Further details on data cleaning and filtering procedures are provided in the Supplementary Information (SI).

We rely on Orbis-Crossborder because it provides narrative descriptions at the project level, which are essential for linking investment activities to green criteria. Other widely used datasets, such as fDi Markets (used by UNCTAD and in recent academic work), do not supply such textual descriptions and therefore cannot support bottom-up, activity-level classification.

To define what counts as "green" beyond projects in the Renewable Energy and Waste (REW) sector, we draw on the EU Taxonomy of Sustainable Activities (EUTSA). The EUTSA was developed over four years by a technical expert group and defines activities considered green within the EU, focusing on those sectors that significantly impact GHG emissions and economic output. This taxonomy not only serves as a financial reporting tool, introducing mandatory disclosure obligations for companies and investors, but also aims to enable the creation of credible financial products that can support the green transition (OECD, 2020).

The EUTSA takes a comprehensive view of what can be considered green by considering the potential contribution toward six environmental objectives: climate change mitigation, climate change adaptation, sustainable use and protection of water and marine resources, transition to a circular economy, pollution prevention and control, and protection and restoration of biodiversity and ecosystems. An activity qualifies as green if it makes a substantial contribution to at least one of these objectives while doing no significant harm to the others (European Commission, 2021).

Critically, the EUTSA distinguishes between activities that are inherently low-carbon or environmentally beneficial through their own performance (such as renewable energy generation or electric vehicle manufacturing), and enabling activities that facilitate the green transition in other sectors (such as manufacturing energy-efficient building components or providing environmental consulting services). Given our endeavour of identifying green activities beyond 'own-performance' green sectors like renewable energy, this expanded conceptualisation of what 'green' can entail constitutes an essential feature of the EUTSA.

For each qualifying activity, the EUTSA provides detailed technical screening criteria that specify thresholds and standards – for example carbon intensity limits for steel production, energy performance requirements for building renovation, or emissions standards for transport equipment. Across the 87 NACE 2-digit codes in our sample, 60 contain at least one green activity as defined by the Taxonomy. This broad sectoral coverage enables us to systematically identify green FDI beyond renewable energy, capturing activities such as green steelmaking, sustainable ammonia production, building renovation, environmental due diligence consulting, and aviation using LEAP engines. Concrete examples of classified projects are provided in the SI, alongside further detail on the Taxonomy structure and criteria.

Before processing, we collapse and combine the EUTSA variables at the NACE 2-digit level, so that the title, description and substantial contribution criteria of each green activity are provided for each code. We also create an alternative set of NACE sector-level green activities based on the EUTSA descriptions only, excluding the more technical substantial contribution criteria, which enables us to test the identification of green projects when using an alternative set of shorter and more generic EUTSA information. These green activity sets are then matched to FDI projects, sourced from Orbis-Crossborder as described above, according to the primary NACE code of the FDI project.

The dataset reports both the capital value (USD) and the number of projects. In the main analysis we use capital value as the unit of measurement, as it best captures the scale of investment flows. All results are replicated using project counts, which are presented in the SI to demonstrate robustness.

Finally, for descriptive purposes we distinguish projects in Renewable Energy and Waste (REW) – corresponding to NACE sections D[11] (Electricity, gas, steam and air conditioning) and E (Water supply; sewerage; waste management and remediation) – from projects in all other sectors. In our methodology, REW projects are automatically classified as green, while all other projects require evaluation through the bottom-up approach described in the following section.

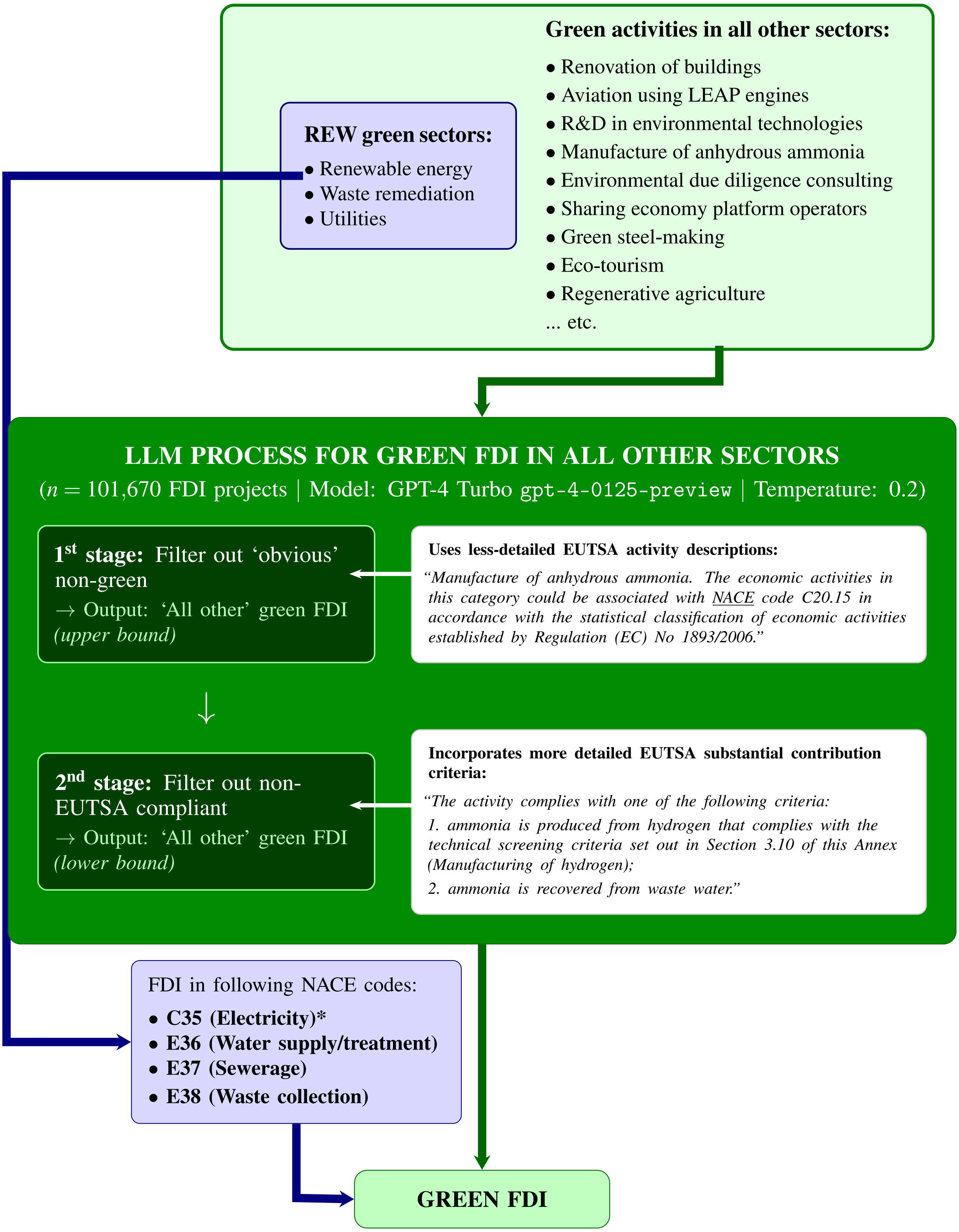
4.2 Processing: Prompting the LLM to classify FDI projects in line with EUTSA criteria

Our process for classifying green FDI is summarised in Figure 10. Green FDI in REW sectors is straightforwardly identified based on each FDI project's sector label, as described in the previous section.

To identify which FDI projects outside of the REW sector nevertheless involve green activities, we employ the GPT-4 LLM (hereafter referred to as "the LLM"). For each non-REW FDI project in our dataset where a textual description is provided, we build a custom prompt combining this description plus the EUTSA text on green activities for the corresponding project sector. Each prompt is submitted to the LLM via OpenAI's API, with temperature parameter set relatively low at 0.2 (in line with guidance for textual analysis tasks; (Renze and Guven,

[11]This sector includes both fossil-fuel and renewable electricity generation. Fossil-fuel investments – including natural gas, which we exclude notwithstanding its 'transitional' EU Taxonomy status – were identified via an additional BvD sector classification field and removed from the green category (details provided in SI and replication code). Nuclear energy is retained despite not being renewable in the strict sense (IRENA, 2024), consistent with the EU Taxonomy's recognition of nuclear as a low-carbon activity.

Figure 10: Process for classifying FDI in REW sectors and green FDI projects in all other sectors.



Source: Own elaboration.
**We keep only electricity generation using renewable energy sources (in line with IRENA definitions), plus nuclear fission* [a]

[a] Although nuclear is not a renewable energy source, we include this out of consistency with the EU Taxonomy's recognition of nuclear as a low-carbon activity.

2024*a*)). This, together with the explicit instruction to return a binary ('Yes'/'No') response with a short justification, means that the LLM outputs were consistently structured and it was not necessary to explicitly enforce this. The provision of a justification for each decision which cites the relevant parts of the EUTSA activities or project description[12] significantly facilitated the human checks which are discussed in the following section.

The processing of FDI projects through our LLM methodology involved two stages, as illustrated within the pale blue box of Figure 10 above.[13] Firstly, every FDI project description is processed with respect to the 'simpler', less technically-detailed EUTSA text. This provides an initial identification of projects where green activities are present according to its description. Secondly, any projects identified as green in this first stage are re-processed through the LLM, but this time with respect to the EUTSA green criteria that incorporates information from the 'substantial contribution criteria' variable. Any projects which 'survive' this second step, in the sense of still being labelled as green, are classified as green FDI in our main classification. The rationale for applying this secondary stage is to ensure that projects are not only associated with broadly defined green activities, but also provide some evidence of meeting the more rigorous technical screening criteria of the EUTSA, thereby reducing false positives and aligning our main classification with the taxonomy's full compliance requirements. Further detail on these stages, including the 'upper bound' classification results, are provided in SI.

4.3 Evaluation: Verifying the accuracy, validity and reproducibility of GPT-4 outputs

We evaluated the classification methodology along three axes: accuracy, validity, and reproducibility. Extended figures, tables, and full prompt texts are provided in the Supplementary Information (SI).

[12]While the nature of the dataset requires that we take these descriptions at face value in order to be able to classify at all, there is of course always the possibility that companies might be misrepresenting their activities by, for instance, including generic key words that merely 'sound' green. We do not view this as a significant concern, given the inherently technical and specific language in EUTSA that forms the basis for classification. Moreover, any overt falsehoods where a company refers to a specific, technically named green activity or technology that in fact does not exist would be uncovered in future audits. The EUTSA is after all a financial accounting tool, and false reporting could be considered a form of fraud, so we do not consider this to be a significant-enough risk to bias our classification.

[13]The two-stage design was adopted primarily for cost efficiency: the first stage uses shorter, less token-intensive prompts based on the more generic taxonomy descriptions, thereby reducing the total cost of processing the full sample of 109,084 projects. Only projects flagged as potentially green at stage one – a subset of the full sample – are then passed to the more expensive stage-two prompts incorporating the detailed substantial contribution criteria. Researchers working with smaller samples may find this two-stage approach unnecessary. The replication package therefore includes an option to bypass stage one and process all projects directly against the fuller taxonomy criteria, which may be preferable where the total classification cost is manageable or where the user wishes to simplify the pipeline.

4.3.1 **Accuracy**

Human evaluation is widely regarded as the gold standard for assessing text classification and generation tasks (Celikyilmaz et al., 2021; Clark et al., 2021; Reiter and Belz, 2009; van der Lee et al., 2021). We therefore conducted an evaluation exercise with 34 postgraduate students trained in applying the EU Taxonomy (EUTSA) criteria. Each evaluator received 10 projects drawn at random from a pool of 340 FDI projects (170 classified as green and 170 as non-green by the model in the first-stage classification). Evaluators first classified the projects *blind* (without seeing the model output). They were then asked to repeat the classification after being shown the model's decision and justification. This design allowed us to measure both baseline agreement and whether the model's rationale improved alignment with human reasoning. Performance was evaluated using the F1 score, a standard measure balancing precision and recall for binary classification (Chang et al., 2023). Across 340 blind evaluations of randomly sampled projects, GPT-4 achieved an F1 score of 0.84 (cf. (Besley et al., 2026), who report human-LLM agreement rates of 81%–86% in LLM-based classification of policy narratives using a structured taxonomy). When evaluators were subsequently shown the model's decision and justification, agreement rose further, yielding an F1 of 0.93. This increase indicates that in cases of initial disagreement, evaluators often recognised the model's reasoning as correct after reviewing its detailed application of the sector-specific EUTSA criteria. These scores compare favourably with mean F1 scores of 0.70–0.81 reported by (Asirvatham et al., 2026) across 1,141 binary classification tasks spanning over 300 text datasets, suggesting that our taxonomy-guided pipeline performs at or above the level typically achieved by GPT-based classifiers on structured labelling tasks. Details of the Qualtrics survey setup, randomisation, and instructions provided to evaluators are included in the SI.

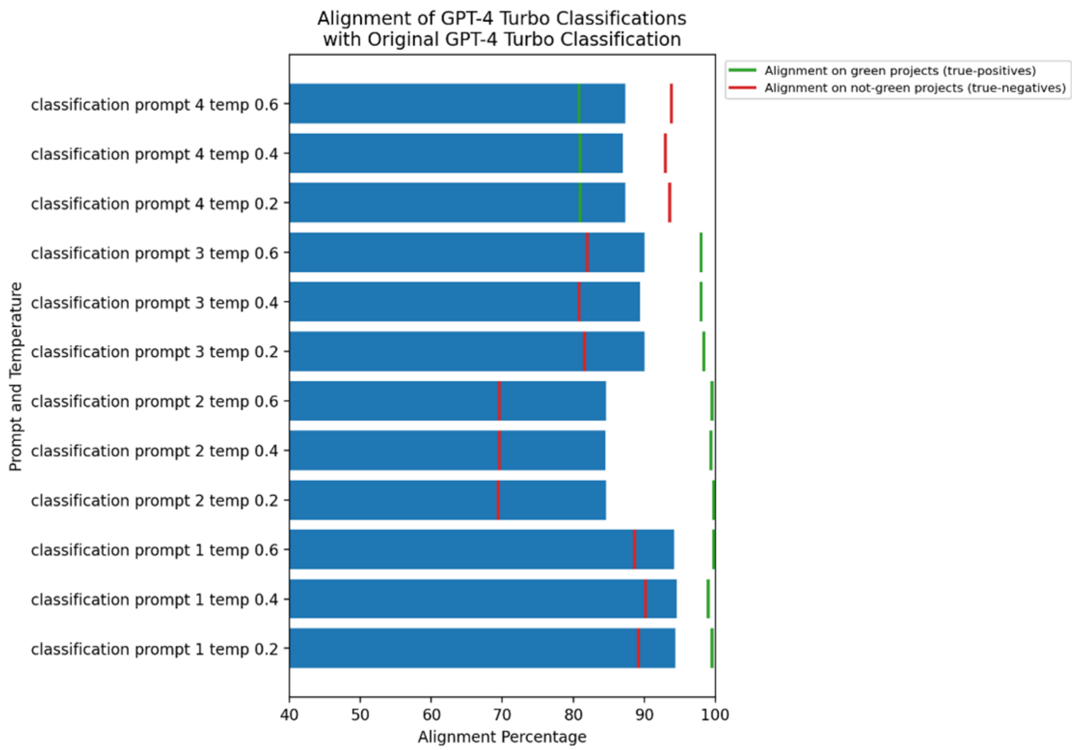
4.3.2 **Validity**

To assess whether the model's decisions were grounded in the EUTSA criteria provided in the prompt, rather than extraneous knowledge, we implemented three complementary checks. First, a simple keyword search confirmed that almost all justifications explicitly referenced the "EU Taxonomy," indicating that the model references the appropriate source in its decision-making. Second, we applied a BLEU similarity analysis (Papineni et al., 2002), comparing each justification against the relevant sectoral EUTSA text. For a sample of 1,000 projects, BLEU scores were systematically higher when using the correct sector text as a reference compared to incorrect sector texts, showing that the model's justifications drew on the appropriate taxonomy definitions. Further detail and results from the BLEU analysis are provided in the SI. Third, we re-processed the same 1,000 projects using "self-reflection" prompts (Asai et al., 2023; Renze and Guven, 2024*b*), which required the model to explicitly review and justify its own prior decision.

4.3.3 **Reproducibility**

Finally, we tested the robustness of outputs to changes in prompt style, parameterisation, and model type. Using the same 1,000 projects, we trialled three alternative prompt formulations (from concise instructions to persona-based prompts) and three temperature settings (0.2, 0.4, 0.6), generating 11 variations on the original setup. Reproducibility was evaluated by measuring the percentage of classifications consistent with the original outputs across all variations. Unsurprisingly, reproducing our results with exactly the same model, prompt and temperature setting yielded the highest agreement, at around 95% (see lowest bar in Figure 11). That said, even when adjusting the prompt, agreement rates stay above 85%. This is in line with evidence from (Asirvatham et al., 2026), who similarly find that the accuracy of labelling results from LLMs does not rely on the exact prompting strategy applied. Critically, the rate of 'true positives' was almost 100% in three out of four prompts, suggesting that our approach is cautious in classifying projects as green; far from 'green-washing', it is likely providing an overall 'lower-bound' of green FDI patterns.

Figure 11: Alignment percentages when re-processing subsample of 1,000 FDI projects through GPT-4o, for 12 prompt-temperature combinations.



Note: Bottom bar = alignment when GPT-4 Turbo was re-tested with the same prompt and temperature as the original.
Source: Own elaboration.

We also re-processed projects across multiple alternative models, including two models by alternative providers (Google and Anthropic) to examine whether performance depends on intrinsic model capabilities or size. These results, shown in the SI, exhibit similar patterns: alignment with the original classification hovers at around 90%, and reaches 95% for true-positives.

Together, these exercises provide a systematic evaluation of the classification pipeline: accuracy through human benchmarking, validity through text-similarity and self-reflection checks, and reproducibility through robustness tests across prompts, parameters, and models. The results confirm that GPT-4 can apply EUTSA criteria to classify FDI projects with high accuracy, valid reasoning, and reproducibility, underscoring the robustness of our methodology and the potential of LLMs as reliable tools for classification, with potential application in geographies beyond EU27+UK and remits beyond FDI specifically. Further details and results are reported in the SI, while the scripts used are available in the replication package.

Data availability

The foreign direct investment (FDI) project-level data used in this study were obtained from Orbis-Crossborder (Bureau van Dijk / Moody's Analytics) under institutional licence. These data are subject to contractual restrictions and cannot be shared publicly. Researchers with authorised access to Orbis-Crossborder can reproduce the dataset by exporting project-level greenfield FDI records using the template and instructions provided in the replication repository (access granted upon request). Information on accessing Orbis-Crossborder is available at: https://www.moodys.com/web/en/us/capabilities/company-reference-data/orbis.html.

The EU Taxonomy of Sustainable Activities (EUTSA) criteria used to construct sector-level classification inputs are publicly available from the European Commission's Taxonomy Compass: https://ec.europa.eu/sustainable-finance-taxonomy/taxonomy-compass/the-compass. The raw Compass export used in this study (February 2025 download) and the curated sector-level taxonomy mappings (strict and basic variants) are archived in the replication repository.

The ENSPRESO renewable energy potential data are publicly available from the European Commission Joint Research Centre at: https://data.jrc.ec.europa.eu/collection/id-00138. OECD innovation indicators are available at: https://www.oecd.org/en/data/datasets/business-innovation-statistics-and-indicators.html.

All publicly shareable input datasets, including the curated taxonomy mappings and external covariates, are archived at https://doi.org/10.5281/zenodo.18702402 and are available upon request. The repository enables full end-to-end reproduction of the classification and analysis pipeline for users with authorised access to Orbis-Crossborder data.

Orbis-Crossborder is a dynamic commercial database that is periodically updated by the provider as new investment information becomes available and as project statuses are revised (e.g., postponements, cancellations, revised investment values, or changes in the USD exchange rate). As a result, exact replication of project counts and aggregate statistics may vary slightly if a new extract is downloaded at a later date, even when using identical geographic and temporal filters. The replication repository therefore includes detailed export instructions and a documented extraction date to facilitate close reproduction. Differences arising from database updates do not reflect instability of the classification methodology but rather retrospective revisions to the underlying source data.

Code availability

All code used to process Orbis-Crossborder exports, harmonise NACE sector mappings, construct taxonomy-guided prompts, query the large language model (LLM), post-process model outputs, construct master datasets, reproduce all analyses and figures, and implement the evaluation exercises described in the manuscript is available upon request via https://doi.org/10.5281/zenodo.18702402 or by contacting the authors directly.

The repository provides a modular Python pipeline with a command-line interface covering all stages from data processing and LLM classification through to descriptive analysis and evaluation. To support partial reproduction without access to licensed Orbis-Crossborder data, the repository additionally ships pre-computed intermediate outputs for the full EU27+UK sample. These pre-loaded files contain only derived evaluation metrics – such as project-level BLEU scores, self-reflection results, and cross-model classification outputs – and do not include any licensed variables or project-level FDI descriptions. Users can therefore reproduce all evaluation results reported in the manuscript – including F1 scores, BLEU faithfulness checks, self-reflection agreement rates, and cross-model reproducibility analyses – directly from these files without requiring access to the underlying Orbis-Crossborder data or additional API calls. Reproduction of the descriptive analyses and figures reported in the Results section, however, requires an authorised Orbis-Crossborder extract, as these are based directly on licensed project-level variables.

LLM classification was executed via the OpenAI API using the model `gpt-4-0125-preview`, with temperature set to 0.2 and a fixed seed (2902). Structured JSON output mode was enforced to standardise response formatting. The repository records all parameters required to reproduce the classification and evaluation procedures, including model identifier, prompt version, temperature setting, worker configuration, and token usage logs. Re-running the classification with identical inputs and parameters yields highly stable outputs, although exact byte-identical reproduction cannot be guaranteed due to potential upstream API or model updates.

Funding

This work was funded by UK Research and Innovation (UKRI) under the UK government's Horizon Europe funding guarantee (grant number 10041284). This work is also part of a project that has received funding from the European Union's Horizon Europe Programme (Grant agreement No. 101061104 – ESSPIN-HORIZON-CL2-2021-TRANSFORMATIONS-01). The research on which this article is based was also supported by the Global School of Sustainability at the London School of Economics.

References

Alfaro, L. and Charlton, A. (2009), 'Intra-industry foreign direct investment', *American Economic Review* **99**(5), 2096–2119.

Amendolagine, V., Barragan, R. and Rabellotti, R. (2023), 'Do green foreign direct investments increase the innovative capability of MNE subsidiaries?', *World Development* **170**, 106342.

Asai, A., Wu, Z., Wang, Y., Sil, A. and Hajishirzi, H. (2023), 'Self-RAG: Learning to retrieve, generate, and critique through self-reflection'. arXiv:2310.11511.

Asirvatham, H., Mokski, E. and Shleifer, A. (2026), 'GPT as a measurement tool', *NBER Working Paper* **34834**.

Besley, T., Brzezinski, A. and Fazzio, J. (2026), 'Economic policy narratives: A taxonomy and application', *LSE Public Policy Review* **4**(1), 1–17.

Bubeck, S., Chandrasekaran, V., Eldan, R., Gehrke, J., Horvitz, E., Kamar, E., Lee, P., Li, Y., Lundberg, S., Nori, H., Palangi, H., Ribeiro, M. and Zhang, Y. (2023), 'Sparks of artificial general intelligence: Early experiments with GPT-4'. arXiv:2303.12712.

Celikyilmaz, A., Clark, E. and Gao, J. (2021), 'Evaluation of text generation: A survey'. arXiv:2006.14799.

Chang, Y., Wang, X., Wang, J., Wu, Y., Yang, L., Zhu, K., Chen, H., Yi, X., Wang, C., Wang, Y., Ye, W., Zhang, Y., Chang, Y., Yu, P. S., Yang, Q. and Xie, X. (2023), 'A survey on evaluation of large language models'. arXiv:2307.03109.

Chen, H., Zhao, Y., Chen, Z., Wang, M., Li, L., Zhang, M. and Zhang, M. (2024), 'Retrieval-style in-context learning for few-shot hierarchical text classification', *Transactions of the Association for Computational Linguistics* **12**, 1214–1231.

Clark, E., August, T., Serrano, S., Haduong, N., Gururangan, S. and Smith, N. A. (2021), All that's 'Human' is not gold: Evaluating human evaluation of generated text, *in* 'Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics', Vol. 1, pp. 7282–7296.
URL: *https://aclanthology.org/2021.acl-long.565*

Cole, M. A., Elliott, R. J. and Fredriksson, P. G. (2006), 'Endogenous pollution havens: Does FDI influence environmental regulations?', *Scandinavian Journal of Economics* **108**(1), 157–178.

Colesanti Senni, C., Schimanski, T., Bingler, J., Ni, J. and Leippold, M. (2024), Combining AI and domain expertise to assess corporate climate transition disclosures, Technical Report 24-92, Swiss Finance Institute.

Crescenzi, R. and Harman, O. (2023), Regional upgrading through the evolution of existing sectors. Manuscript, London School of Economics.

Crescenzi, R. and Harman, O. (2026), Internationalisation, FDI and Global Value Chains in green upgrading. Manuscript, London School of Economics.

Dierdorff, E. C., Norton, J. J., Drewes, D. W., Kroustalis, C. M., Rivkin, D. and Lewis, P. (2009), Greening of the world of work: Implications for O*NET-SOC and new and emerging occupations, Technical report, U.S. Department of Labor.

Eskeland, G. S. and Harrison, A. E. (2003), 'Moving to greener pastures? Multinationals and the pollution haven hypothesis', *Journal of Development Economics* **70**(1), 1–23.

European Commission (2020), Taxonomy report: Technical annex, Technical report, Publications Office of the European Union.

European Commission (2021), EU taxonomy climate delegated act, Technical report, Publications Office of the European Union.

European Commission: Directorate-General for Regional and Urban Policy (2024), Forging a sustainable future together – Cohesion for a competitive and inclusive Europe – Report of the High-Level Group on the Future of Cohesion Policy, Technical report, Publications Office of the European Union.

Ghodsi, M. et al. (2025), International capital and low-carbon transformation. Manuscript.

Griffin, P. W. and Hammond, G. P. (2021), 'The prospects for 'green steel' making in a net-zero economy: A UK perspective', *Joule* **5**(3), 531–541.

Guu, K., Lee, K., Tung, Z., Pasupat, P. and Chang, M.-W. (2020), REALM: Retrieval-Augmented Language Model Pre-Training, *in* 'Proceedings of the 37th International Conference on Machine Learning (ICML)'. arXiv:2002.08909.

Hascic, I. and Migotto, M. (2015), Measuring environmental innovation using patent data, OECD Environment Working Papers 89, OECD Publishing.

Hertwich, E. G. and Wood, R. (2018), 'The growing importance of Scope 3 greenhouse gas emissions from industry', *Environmental Research Letters* **13**(10), 104013.

Intergovernmental Panel on Climate Change (IPCC) (2022), *Climate Change 2022: Mitigation of Climate Change*, Cambridge University Press.

International Energy Agency (IEA) (2020), *Iron and steel technology roadmap: Towards more sustainable steelmaking*, IEA.

Javorcik, B. S. (2004), 'Does foreign direct investment increase the productivity of domestic firms? In search of spillovers through backward linkages', *American Economic Review* **94**(3), 605–627.

Knutsson, P. and Ibarlucea Flores, P. (2022), Trends, investor types and drivers of renewable energy FDI, OECD Working Papers on International Investment 2022/02, OECD Publishing.

Krahmer, E. and Theune, M., eds (2010), *Empirical Methods in Natural Language Generation: Data-Oriented Methods and Empirical Evaluation*, Springer.

Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W.-t., Rocktäschel, T., Riedel, S. and Kiela, D. (2020), Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks, *in* 'Advances in Neural Information Processing Systems', Vol. 33, pp. 9459–9474. arXiv:2005.11401.

Li, X. and Gallagher, K. P. (2022), 'Assessing the climate change exposure of foreign direct investment', *Nature Communications* **13**, 1451.

Li, Z., Li, C., Zhang, M., Mei, Q. and Bendersky, M. (2024), Retrieval Augmented Generation or Long-Context LLMs? A Comprehensive Study and Hybrid Approach, *in* 'Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing: Industry Track', Association for Computational Linguistics, pp. 881–893.

Liu, N. F., Lin, K., Hewitt, J., Paranjape, A., Bevilacqua, M., Petroni, F. and Liang, P. (2024), 'Lost in the Middle: How Language Models Use Long Contexts', *Transactions of the Association for Computational Linguistics* **12**, 157–173.

London Stock Exchange Group (LSEG) (2024), Green Revenues Data Model Methodology, Technical report, LSEG.
URL: *https://www.lseg.com/en/sustainable-finance/green-economy-mark/green-revenues*

Lucarelli, C., Mazzoli, C., Rancan, M. and Severini, S. (2023), 'The impact of EU taxonomy on corporate investments', *Journal of Financial Management, Markets and Institutions* **11**(1), 2350004.

Marin, G. and Vona, F. (2019), 'Climate policies and skill-biased employment dynamics: Evidence from EU countries', *Journal of Environmental Economics and Management* **98**, 102253.

OECD (2020), *Developing Sustainable Finance Definitions and Taxonomies*, Green Finance and Investment, OECD Publishing.

OECD (2024*a*), Business innovation statistics and indicators, Technical report, OECD Publishing.
URL: *https://www.oecd.org/en/data/datasets/business-innovation-statistics-and-indicators.html*

OECD (2024*b*), Cross-border investment into low-carbon infrastructure: An empirical glance, OECD Working Papers on International Investment 2024/01, OECD Publishing.

Papineni, K., Roukos, S., Ward, T. and Zhu, W. J. (2002), BLEU: a method for automatic evaluation of machine translation, *in* 'Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics', pp. 311–318.

Pavlínek, P. (2023), 'Transition of the automotive industry towards electric vehicle production in the east European integrated periphery', *Empirica* **50**(1), 35–73.

Qin, C., Zhang, A., Zhang, Z., Chen, J., Yasunaga, M. and Yang, D. (2023), 'Is ChatGPT a general-purpose natural language processing task solver?'. arXiv:2302.06476.

Reiter, E. (2018), 'A structured review of the validity of BLEU', *Computational Linguistics* **44**(3), 393–401.

Reiter, E. and Belz, A. (2009), 'An investigation into the validity of some metrics for automatically evaluating NLG systems', *Computational Linguistics* **35**(4), 529–558.

Renze, M. and Guven, E. (2024*a*), The effect of sampling temperature on problem solving in large language models, *in* 'Findings of the Association for Computational Linguistics: EMNLP 2024'. arXiv:2402.05201.

Renze, M. and Guven, E. (2024*b*), 'Self-reflection in LLM agents: Effects on problem-solving performance'. arXiv:2405.06682.

Rogelj, J., Shindell, D., Jiang, K., Fifita, S., Forster, P., Ginzburg, V., Handa, C., Kheshgi, H., Kobayashi, S., Kriegler, E., Mundaca, L., Séférian, R., Vilariño, M. V. and Zickfeld, K. (2018), Mitigation pathways compatible with 1.5°C in the context of sustainable development, *in* V. Masson-Delmotte et al., eds, 'Global Warming of 1.5°C: An IPCC Special Report', Cambridge University Press, pp. 93–174.

UNCTAD (2023), *World Investment Report 2023: Investing in Sustainable Energy for All*, United Nations Conference on Trade and Development.

UNCTAD (2024), *Global Economic Fracturing and Shifting Investment Patterns: A Diagnostic of Ten FDI Trends and Their Development Implications*, United Nations Conference on Trade and Development.

van der Lee, C., Gatt, A., van Miltenburg, E. and Krahmer, E. (2021), 'Human evaluation of automatically generated text: Current trends and best practice guidelines', *Computer Speech and Language* **67**, 101151.

Wang, M. and Wong, M. C. S. (2009), What drives economic growth? The case of cross-border M&A and greenfield FDI activities, Economics Faculty Research and Publications 108, Marquette University, Department of Economics.
URL: *https://epublications.marquette.edu/econ$_f$ac*/108

World Steel Association (2021), *Sustainable Steel: Indicators 2021*, World Steel Association.

Zou, Y., Shi, M., Chen, Z., Deng, Z., Lei, Z., Zeng, Z., Yang, S., Tong, H., Xiao, L. and Zhou, W. (2025), 'ESGReveal: An LLM-based approach for extracting structured data from ESG reports', *Journal of Cleaner Production* **489**, 144572.

Supplementary Information

SI.1. Data Processing

1.1 Orbis-crossborder data

The data on inward FDI projects is taken from the Orbis-crossborder database, by Bureau van Dijk/Moody's Analytics. This contains information on global greenfield investment plus merger & acquisitions (M&A) deals. We focus on the former, which BvD/Moody's Analytics defines as capital investment by a company in a national market other than its own. As explained by the data provider,[14] this includes the opening, expansion, co-location and relocation of data centres, sales offices, shared service centres, headquarters, distribution centres, experience centres, and research and development centres.

The rationale for excluding M&A deals aligns with our conceptual framing of FDI as a dual mechanism for financing green technologies and facilitating their cross-border diffusion. The spillovers literature consistently finds that greenfield investments – which involve establishing new productive capacity – generate stronger technology transfer, knowledge spillovers, and productivity gains in host economies compared to M&As, which primarily represent ownership changes of existing assets (Javorcik, 2004; Wang and Wong, 2009). Greenfield projects are more likely to introduce new technologies, management practices, and production processes that can diffuse to domestic firms through labour mobility, supplier linkages, and demonstration effects (Alfaro and Charlton, 2009).

To build our dataset, we select the 109,084 projects from the Orbis-crossborder database that met the following criteria as at March 2025:

- The "Project status" is listed as "Announced", "Completed – assumed" or "Completed – confirmed" – thereby excluding projects which listed under "Rumoured", or those that have been announced but subsequently cancelled or postponed. It should be noted that the first category makes up under 5% of projects downloaded. Moreover, BvD changes the status of projects if an announcement is made to cancel or postpone a particular investment, meaning that we are not considering investments that ultimately do not take place.

- The "Project type" is listed as 'new' (77% of projects) or any of the following: expansions, re-locations and co-locations, which together make up the remaining 23%. The latter are not always considered greenfield FDI as such, given that they may involve a company that

[14] See data collection information by BvD/Moody's Analytics with respect to its Orbis-crossborder product: https://help.bvdinfo.com/LearningZone/Products/cbii4.1/Content/I_Data/Collection.htm#GreenfieldFDIProjects

is already established in some form in that location. As such, green FDI results are also shown for a subsample which keeps only "new" FDI projects.

- The project location, i.e. what country or region the investment is actually into, is any EU member state (EU27) or the UK.
- The year in which the investment was recorded by BvD, as reported in the variable "Project status date", is between 2013 and 2024 inclusive (the Orbis-crossborder database does not provide this information prior to 2013).
- The sector in which the FDI project took place is neither of the following: NACE segment O (Public administration and defence; compulsory social security), NACE segment P (Education), NACE segment Q (Human health and social work activities) or segment S (Other services activities).

The below tables provide some summary statistics of these investments, specifically how the investment value and job creation figures vary by year:

Table S1: Summary statistics, in terms of FDI value and job creation, for described Orbis-crossborder sample of all FDI projects between 2013–2024 into EU27+UK

Year	Num. jobs created	USD value (M)
2013	476,223	231,523
2014	516,731	231,579
2015	578,251	225,459
2016	520,784	164,620
2017	683,550	233,341
2018	740,069	264,953
2019	826,379	281,491
2020	473,245	185,222
2021	520,933	214,411
2022	403,921	198,986
2023	634,723	312,073
2024	322,371	206,207

Source: Orbis crossborder by Bureau van Dijk / Moody's Analytics (2025) pt

It should be noted that a small number of these 109,084 investment projects actually take place in more than one region or country – for example, an MNE may decide to open a sales office in both Madrid and Lombardy at the same time – but are in fact listed as a single investment. We separate these investments out, so that we are left with 109,084 investment-region pairs.

Figure S1: Examples of the text description for each inward FDI investment in the Orbis-Crossborder database (2013–2024, EU27+UK)

A) A software development centre in Northern Ireland

Project history

On 18/07/24 it was announced that [Company], a digital product development service provider, has opened a software development centre in [Location] to serve the domestic market. The new centre will create about 15 new jobs over the next two years across human resources, business development and software engineering. The centre offers digital engineering expertise, design-led experiences, data services, and AI-powered product and platform development services. The opening was motivated by domestic market potential, market access and skilled workforce availability. No further information has been disclosed.

Project rationale

On 18/07/24 [Name], Managing Director – Ireland, [Company] said: "This expansion in Northern Ireland cements [Company's] commitment to businesses on the island of Ireland, while also representing a significant step in our ambitious growth plans. This new presence builds on the success we have steered from our [Location] and is a strong validation of the skills, expertise and dedication of the entire [Company] team in Ireland. As a champion of regional investment and the advantages that it can deliver, [Location] is an ideal base from which we can expand our business. We are tapping into the rich economy of Northern Ireland and in doing so, this move will enable us to roll out our services to even more sectors and industries. We anticipate many opportunities for growth in [Location] and across Northern Ireland and we look forward with excitement to the next chapter and continued success on the island of Ireland."

B) A Cathode Active Material (CAM) manufacturing plant in Finland

Project history

On 16/05/24 it was reported that [Company], a lithium-ion battery material manufacturer, plans to open a cathode active material (CAM) manufacturing plant in [Location], Finland. The company has submitted an environmental permit application to the local authority. The plant will occupy 27.8 hectares and produce roughly 60,000 tonnes of materials per year. About 300 jobs will be generated, motivated by market access, supply chain and location attractiveness. The company is a joint venture between [Company A] and [Company B]. No further details were disclosed. On 10/12/24 it was announced that [Company A] owns 70 percent and [Company B] 30 percent. Also, the plant is expected to start operations in 2026. No further details were disclosed.

Project rationale

On 16/05/24 Mr [Name] said "The CAM factory in [Location] is a significant investment project for [Company] and we see Finland as a good location for high-quality industrial operations. Our goal is to accelerate the green transition in Europe by leveraging our expertise in battery materials production through nearly three decades of long-term R&D efforts." On 16/05/24 Mr [Name] said "The establishment of the joint venture is an important step forward in building the Finnish battery value chain. With the plant, new, modern industry is emerging in Finland to help electrify European transport and mitigate climate change. The future mill will also create jobs and significant economic impacts both in [Location] and in Finland as a whole." On 16/05/24 Mr [Name], Battery Value Chain Director, [Company] said "In Finland, we have good opportunities to develop transparent and responsible raw material production. The battery materials industry would increase the degree of added value of domestic production and increase the value of exports. Concentrating production in Finland would also shorten the distances in the production chain and thus reduce the carbon footprint of the end product."

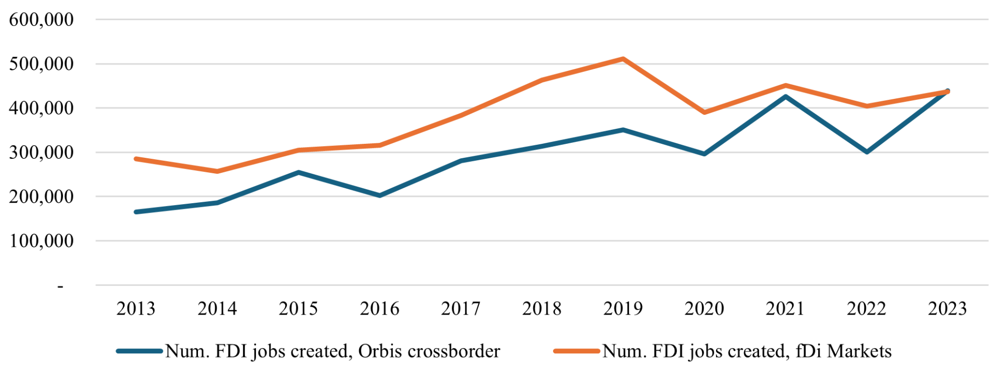
Source: Orbis crossborder by Bureau van Dijk / Moody's Analytics (2025)

For each investment-region pair, we download information on the following key variables:

- The NACE sector of the investment
- The project business function, which can include "Sales Office", "R&D centre", "Manufacturing" or "Data Centre" (among others)
- The estimated capital value of the investment
- The number of jobs expected to be created through the investment
- Text describing the project, both in terms of the project history and/or the rationale, collated by BvD/Moody's from corresponding news articles or official press releases,[15] available for 89% of investments (see examples in Figure S1 above).

The NACE sector of the investment enables us to classify each project to the green activities listed for each sector within the EU Taxonomy of Sustainable Activities database, as described below. Moreover, as described ahead, the textual description of the FDI project is essential in our methodology for identifying green FDI projects with support from LLMs. Other widely used datasets, such as fDi Markets (used by UNCTAD and in recent academic work), do not supply such textual descriptions and therefore cannot support bottom-up, activity-level classification. This is the key factor behind our decision to use Orbis-crossborder data instead of fDi Markets. In any case, Figures S2 and S3 below demonstrate that both data sources show broadly consistent patterns in terms of aggregate investment value and job creation over the 2013–2024 period.

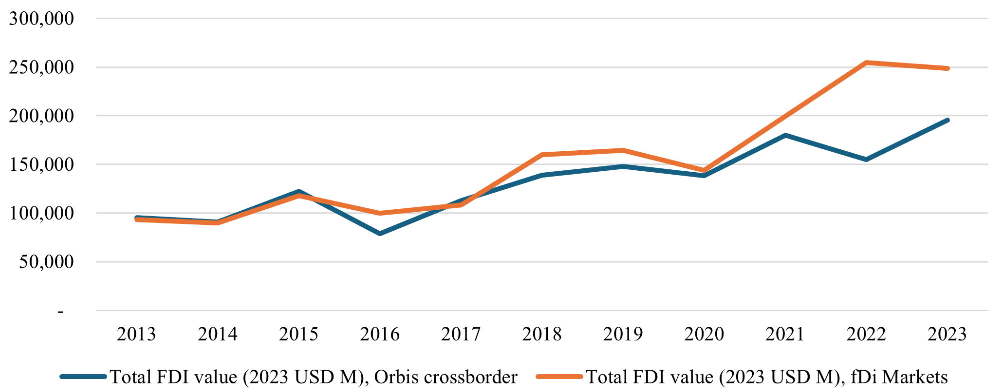
Figure S2: Num. jobs created through new FDI projects in EU27+UK, Orbis-crossborder vs fDi Markets, 2013–2023



Source: Own elaboration using data from Orbis-crossborder, fDi Markets (2025).

[15] As reported here: https://help.bvdinfo.com/LearningZone/Products/cbii4.1/Content/I_Data/Collection.htm#GreenfieldFDIProjects

Figure S3: Total value of FDI (in 2023 USD) through new FDI projects in EU27+UK, Orbis-crossborder vs fDi Markets, 2013–2023

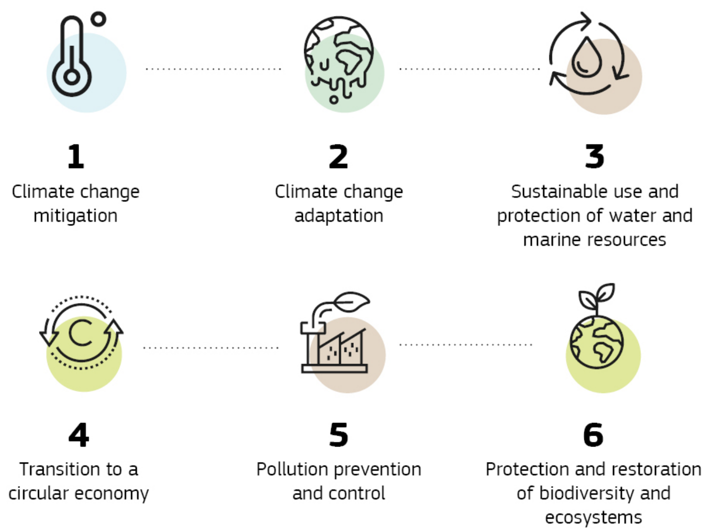


Source: Own elaboration using data from Orbis-crossborder, fDi Markets (2025).

1.2 The EU Taxonomy of Sustainable Activities

To determine which FDI projects are green outside of the REW sector, we use the full list of green activities identified across all NACE sectors, which is available to download from the EU Commission website.[16] We consider activities which are green by their 'own-performance', plus those that are 'enabling' or 'transitional'. Moreover, we consider activities that contribute to any of the six environmental objectives set by the EU Commission in its definition of the dimensions to 'green', as shown in Figure S4 below.

Figure S4: The six possible environmental objectives that an activity must contribute towards in order to be considered green in the EUTSA



Source: EU Commission (2023)

Table S2 shows how many activities in total are defined across each of these six objectives,

[16] https://ec.europa.eu/sustainable-finance-taxonomy/taxonomy-compass

together with an example of each. 60 out of 87 possible 2-digit NACE codes have at least one such green activity, in the sense of contributing to (at least) one of the six stated environmental objectives (while not doing harm to any of the others).

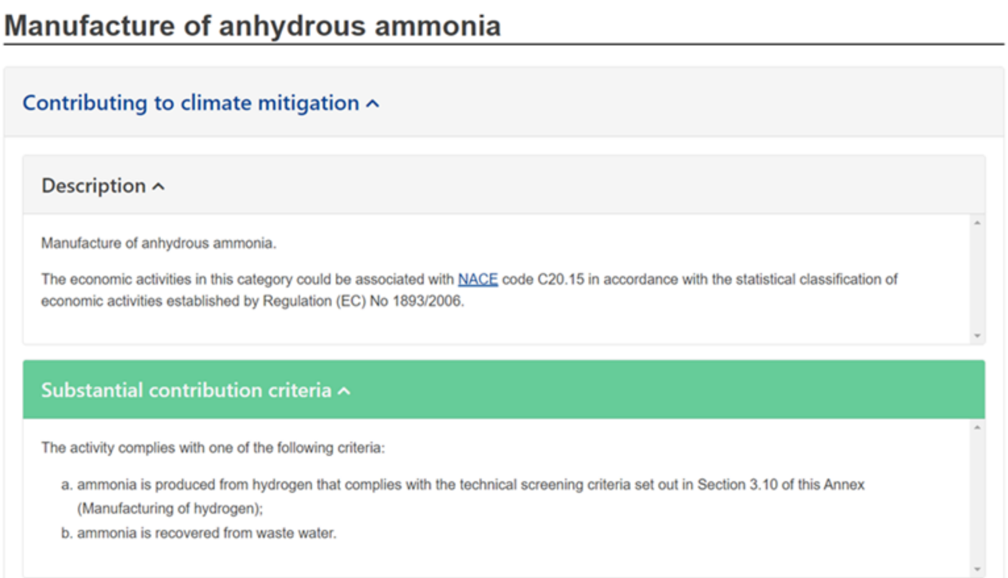
Table S2: Number of possible 'green' activities (across all activity types) by environmental objective, plus examples

Objective	Num. individual activities	Examples
Climate change mitigation	101	Manufacturing of anhydrous ammonia, NACE C20
Climate change adaptation	106	Engineering activities and related technical consultancy dedicated to adaptation to climate change, NACE M71
Water	6	Urban waste water treatment, NACE E37; F42
Circular economy	21	Sale of spare parts, NACE C26, C27, C28, C31, G46, G47
Pollution prevention	6	Remediation of legally non-conforming landfills and abandoned or illegal waste dumps, NACE E38, E39, F42
Biodiversity	2	Hotels, holiday, camping grounds and similar accommodation, NACE I55

Source: EU Commission (2023)

An example green activity is shown in Figure S5 below: within NACE sector C20 (Manufacturing of chemicals), one possible 'green' activity is the manufacture of anhydrous ammonia through renewable-energy powered water electrolysis, producing what is a vital component for fertiliser (among other things) without carbon emissions and therefore contributing to climate change mitigation.

Figure S5: Example of a 'green activity' within NACE sector C20 (Manufacturing of Chemicals), according to the EUTSA



Source: EU Commission (2023)

For each of these activities, the EUTSA provides an activity title, a description of the activity, plus a list of 'substantial contribution' or screening criteria that must be adhered to. Table S3 illustrates the level of technical specificity embedded in the EUTSA framework. These screening criteria ensure that activities classified as 'green' meet quantifiable environmental performance standards rather than relying on broad sectoral categorisations.

Table S3: Technical Screening Criteria – Illustrative Examples

NACE Sector	Green Activity	Technical Screening Criteria (Substantial Contribution)
C24 – Manufacture of basic metals	Manufacture of iron and steel	Direct CO_2 emissions below 1.331 tCO_2e per tonne of crude steel produced when using electric arc furnace (EAF) route. For integrated steelmaking: emissions $\leq$1.484 tCO_2e per tonne using best available techniques, with a plan to reduce to $\leq$0.266 tCO_2e by 2050.
C24 – Manufacture of basic metals	Manufacture of aluminium	Greenhouse gas emissions from the production process below 1.484 tCO_2e per tonne of aluminium produced. Production using scrap-based processes or renewable electricity.
J62 – Computer programming, consultancy and related activities	Data-driven solutions for GHG emissions reductions	The activity results in substantial GHG emission reductions in other sectors through: (a) deployment of ICT solutions that enable significant reduction in GHG emissions; (b) the ICT solution is shown to deliver net GHG emission savings compared to best-performing alternative solutions/technologies.
J62 – Computer programming, consultancy and related activities	Data processing, hosting and related activities	Power Usage Effectiveness (PUE) of data centres $\leq$1.50 for new facilities (effective 2025) and $\leq$1.80 for existing facilities. Implementation of best practices in energy-efficient cooling, waste heat recovery, and direct use of renewable electricity.
C21 – Manufacture of basic pharmaceutical products and pharmaceutical preparations	Manufacture of pharmaceutical active ingredients or intermediates	Production processes achieve at least 10% improvement in energy efficiency and/or significant reduction in water consumption and waste generation compared to industry benchmarks. Facilities implement closed-loop systems and eliminate or substitute hazardous substances where technically feasible.
M72 – Scientific research and development	Research, development and innovation for direct CO_2 capture or other substantial GHG reductions	The research activity is dedicated to: (a) developing technologies that enable substantial life-cycle GHG emission reductions in other sectors; (b) improving energy efficiency; (c) increasing the use of renewable energy; or (d) direct air capture of CO_2. Activities include R&D for low-carbon technologies with clear pathway to market deployment.
M72 – Scientific research and development	Research and development on climate change adaptation solutions	The activity develops knowledge, products or technologies that substantially enhance adaptive capacity, strengthen resilience, or reduce vulnerability to climate change impacts in specific sectors or regions. Includes R&D on flood defences, drought-resistant crops, or climate-resilient infrastructure design.

Source: EU Commission, 2023.

For example, in the metals sector (C24), the Taxonomy distinguishes between steel production methods based on precise carbon intensity thresholds: Electric Arc Furnace production must emit less than 1.331 tCO_2e per tonne, while traditional integrated steelmaking faces stricter requirements with a pathway to deep decarbonisation by 2050. Similarly, data centres (J62) must meet specific Power Usage Effectiveness benchmarks, and pharmaceutical manufacturing (C21) must demonstrate measurable improvements in resource efficiency. This granularity enables our LLM-based approach to evaluate FDI projects against rigorous, sector-specific environmental standards.

After downloading the full list of green activities and screening criteria, we collapse the relevant text at the NACE 2-digit level, combining the title, description, and substantial contribution criteria for each activity within a sector. Because some NACE codes contain numerous green activities with detailed criteria, the resulting text can be extremely lengthy and repetitive. To improve efficiency and model performance, we process this raw text through GPT-4o to create condensed versions that eliminate redundancy while preserving all essential information on activities and criteria. Manual checks confirmed that no critical details were lost in this cleaning process. This step reduces both processing costs and the risk of LLM hallucination, which increases with extremely long prompts as models struggle to retrieve information from the middle of their input (Liu et al., 2024).

We also create an alternative set of NACE sector-level green activities based on the EUTSA descriptions only (i.e., excluding the 'substantial contribution' criteria), providing a shorter and less technically-detailed version of the EUTSA activities for each sector. Table S4, below, provides a comparison of these two 'clean' EUTSA texts for one sector (NACE J63 – Information Service Activities), with the left-hand column showing the 'simpler' text based on just the activity descriptions, and the right-hand column showing the 'detailed' text based on both the activity descriptions and the corresponding substantial contribution criteria. It can be seen how the detailed text provides more specific criteria and/or technical standards that must be adhered to, therefore constituting the stricter and more robust of the two EUTSA text options that we generate for each NACE 2-digit sector (or 2-digit sector group). The full matrix of 'cleaned' EUTSA text(s) for each NACE 2-digit sector (or 2-digit sector group) which was ultimately used for the LLM processing (described ahead) is available upon request.

Finally, we match these sector-specific EUTSA texts to each FDI project in our dataset on the basis of the NACE code the investment falls under.

Table S4: Comparison of 'simple' EUTSA text and 'detailed' EUTSA text obtained for the same sector (NACE 2-digit code J63 – Information service activities)

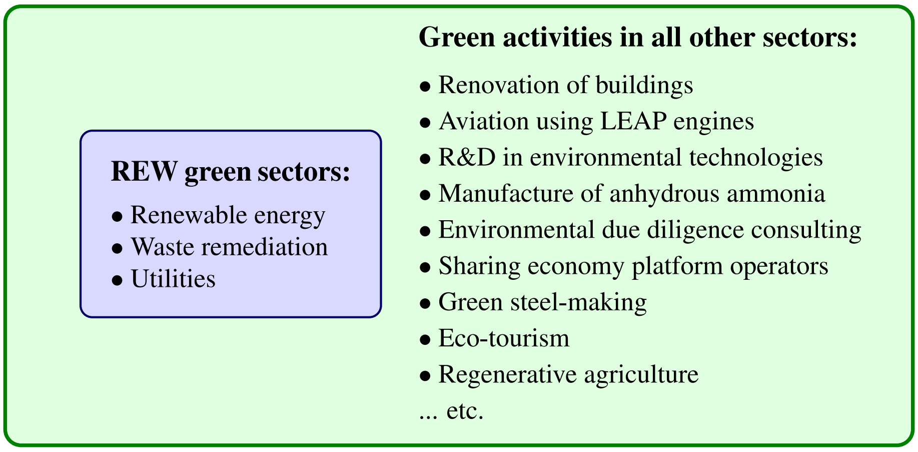
EUTSA text for LLM: based on description of activities	EUTSA text for LLM: based on description of activities *and* substantial contribution criteria
Sector: J63 – Information service activities	
The EU Taxonomy for Sustainable Activities identifies several green and sustainable activities within the Information service activities sector, particularly focusing on data processing and information technology (IT) applications. These activities can be summarized as follows: 1. Data Centres and Edge Computing: Activities involve the storage, manipulation, management, movement, control, display, switching, interchange, transmission, or processing of data through data centres, as well as through edge computing technologies. 2. Data-Driven Solutions for GHG Emissions Reductions: Development and use of ICT solutions designed to collect, transmit, store, model, and utilise data primarily to enable reductions in greenhouse gas (GHG) emissions. These technologies include decentralized distributed ledger technologies, the Internet of Things (IoT), 5G networks, and Artificial Intelligence (AI). 3. IT/OT Data-Driven Solutions: This entails the production, development, installation, deployment, maintenance, and repair of software and IT or operational technology (OT) systems, including AI-based solutions. Such systems are for: – Remote monitoring and predictive maintenance, including collecting, processing, and storing data from in-use equipment or infrastructure. – Analysing operational performance and providing maintenance recommendations. – Tracking and tracing materials, products, and assets to support circularity in value chains. – Lifecycle assessment software to support product reporting. – Eco-design software for products, equipment, and infrastructure. – Supplier management software aiding in green material procurement. – Lifecycle performance management software for monitoring circularity performance throughout product lifecycles. 4. Marketplace for Reuse: Developing and operating online marketplaces and classifieds for trading second-hand goods for reuse, focusing on products from various manufacturing sectors listed under specific NACE codes. These platforms facilitate B2B, B2C, and C2C sales of second-hand goods, excluding wholesale or retail trade. The classification of these activities demonstrates the EU's commitment to supporting the transition to a sustainable economy by promoting practices that reduce the environmental impact of data management and technology utilization.	*EU Taxonomy: Green Activities in Information Service Activities 1. Data Processing, Hosting, and Storage: – Must follow best practices for energy efficiency, verified independently every three years. – Use of refrigerants in cooling systems with a GWP of no more than 675. – Implement climate risk reduction solutions, prioritizing nature-based solutions, with effectiveness monitored against specific indicators. 2. Data-Driven GHG Emission Reduction Solutions: – Involves ICT solutions like distributed ledger technologies, IoT, 5G, and AI to reduce GHG emissions. – Must demonstrate significant life-cycle GHG savings over alternatives, verified by an independent third party. 3. IT/OT Data-Driven Solutions: – For manufacturing, developing, installing, deploying, maintaining, or repairing software and IT/OT systems aimed at operational efficiency, product life extension, and circular economy support. – Requires capabilities for remote monitoring, predictive maintenance, lifecycle assessment, eco-design, green procurement, and lifecycle performance management. – Emphasizes durability, recyclability, and use of secondary materials, aligning with EU waste regulations and circular economy goals. 4. Marketplaces for Second-Hand Goods: – Development and operation of platforms for trading used products, materials, or components, supporting B2B, B2C, and C2C transactions. – Servers and data storage must comply with environmental directives, manage waste with a focus on reuse and recycling, and ensure proper treatment or recycling of end-of-life equipment according to EU directives.*

Source: Own elaboration, based on EU Taxonomy of Sustainable Activities (EU Commission, 2023)

1.3 LLM processing

Our classification of green FDI is based on two fundamental types of green FDI: those that fall within 'core' Renewable Energy & Waste (REW) sectors, plus those that involve green activities but fall outside of these (Beyond-Energy Green FDI). Figure S6, below, provides some examples of specific activities within both REW sectors and all other sectors, according to the EUTSA.

Figure S6: Distinction between Renewable Energy & Waste sectors in the 'core' and green activities in other sectors, with examples



Source: Own elaboration.

Identifying the first group is fairly straightforward: these are the FDI projects in our dataset that fall into the following NACE segments:

D – Electricity, gas, steam and air conditioning supply[17]

E – Water supply; sewerage; waste management and remediation activities

The identification of the Renewable Energy & Waste (hereafter REW) sector is based upon a literature review which highlighted that much previous research into green or sustainable FDI has focused on renewable energy and waste remediation (e.g., UNCTAD 2023), as discussed in the introduction.

[17]This sector includes both fossil-fuel based and renewable electricity generation and supply. Any investments involving fossil fuels – including natural gas – were identified through an additional sector field available in the dataset (BvD sector), which distinguishes between fossil fuel and renewable energy generation, and were subsequently excluded from the green category despite technically falling within NACE segment D. Natural gas is excluded notwithstanding its 'transitional' classification in the EU Taxonomy, given the contested nature of that designation and its incompatibility with long-run climate neutrality. Nuclear energy projects, by contrast, are retained in the green category. While nuclear is not renewable in the strict sense, this treatment is consistent with the EU Taxonomy's own recognition of nuclear as a low-carbon activity and with our primary objective of identifying investment flows that do not contribute to greenhouse gas emissions. The classification of remaining NACE D projects into specific energy sub-types (solar, wind, hydroelectric, biomass, geothermal, and other renewables) was performed through keyword searches on project descriptions and headlines, as detailed in the replication code.

To identify which of these investments are green but fall outside of the REW sectors, we employ the GPT-4 LLM (hereafter referred to as "the LLM") to determine whether each investment aligns with the activities listed in EUTSA. This is done by providing the LLM with two key pieces of information. Firstly, the textual description of each investment in our dataset, which is built from two text fields available in the Orbis-crossborder data, namely "Project comments" and "Project rationale". Secondly, we extract and provide the text on the activities considered 'green' in that particular NACE sector as per the EUTSA.

These two pieces of information, the textual description plus the corresponding EUTSA text on green activities, make up the bulk of the context provided to our LLM agent, GPT-4 (the particular models used are detailed ahead). The specific prompt used, which enables us to identify FDI projects involving green activities beyond the REW sectors, is shown in Figure S7, below.

Figure S7: Prompt used for evaluating EUTSA-alignment of each FDI project

> "You are an expert, systematic classifier of whether foreign investments made by companies can be considered 'green' investments. To do this, I will give you two pieces of information. Firstly, ahead is some information describing a particular foreign investment made by a company: *{invest_desc}*. Secondly, the following describes the kinds of activities by companies operating in the same sector that are considered 'green' by the EU Taxonomy on Sustainable Activities: *{eutsa_green_criteria}*. Based strictly on the second piece of information – the criteria for green activities in that particular sector – would you say that the investment described can be considered 'green'? Please do not infer anything about the investment. Base your decision exclusively on the description of the investment that is given. It is essential that you provide your response in the following format: Respond only with 'Yes' or 'No'."

Source: Own elaboration.

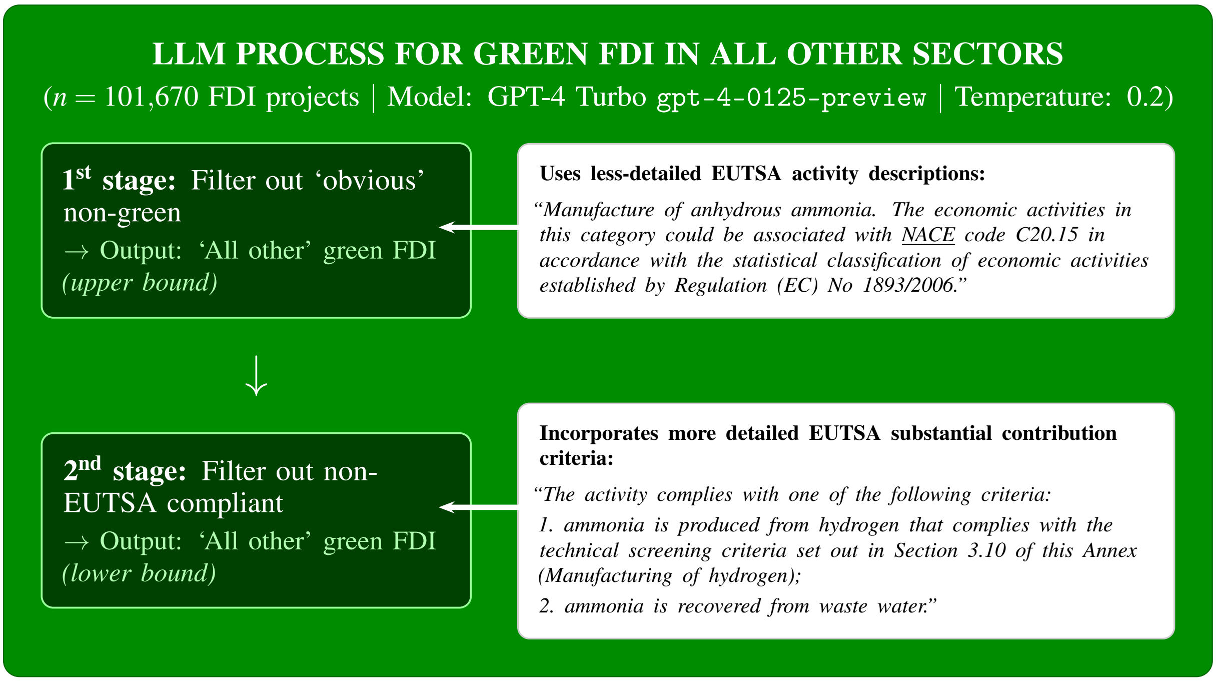
As can be seen, the prompt is designed to instruct the model to base its decision exclusively on the provided information, rather than inferring anything about the investment or sourcing related but irrelevant information from its own corpus. Instead, the FDI project description and relevant EUTSA text are directly integrated into the prompt text, under '*invest_desc*' and '*EUTSA_green_criteria*' (respectively). This means that each prompt is unique and bolstered with curated knowledge regarding the FDI project in question plus the corresponding green activities that are applicable as per the EUTSA. While the specific LLMs used (GPT-4 Turbo,[18] from the GPT-4 family of OpenAI models), are pre-trained on data up until December 2023 (by which point plenty of documentation regarding the EUTSA and its criteria was available online for parametrization into the model), providing the EUTSA text for the LLM to draw on in this way has been shown to improve its performance in terms of accuracy and consistency (Chen et al., 2024; Guu et al., 2020). This is because, while LLMs like GPT-4 may well store

[18]Specifically, the `gpt-4-0125-preview` version.

the relevant factual knowledge on topics such as green activities within a given sector, without explicit direction it is unlikely to reliably and accurately retrieve the appropriate text from what is a vast corpus of parametrised knowledge (Guu et al., 2020). We do not employ a Retrieval Augmented Generation (RAG) approach, whereby the LLM relies on semantic search to retrieve relevant segments from a large corpus (in our case, the full EUTSA documentation) (Asai et al., 2023; Lewis et al., 2020). The main reason is the quality of vector search retrieval typical in RAG systems is consistently worse than in-context learning for advanced LLMs, as long as the knowledge is small enough to fit comfortably within an LLM's context limit (Li et al., 2024), which applies in our case as each definition is less than 1,000 words.

The temperature parameter, which regulates the model's propensity for 'creativity', is set relatively low at 0.2, in line with guidance for textual analysis tasks (Renze and Guven, 2024*a*). This, together with the explicit instruction to provide a binary 'Yes' or 'No' response as to whether the FDI project can be considered green (plus its justification), means that the LLM outputs were consistently structured and it was not necessary to explicitly enforce this. Every single FDI project that was processed was unambiguously labelled as green or not green, making the subsequent coding up and re-incorporation into our dataset straight-forward and uncontentious. The provision of a justification for each decision which cites the relevant parts of the EUTSA activities or project description, moreover, significantly facilitated the human checks which are discussed in the following section.

Figure S8: Prompt processing stages & GPT-4 models for identification of green FDI in non-REW sectors



Source: Own elaboration.

The processing of FDI projects with the above prompt and parameters involves two stages, as illustrated in Figure S8 above. Firstly, every FDI project description is processed with respect to the ‘simpler’, less technically-detailed EUTSA text. This provides an initial identification of projects where green activities are present according to its description. Secondly, any projects identified as green in this first stage are re-processed through the LLM, but this time with respect to the EUTSA green criteria that incorporates information from the ‘substantial contribution criteria’ variable. Any projects which ‘survive’ this second step, in the sense of still being labelled as green, are classified as green FDI in our main classification (although the classification which results after just the first stage, which we term an ‘upper bound’ estimate of green FDI, are provided in Additional Results, below).

1.4 Definition of sectors

The sectoral classification used in our analysis and descriptives is based on NACE 2-digit sectors (or aggregations of NACE 2-digit sectors), with one exception for NACE code 35 (as explained below), giving a total of 39 sectors. These are shown in Table S5.

In most cases, we collapse the NACE 2-digit sectors into their lettered sections to create our own sectors. In a few cases, we group together 2-digit codes or consider them on their own, in order to provide the level of detail we want for each sector. The exact crosswalk for this can be seen below, but the key points are as follows:

- We largely keep manufacturing 2-digit codes at that level, except in a select few cases where we merge together very closely related 2-digit codes (e.g., food, drink and tobacco)
- We ‘split out’ NACE code 72 (Scientific research) from Section M and give it its own category
- We ‘split out’ NACE codes 61, 62 and 63 from J (Information and Communication) and give each its own sector, while the remaining three codes in J (58, 59, 60) are grouped together

Because NACE code 35 (Electricity, gas, steam and air conditioning supply) includes both renewable and fossil fuel energy generation, whereas for the purposes of this research this distinction is key, we create 6 new categories into which we sort any investments in this NACE code:

1. Energy – coal, oil & gas
2. Energy – nuclear
3. Energy – hydroelectric
4. Energy – solar
5. Energy – wind

6. Energy – other renewable (incl. geothermal, biomass and marine/ocean/tidal)

To do this classification, we undertake the following two steps:

1. Use Orbis-crossborder 'BvD sectors', which are provided together with NACE codes and do distinguish between renewable electricity generation and non-renewable, to identify investments that are "Energy – coal, oil & gas"
2. Apply a keyword search for 'nuclear', 'hydroelectric', 'solar', 'wind' and 'geothermal'/'biomass'/'marine'/'ocean'/'tidal' within each investment's project description in order to sort the remainder into the corresponding renewable categories.

Table S5 below summarises the composition of each sector used in our analysis, in terms of its correspondence to the NACE 2-digit classification.

Table S5: Sectors used in analysis, plus correspondence to NACE classification

Sector	**Correspondence to NACE classification**	**Comments**
Accommodation, restaurants & bars (incl. tourism)	SECTION I – 55, 56	Whole section
Administrative services & auxiliary business support (incl. call centres, security & rental services)	77, 78, 79, 81, 82	All of section N
Agriculture, forestry & fishing	SECTION A – 01, 02, 03	Whole section
Arts, entertainment, recreation, natural parks & sport	SECTION R – 90, 91, 92, 93	Whole section
Construction	SECTION F – 41, 42, 43	We are matching the FDI Markets sub-sectors to SECTION F directly in this case
Energy generation & distribution – coal, oil & gas	SECTION D – 35	Exceptional case where we split a NACE 2-digit code
Energy generation & distribution – nuclear	SECTION D – 35	Exceptional case where we split a NACE 2-digit code
Energy generation & distribution – hydroelectric	SECTION D – 35	Exceptional case where we split a NACE 2-digit code

Continued on next page

Sector	Correspondence to NACE classification	Comments
Energy generation & distribution – wind	SECTION D – 35	Exceptional case where we split a NACE 2-digit code
Energy generation & distribution – solar	SECTION D – 35	Exceptional case where we split a NACE 2-digit code
Energy generation & distribution – other renewable (incl. biomass, geothermal and marine/ocean/tidal)	SECTION D – 35	Exceptional case; also bunches together: other renewable, geothermal, biomass and marine
Financial services & insurance	SECTION K – 64, 65, 66	Whole section
Publishing (books, periodicals & digital), motion picture/sound recording & broadcasting	SECTION J – 58, 59, 60	
Telecommunications (wired, satellite & other)	SECTION J – 61	
Computer programming, IT consulting and IT management	SECTION J – 62	
Information service activities (incl. data processing, hosting and storage)	SECTION J – 63	
Manufacturing – basic metals	24	
Manufacturing – building materials	23	
Manufacturing – cars, trucks and other automobiles (incl. parts)	29	
Manufacturing – chemicals & chemical products (excl. pharmaceutical)	20	
Manufacturing – coke and refined petroleum products (excl. plastic & rubber)	19	
Manufacturing – computers, electronic components & electrical devices	26, 27	

Continued on next page

Sector	Correspondence to NACE classification	Comments
Manufacturing – fabricated metal products	25	
Manufacturing – food, drink & tobacco products	10, 11, 12	
Manufacturing – furniture & wood products	16, 31	
Manufacturing – machinery & industrial equipment	28	
Manufacturing – medicines & pharmaceutical products	21	
Manufacturing – non-automobile transport (incl. aircraft, ships & military equipment)	30	
Manufacturing – other (incl. medical devices)	32	
Manufacturing – paper, paper products & printed material	17, 18	
Manufacturing – rubber & plastic products	22	
Manufacturing – textile, footwear, apparel & leather	13, 14, 15	
Mining & quarrying	SECTION B – 05, 06, 07, 08, 09	Whole section
Professional & technical services (excl. financial, scientific & software)	69, 70, 71, 73, 74	All of section M but 72, which has its own category, and 79
Real estate activities	SECTION L – 68	Whole section
Retail/wholesale trade & repairs	SECTION G – 45, 46, 47	Matching FDI Markets subsectors to SECTION G
Scientific research & development (incl. biotechnology)	72	
Transportation (incl. air travel), logistics & storage	SECTION H – 49, 50, 51, 52, 53	Whole section
Water, sewage & waste management	SECTION E – 36, 37, 38, 39	Matching FDI Markets subsectors to SECTION E

1.5 Definition of sub-national regions

Where we show results at the regional level, these regions are based on the 2021 NUTS classification. However, given the varying administrative geography across the EU, with the main functional regions in certain countries being NUTS1 in some countries but NUTS2 in (most) others, we decide to identify regions to reflect this heterogeneity. The table below shows, for each country in our sample, what NUTS level was used.

Table S6: NUTS level used in regional analysis for each country in EU27+UK

NUTS level	Country
NUTS1	Belgium, France, Germany, United Kingdom.
NUTS2	Austria, Bulgaria, Croatia, Cyprus*, Czechia, Denmark, Estonia*, Finland, Greece, Hungary, Italy, Ireland, Latvia*, Lithuania, Luxembourg*, Malta*, Netherlands, Poland, Portugal, Romania, Slovakia, Slovenia*, Spain, Sweden.

**In the NUTS2021 classification, these countries have a single NUTS2 region.*

Table S7 lists all NUTS regions across EU27+UK countries that received at least one greenfield FDI project between 2013 and 2024. The dataset covers 208 of the 209 NUTS 2021 regions, representing 99.5% regional coverage. The only region without recorded FDI is FI20 (Åland), a small autonomous archipelago of Finland with a population of approximately 30,000. Note that French and Dutch overseas territories (outermost regions) are not included in this analysis, as they fall outside the NUTS classification system applied to continental Europe.

Table S7: NUTS Regions Represented in FDI Dataset (2013–2024)

Country	NUTS Regions
Austria	AT11 (Burgenland); AT12 (Niederösterreich); AT13 (Wien); AT21 (Kärnten); AT22 (Steiermark); AT31 (Oberösterreich); AT32 (Salzburg); AT33 (Tirol); AT34 (Vorarlberg)
Belgium	BE1 (Région de Bruxelles-Capitale); BE2 (Vlaams Gewest); BE3 (Région wallonne)
Bulgaria	BG31 (Severozapaden); BG32 (Severen tsentralen); BG33 (Severoiztochen); BG34 (Yugoiztochen); BG41 (Yugozapaden); BG42 (Yuzhen tsentralen)
Cyprus	CY0 (Cyprus)
Czechia	CZ01 (Praha); CZ02 (Střední Čechy); CZ03 (Jihozápad); CZ04 (Severozápad); CZ05 (Severovýchod); CZ06 (Jihovýchod); CZ07 (Střední Morava); CZ08 (Moravskoslezsko)

Country	NUTS Regions
Germany	DE1 (Baden-Württemberg); DE2 (Bayern); DE3 (Berlin); DE4 (Brandenburg); DE5 (Bremen); DE6 (Hamburg); DE7 (Hessen); DE8 (Mecklenburg-Vorpommern); DE9 (Niedersachsen); DEA (Nordrhein-Westfalen); DEB (Rheinland-Pfalz); DEC (Saarland); DED (Sachsen); DEE (Sachsen-Anhalt); DEF (Schleswig-Holstein); DEG (Thüringen)
Denmark	DK01 (Hovedstaden); DK02 (Sjælland); DK03 (Syddanmark); DK04 (Midtjylland); DK05 (Nordjylland)
Estonia	EE0 (Estonia)
Greece	EL30 (Attiki); EL41 (Voreio Aigaio); EL42 (Notio Aigaio); EL43 (Kriti); EL51 (Anatoliki Makedonia, Thraki); EL52 (Kentriki Makedonia); EL53 (Dytiki Makedonia); EL54 (Ipeiros); EL61 (Thessalia); EL62 (Ionia Nisia); EL63 (Dytiki Ellada); EL64 (Sterea Ellada); EL65 (Peloponnisos)
Spain	ES11 (Galicia); ES12 (Principado de Asturias); ES13 (Cantabria); ES21 (País Vasco); ES22 (Comunidad Foral de Navarra); ES23 (La Rioja); ES24 (Aragón); ES30 (Comunidad de Madrid); ES41 (Castilla y León); ES42 (Castilla-La Mancha); ES43 (Extremadura); ES51 (Cataluña); ES52 (Comunidad Valenciana); ES53 (Illes Balears); ES61 (Andalucía); ES62 (Región de Murcia); ES63 (Ciudad Autónoma de Ceuta); ES64 (Ciudad Autónoma de Melilla); ES70 (Canarias)
Finland	FI19 (Länsi-Suomi); FI1B (Helsinki-Uusimaa); FI1C (Etelä-Suomi); FI1D (Pohjois- ja Itä-Suomi)
France	FR1 (Île-de-France); FRB (Centre-Val de Loire); FRC (Bourgogne-Franche-Comté); FRD (Normandie); FRE (Hauts-de-France); FRF (Grand Est); FRG (Pays de la Loire); FRH (Bretagne); FRI (Nouvelle-Aquitaine); FRJ (Occitanie); FRK (Auvergne-Rhône-Alpes); FRL (Provence-Alpes-Côte d'Azur); FRM (Corse)
Croatia	HR02 (Panonska Hrvatska); HR03 (Jadranska Hrvatska); HR05 (Grad Zagreb); HR06 (Sjeverna Hrvatska)
Hungary	HU11 (Budapest); HU12 (Pest); HU21 (Közép-Dunántúl); HU22 (Nyugat-Dunántúl); HU23 (Dél-Dunántúl); HU31 (Észak-Magyarország); HU32 (Észak-Alföld); HU33 (Dél-Alföld)
Ireland	IE04 (Northern and Western); IE05 (Southern); IE06 (Eastern and Midland)

Country	NUTS Regions
Italy	ITC1 (Piemonte); ITC2 (Valle d'Aosta); ITC3 (Liguria); ITC4 (Lombardia); ITF1 (Abruzzo); ITF2 (Molise); ITF3 (Campania); ITF4 (Puglia); ITF5 (Basilicata); ITF6 (Calabria); ITG1 (Sicilia); ITG2 (Sardegna); ITH1 (Provincia Autonoma di Bolzano); ITH2 (Provincia Autonoma di Trento); ITH3 (Veneto); ITH4 (Friuli-Venezia Giulia); ITH5 (Emilia-Romagna); ITI1 (Toscana); ITI2 (Umbria); ITI3 (Marche); ITI4 (Lazio)
Lithuania	LT01 (Sostinės regionas); LT02 (Vidurio ir vakarų Lietuvos regionas)
Luxembourg	LU0 (Luxembourg)
Latvia	LV0 (Latvia)
Malta	MT0 (Malta)
Netherlands	NL11 (Groningen); NL12 (Friesland); NL13 (Drenthe); NL21 (Overijssel); NL22 (Gelderland); NL23 (Flevoland); NL31 (Utrecht); NL32 (Noord-Holland); NL33 (Zuid-Holland); NL34 (Zeeland); NL41 (Noord-Brabant); NL42 (Limburg)
Poland	PL21 (Małopolskie); PL22 (Śląskie); PL41 (Wielkopolskie); PL42 (Zachodniopomorskie); PL43 (Lubuskie); PL51 (Dolnośląskie); PL52 (Opolskie); PL61 (Kujawsko-pomorskie); PL62 (Warmińsko-mazurskie); PL63 (Pomorskie); PL71 (Łódzkie); PL72 (Świętokrzyskie); PL81 (Lubelskie); PL82 (Podkarpackie); PL84 (Podlaskie); PL91 (Warszawski stołeczny); PL92 (Mazowiecki regionalny)
Portugal	PT11 (Norte); PT15 (Algarve); PT16 (Centro); PT17 (Área Metropolitana de Lisboa); PT18 (Alentejo); PT20 (Região Autónoma dos Açores); PT30 (Região Autónoma da Madeira)
Romania	RO11 (Nord-Vest); RO12 (Centru); RO21 (Nord-Est); RO22 (Sud-Est); RO31 (Sud – Muntenia); RO32 (București – Ilfov); RO41 (Sud-Vest Oltenia); RO42 (Vest)
Sweden	SE11 (Stockholm); SE12 (Östra Mellansverige); SE21 (Småland med öarna); SE22 (Sydsverige); SE23 (Västsverige); SE31 (Norra Mellansverige); SE32 (Mellersta Norrland); SE33 (Övre Norrland)
Slovenia	SI0 (Slovenia)
Slovakia	SK01 (Bratislavský kraj); SK02 (Západné Slovensko); SK03 (Stredné Slovensko); SK04 (Východné Slovensko)

Country	NUTS Regions
United Kingdom	UKC (North East); UKD (North West); UKE (Yorkshire and The Humber); UKF (East Midlands); UKG (West Midlands); UKH (East of England); UKI (London); UKJ (South East); UKK (South West); UKL (Wales); UKM (Scotland); UKN (Northern Ireland)

Source: Own elaboration.

1.6 Calculation of Regional Renewable Energy Potential

To assess whether regions with higher renewable energy and waste (REW) green FDI are those naturally endowed with renewable energy resources, we construct a composite measure of regional renewable energy generation potential. This metric combines wind and solar power capacity factors at the NUTS2 regional level, drawing on technical data from the European Commission's Joint Research Centre.

We use the ENSPRESO (ENergy System Potential for Renewable Energy Sources) dataset, developed by the JRC, which provides estimates of potential electricity generation capacity from renewable sources across European regions. The dataset is available at: https://data.jrc.ec.europa.eu/collection/id-00138.

The ENSPRESO data covers multiple renewable energy sources including onshore wind, offshore wind, solar photovoltaic (PV), concentrated solar power (CSP), and biomass. For our analysis, we focus on wind (both onshore and offshore) and solar (both PV and CSP) as the primary sources of renewable energy relevant to large-scale FDI projects. We exclude biomass from our composite measure as our interest centres on the main utility-scale renewable energy technologies.

1.6.1 Capacity Factor Extraction

The ENSPRESO dataset provides various capacity estimates based on different technical assumptions regarding technology specifications, efficiency parameters, land availability constraints, and climatic conditions. To ensure consistent measurement across regions, we extract capacity factors as follows.

Wind power potential: For each NUTS2 region, we take the average 'Capacity Factor' for the top 50% of land parcels most suitable for wind power generation, based on wind pattern analysis. This approach focuses on the areas within each region where wind conditions are most favourable for electricity generation, providing a realistic upper bound on regional wind potential. The capacity factor represents the ratio of actual electricity generation to the theoretical maximum if turbines operated at full capacity continuously.

Solar power potential: For each NUTS2 region, we extract the 'Capacity Factor' based on annual solar irradiation, assuming a performance ratio of 0.75 and a south-facing panel orientation at 45° angle (variable code: AFA_IRRG45S in the ENSPRESO dataset). These technical parameters reflect standard installations for utility-scale solar projects in Europe.

1.6.2 Regional Aggregation and Harmonisation

Our analysis uses a customised regional classification that combines NUTS1 and NUTS2 regions depending on the administrative structure of each country (see Table S7 for the complete list of regions). This approach recognises that the key subnational administrative unit varies across EU member states – NUTS2 for most countries, but NUTS1 for France, Germany, Belgium, and the United Kingdom.

To align the ENSPRESO NUTS2 data with our regional classification, we apply the following procedures.

Aggregation from NUTS2 to NUTS1: Where our classification uses NUTS1 regions (France, Germany, Belgium, UK), we calculate area-weighted averages of capacity factors across constituent NUTS2 regions. Specifically, for region i composed of NUTS2 sub-regions j, the aggregated capacity factor CF_i is:

$$\mathrm{CF}_i = \frac{\sum_j \left(\mathrm{CF}_j \times \mathrm{Area}_j\right)}{\sum_j \mathrm{Area}_j} \tag{1}$$

where Area_j is the land area in square kilometres of NUTS2 region j. This weighting ensures that larger NUTS2 regions contribute proportionally more to the aggregate capacity factor, reflecting their greater potential for renewable energy installations.

Handling NUTS classification changes: For several smaller countries (Ireland, Lithuania, Slovenia), the NUTS2 boundaries used in ENSPRESO differ from the NUTS 2021 classification employed in our FDI dataset. In these cases, we calculate national-level average capacity factors and assign these values uniformly to all regions within each country. Given that these are relatively small countries with less pronounced internal climatic variation, this simplification introduces minimal measurement error.

For Croatia, where two NUTS2 regions remained unchanged between classifications while two new regions were created, we assign the national average capacity factor to the two new regions.

1.6.3 Construction of Composite Renewable Energy Potential Score

To create a single metric capturing overall renewable energy potential, we combine the normalised capacity factors for wind and solar power:

1. **Wind potential composite**: For each region, we calculate the average capacity factor across onshore and offshore wind potential.

2. **Percentile transformation**: We transform both the wind composite and solar capacity factor into percentile ranks across all EU27+UK regions. A region at the 100th percentile has the highest capacity factor (i.e., the best natural conditions) among all regions, while a region at the 0th percentile has the lowest.

3. **Composite score**: We sum the wind percentile rank and solar percentile rank for each region, yielding a score ranging from 0 (worst conditions for both wind and solar) to 200 (best conditions for both).

4. **Re-normalisation**: We rescale the composite score to a 0–100 scale by dividing by 2, facilitating interpretation. A score of 100 indicates a region is in the top percentile for both wind and solar potential, while a score of 0 indicates bottom percentile for both.

This composite measure allows us to identify regions that are particularly well-endowed with renewable energy resources – either through exceptional solar irradiation, favourable wind patterns, or both – and compare these endowments with observed patterns of REW green FDI inflows.

1.7 Innovation Intensity by Manufacturing Sector

To examine whether green FDI flows disproportionately towards more innovative manufacturing sectors, we construct a measure of sector-level innovation intensity using data from the OECD Business Innovation Statistics and Indicators database, available at: `https://www.oecd.org/en/data/datasets/business-innovation-statistics-and-indicators.html`

We use the indicator *INN_OSLO_IND_XFRMTOT_IND*, which reports the share of firms classified as innovative (as a percentage of total firms) based on the Oslo Manual framework. The data are reported at the industry level (approximately NACE 2-digit) for individual OECD member countries. For each industry, we calculate a simple average of this indicator across all OECD countries for which data are available, yielding a single cross-country mean innovation intensity score per industry. This averaging approach smooths over country-specific idiosyncrasies in innovation patterns and provides a measure of the typical innovation intensity associated with each manufacturing activity across advanced economies.

To align the OECD industry classification with the aggregated survey-sector classification used in our FDI dataset, we map OECD industries onto our sector categories. Both classifications are based on NACE 2-digit codes, so the mapping is generally straightforward. Where the mapping is not one-to-one – that is, where multiple OECD industries correspond to a single aggregated sector in our classification, or vice versa – we calculate simple means of the innovation intensity scores across the relevant OECD industries. The resulting dataset contains one observation per manufacturing sector with the corresponding OECD-average share of innovative firms. We restrict the analysis to manufacturing sectors, where the majority of beyond-energy green FDI is seen to flow.

SI.2. Benchmarking Against Keyword-Based Classification

To assess the value-added of our EUTSA-guided LLM approach, we conducted a parallel classification of non-REW green FDI using a traditional keyword-based method. This comparison serves two purposes: first, to demonstrate the limitations of simpler text-matching approaches that have been used in related domains; and second, to quantify how much green investment activity would remain undetected without the more sophisticated framework we propose.

To ensure a fair comparison, we used the LLM itself (GPT-4o) to identify the 25 most commonly used green-related terms in FDI project descriptions. We prompted the model as follows:

> *"Based on your knowledge of sustainability and green investment terminology, identify the 25 most common words or short phrases that would appear in descriptions of environmentally-focused foreign direct investment projects. Focus on terms related to: climate change mitigation, renewable energy, energy efficiency, waste reduction, circular economy, pollution prevention, and sustainable practices. Provide terms that would be suitable for a keyword search."*

This approach ensured that our keyword list reflected terminology actually used in the green investment space, rather than terms we might subjectively select. The resulting 25 keywords are shown in Table S8.

A project was classified as green under the keyword approach if its description contained at least one of the 25 keywords. The search was case-insensitive and used partial string matching for terms with asterisks (e.g., "energy efficien" matches both "energy efficient" and "energy efficiency"). Additionally, hyphens were removed so as to ensure matches with terms like "zero waste" or "pollution reducing" which may often be hyphenated. The python script used to do this is available as part of the replication package.

Figure S9 presents a cross-tabulation comparing the keyword-based classification with our EUTSA-LLM approach for beyond-energy sectors. The results reveal significant discrepancies between the two methods.

The comparison reveals three distinct categories:

1. **Agreement (2.6% of all non-REW FDI)**: Projects identified as green by both methods. These represent investments with both green-related terminology and evidence of qualifying activities under EUTSA criteria.

Table S8: Keywords Used for Benchmark Classification

Keywords (1–13)	Keywords (14–25)
green	nature based
sustainable	pollution reducing
eco friendly	carbon capture
environmental	biodegradable
low carbon	climate friendly
zero carbon	climate smart
clean tech	climate change
green tech	climate resilien*
renewable	organic
alternative energy	sustainably sourced
net zero	zero waste
energy efficien*	low waste
recycl*	

Note: Asterisk (*) indicates partial string matching was used (e.g., "recycl" matches "recycling," "recyclable," "recycled", etc.).
Source: Keywords generated via GPT-4o prompt; own compilation.

Figure S9: Share of inward FDI that is green in non-REW sectors: EUTSA LLM-based approach vs. simple keyword search

		1. Green in EUTSA – LLM approach	
		NO	**YES**
2. Green in keyword-search approach	**NO**	88.0%	4.6%
	YES	2.6%	2.6%

Total %, keyword: 7.2%

Total %, EUTSA – LLM: 7.4%

Source: Own elaboration, based on data from Orbis-crossborder / Bureau van Dijk (2025)

2. **"Hidden green" projects (4.7% of all non-REW FDI)**: Projects classified as green by the EUTSA-LLM approach but not by keywords. These missed projects represent nearly two-thirds (64%) of the beyond-REW green FDI identified by our EUTSA-LLM approach. This blind spot is particularly problematic in technical or industrial sectors where project descriptions focus on engineering specifications, production processes, or business functions rather than environmental attributes. For instance, investments in electric arc furnace steelmaking, LEAP engine manufacturing, or building renovation using specific insulation technologies may qualify as green under technical criteria without ever mentioning terms like "sustainable" or "low-carbon."

3. **Potential greenwashing (4.6% of all non-REW FDI)**: Projects flagged by keywords but not meeting EUTSA criteria. These descriptions contain green-related terms but lack evidence of qualifying activities according to technical standards. This is a potential indicator of greenwashing, where companies use environmental rhetoric without substantive green practices. This limitation is especially concerning given the policy importance of accurately tracking and incentivizing genuine green investment.

Figure S10 illustrates both categories of misclassification with concrete examples from our dataset: an investment in a LEAP engine manufacturing plant in France, which includes no keywords but nevertheless involves green activities, and another by Schenker in Poland where keywords are mentioned but without substantive evidence of specific green activities taking place.

The EUTSA-LLM approach addresses both limitations by evaluating projects against specific technical criteria rather than relying on the presence of sustainability-related terms. By grounding classification in detailed screening criteria – such as carbon intensity thresholds for steel production, Power Usage Effectiveness limits for data centres, or energy performance standards for building renovation – our method can identify green activities regardless of how they are described, while filtering out projects that use green terminology without qualifying activities.

It should be noted that the keyword approach identifies a larger total share of projects as green (7.2% vs. 7.4%), which might superficially suggest comparable performance. However, this near-equivalence in aggregate numbers masks substantial disagreement at the project level: only 35% of projects classified as green by our EUTSA-LLM method are also captured by keywords (2.6% out of 7.4%), while keywords flag an entirely different set of projects as potential false positives. This low overlap rate (35%) indicates that the two methods are fundamentally measuring different phenomena – keyword searches identify projects with green-related language, while the EUTSA-LLM approach identifies projects with green-qualifying activities.

Figure S10: Two example cases where a keyword search erroneously classifies projects: 'hidden' green (left) and 'potential greenwashing' (right).

FDI project #1

Is the following investment, in the **"non-automobile manufacturing sector"**, green?

*"On [date] it was announced that [Company A], a [City], [Country]-based bearing and engineering products manufacturer, has expanded its aircraft engine parts manufacturing plant in [City], [Country]. A new production line has been installed, **entirely focused on LEAP engines**, to expand production capabilities…"*

Here are the activities considered green within this sector, according to the EUTSA:

"…

3. *Manufacturing of Aircraft: This encompasses the production and servicing of aircraft and their components, with activities such as manufacturing, repairing, maintaining, overhauling, retrofitting, designing, repurposing, and upgrading, ensuring a reduction in environmental impact. … etc."*

LLM response

"Yes – it can be considered green based on the information provided.

*Justification: The investment description states that a new production line has been installed, entirely focused on LEAP engines. **LEAP engines are known for their fuel efficiency and reduced emissions compared to older engine models.** Therefore, the investment aligns with the EU taxonomy's criteria for green activities in the non-automobile transport manufacturing sector, specifically under the "Manufacturing of Aircraft" category…"*

FDI project #2

Is the following investment, in the **"Transportation, logistics & storage"** sector, green?

*"On [date] it was announced that [Company B], a [City], [Country]-based logistics service provider is to open a new distribution centre in [City], [Country]… [Executive], CEO [Region] at [Company B], said: "…we will also start construction of a modern and **environmentally friendly** facility in the [Region]."*

Here are the activities considered green within this sector, according to the EUTSA:

"…

1. *Transmission and distribution of renewable and low-carbon gases…*
2. *Freight rail transport, which involves the operation of mainline and short line freight railroads…*
3. *Passenger interurban rail transport … etc."*

LLM response

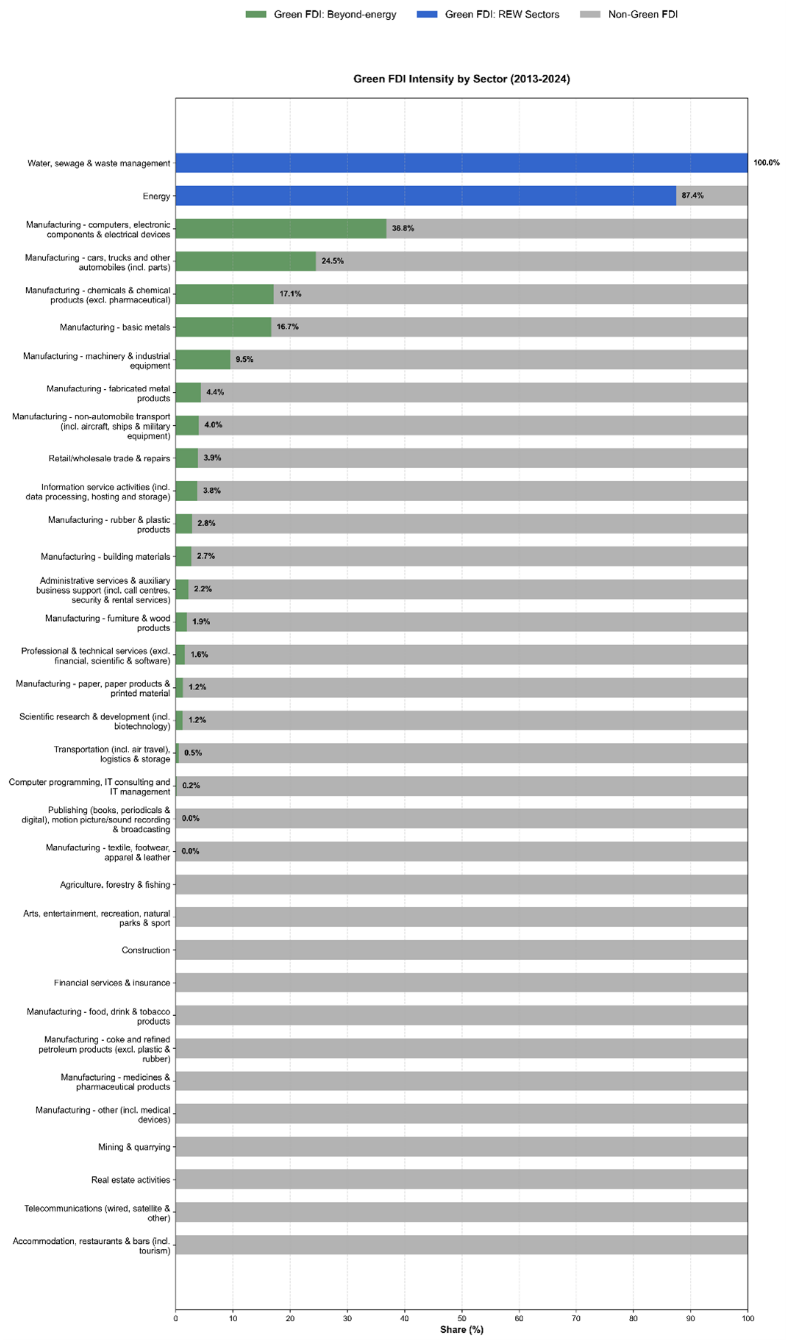
"No – it cannot be considered green based on the information provided.

*Justification: The EU taxonomy for sustainable finance lists activities considered green, such as the transmission and distribution of renewable and low-carbon gases, transport of captured CO2, various forms of passenger and freight transport etc. **The provided description of [Company B]'s investment does not explicitly align with these activities.** It does not mention any specific efforts towards supporting green and low-carbon transportation, such as retrofitting for low-carbon gases, or construction/modernization of infrastructure specifically designed to reduce environmental impact."*

Source: Own elaboration using data from Orbis-crossborder / Bureau van Dijk (2025)

SI.3. Additional Descriptives

Figure S11: % of inward FDI into EU27+UK that is 'green' (FDI cap. value, 2013–2024), by sector



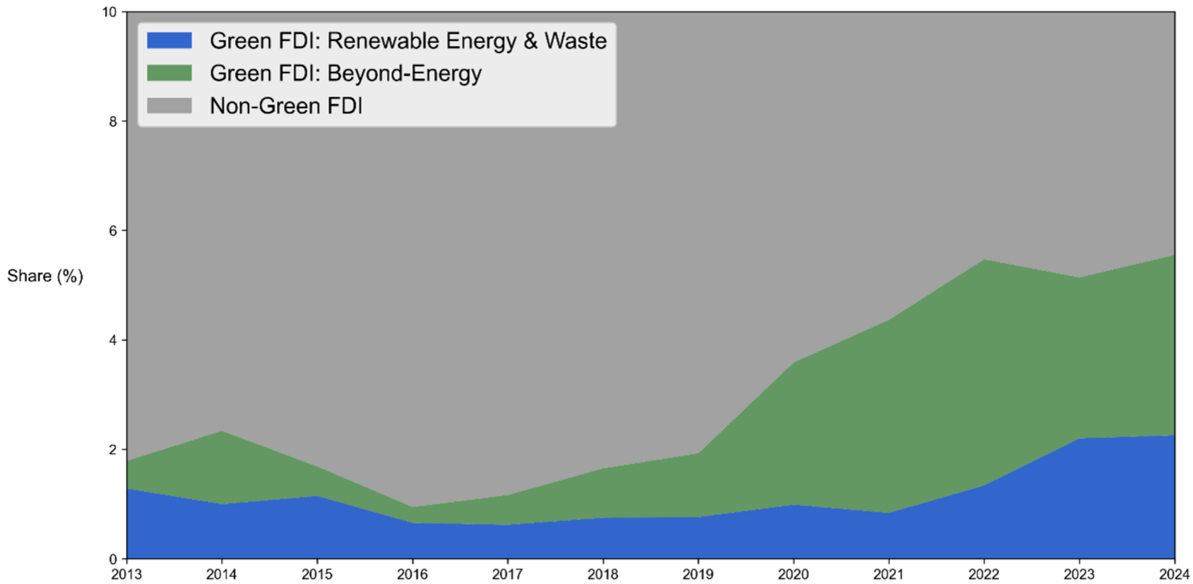
Source: Own elaboration with data from Orbis-crossborder / Bureau van Dijk (2025)

3.1 Green FDI descriptives: main approach – FDI project counts

Figure S12: Share of inward FDI into EU27+UK that is 'green' (Num. projects, 2013–2024)

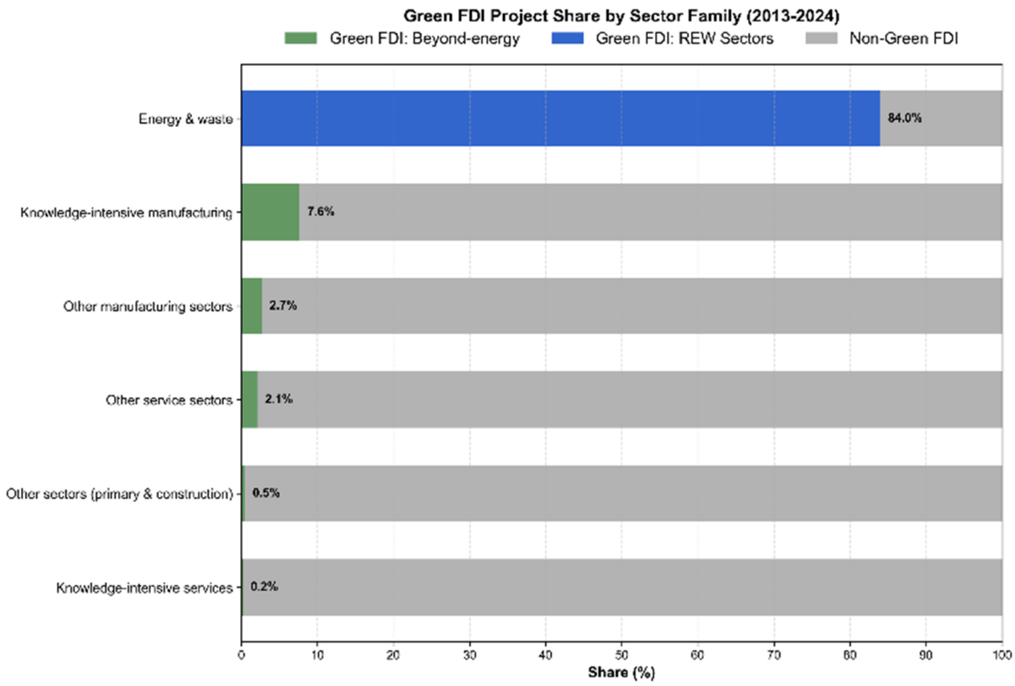
Source: Own elaboration using data from Orbis-crossborder / Bureau van Dijk (2025)

Figure S13: Share of inward FDI into EU27+UK that is 'green' (Num. projects, 2013–2024), by year



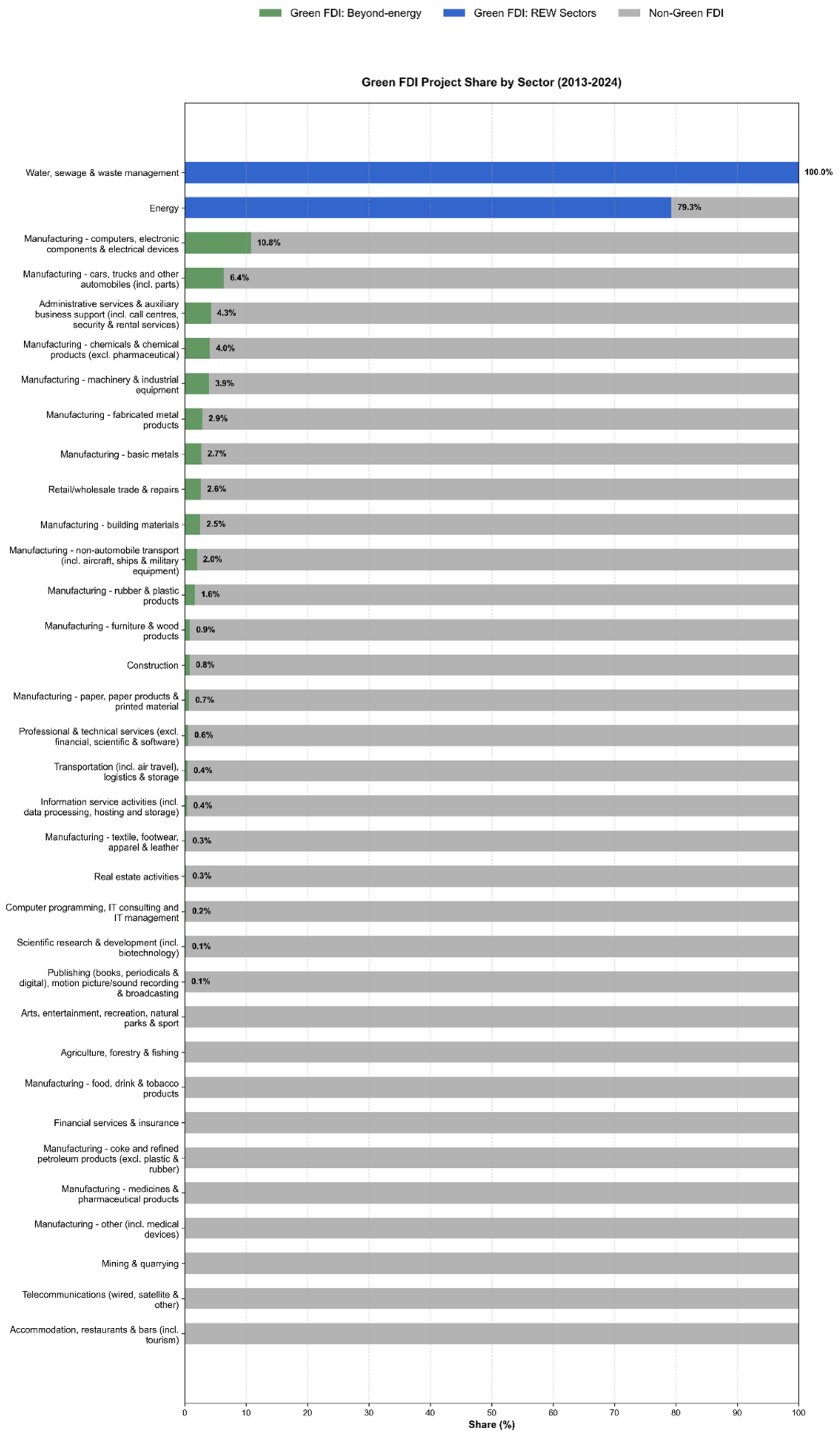
Source: Own elaboration using data from Orbis-crossborder / Bureau van Dijk (2025)

Figure S14: Share of inward FDI into EU27+UK that is 'green' (Num. projects, 2013–2024), by sector family



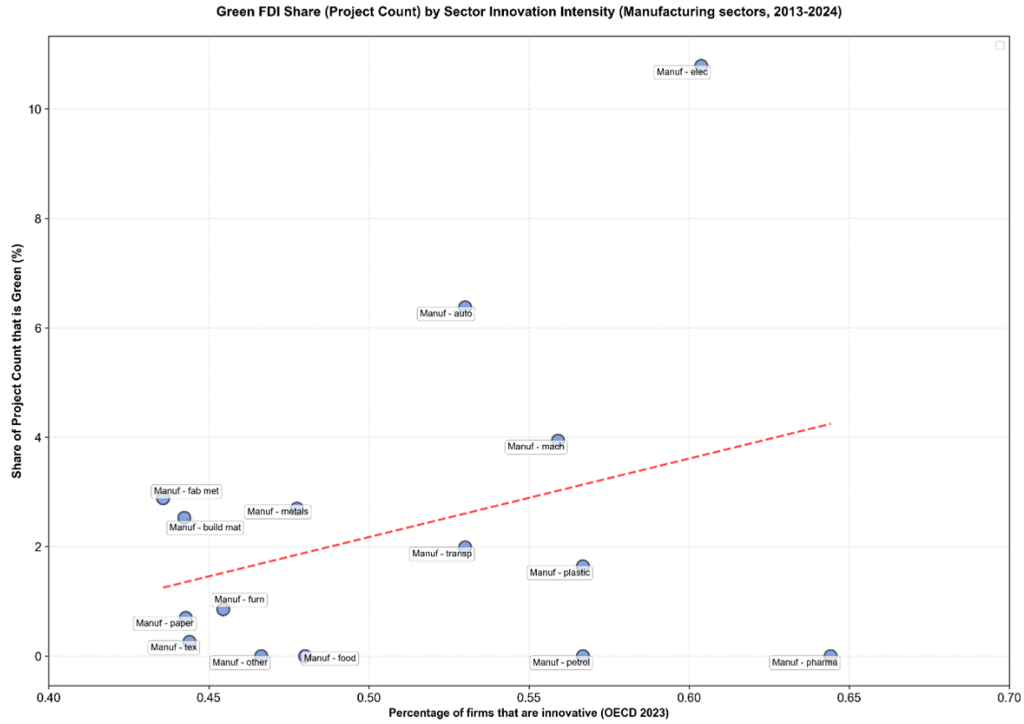
Source: Own elaboration using data from Orbis-crossborder / Bureau van Dijk (2025)

Figure S15: Share of inward FDI into EU27+UK that is 'green' (Num. projects, 2013–2024), by sector



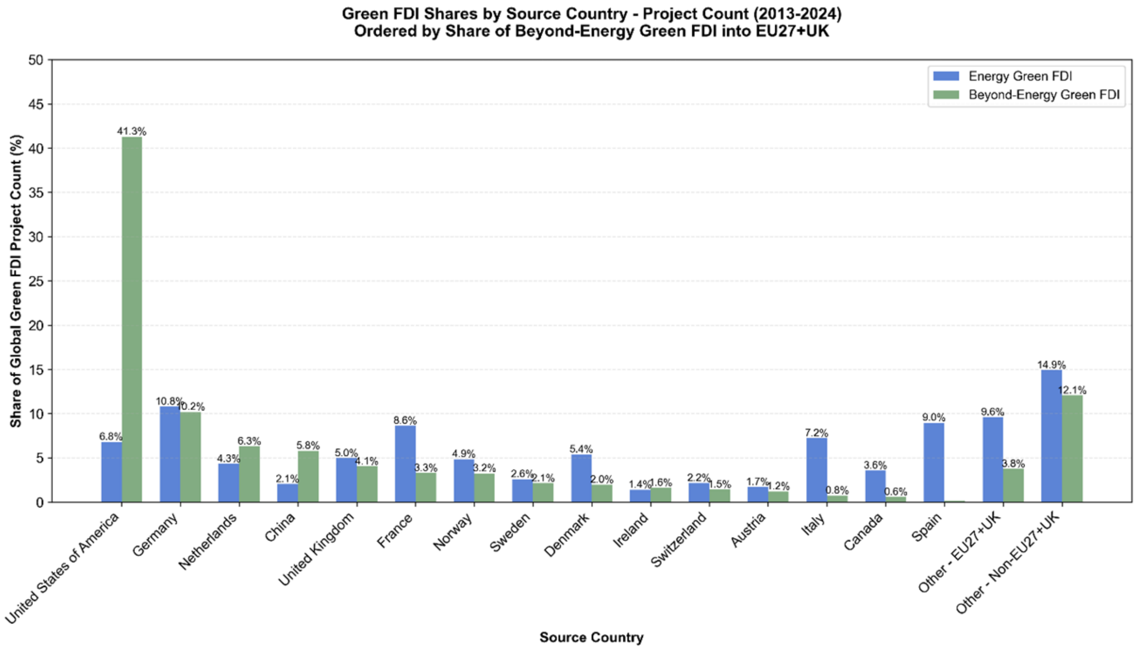
Source: Own elaboration using data from Orbis-crossborder / Bureau van Dijk (2025)

Figure S16: Share of inward FDI into EU27+UK that is 'green' (Num. projects, 2013–2024) in each manufacturing sector vs. share of innovative firms in sector (2023)



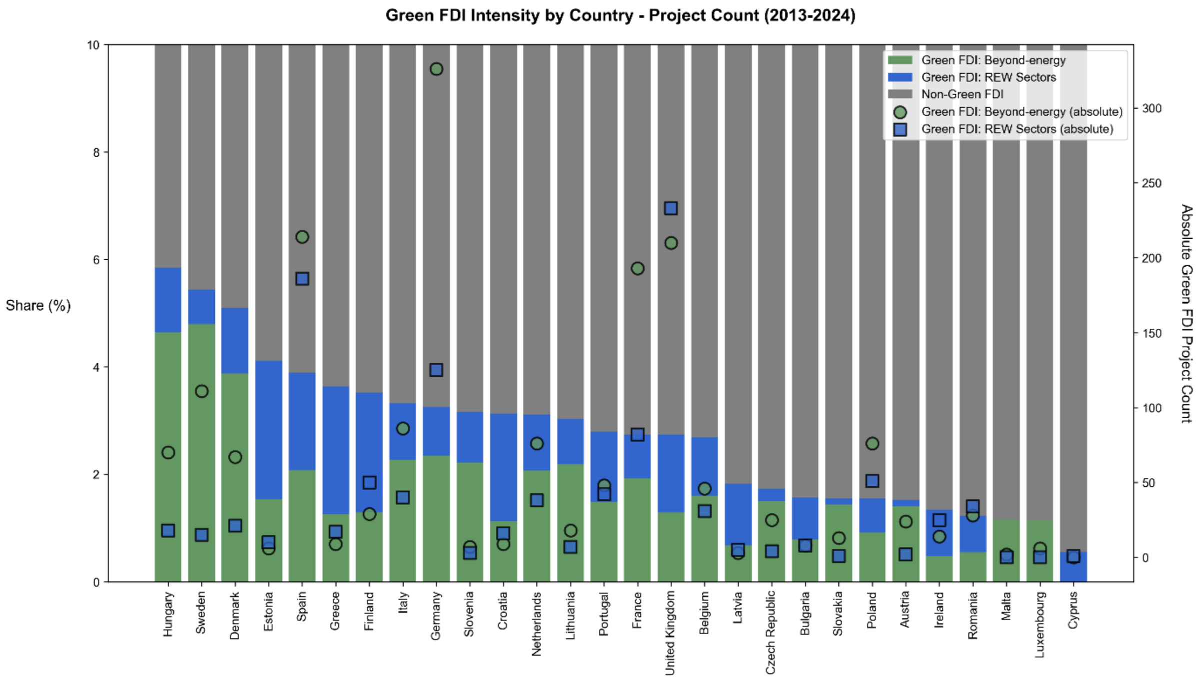
Source: Own elaboration using data from Orbis-crossborder / Bureau van Dijk (2025); OECD, 2024.

Figure S17: Share of inward FDI into EU27+UK that is 'green' (Num. projects, 2013–2024) by country of origin, for both energy and beyond-energy green FDI, top 15 source countries



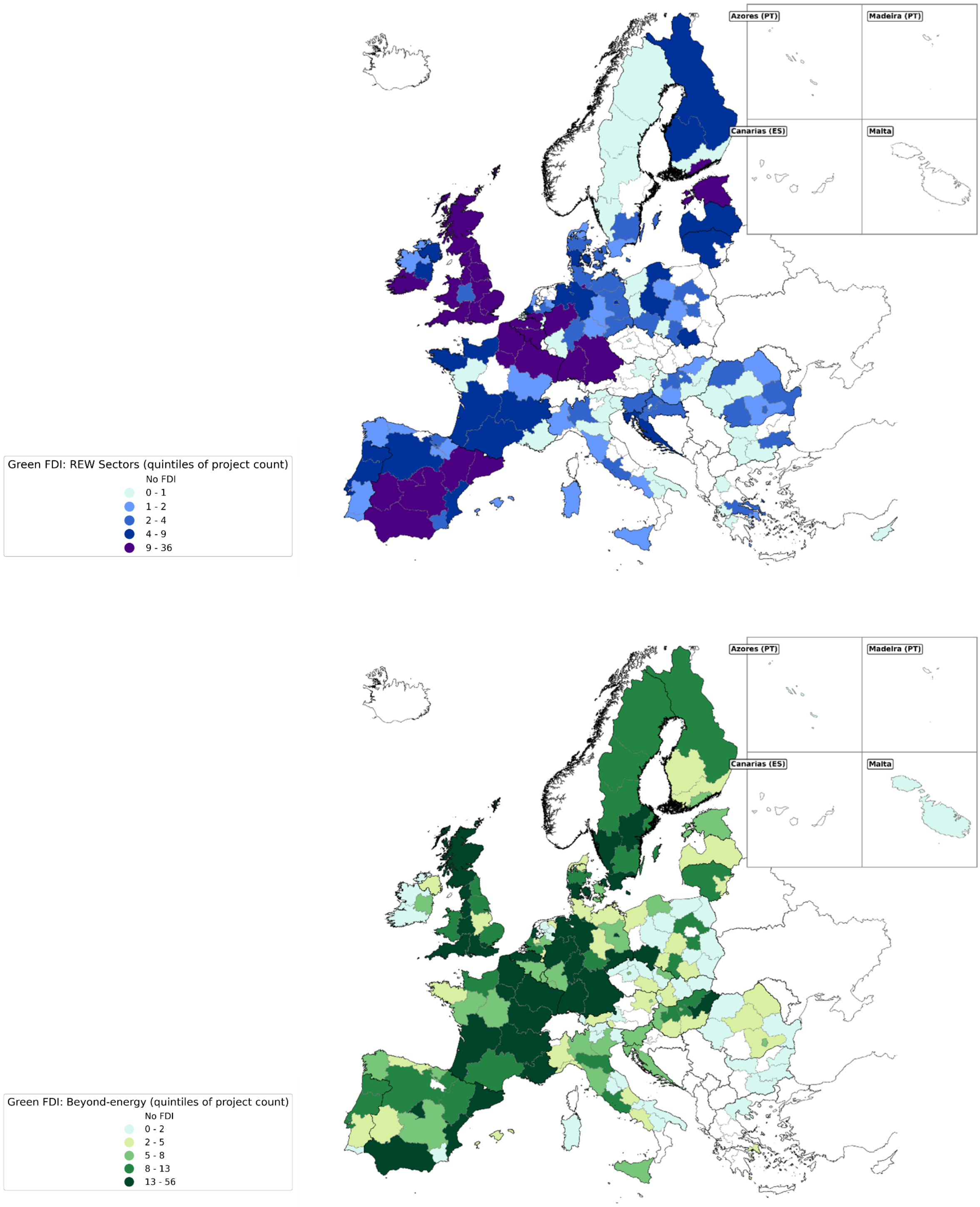
Source: Own elaboration using data from Orbis-crossborder / Bureau van Dijk (2025)

Figure S18: Number and share of inward FDI projects into EU27+UK that are 'green' (Num. projects, 2013–2024), by country



Source: Own elaboration using data from Orbis-crossborder / Bureau van Dijk (2025)

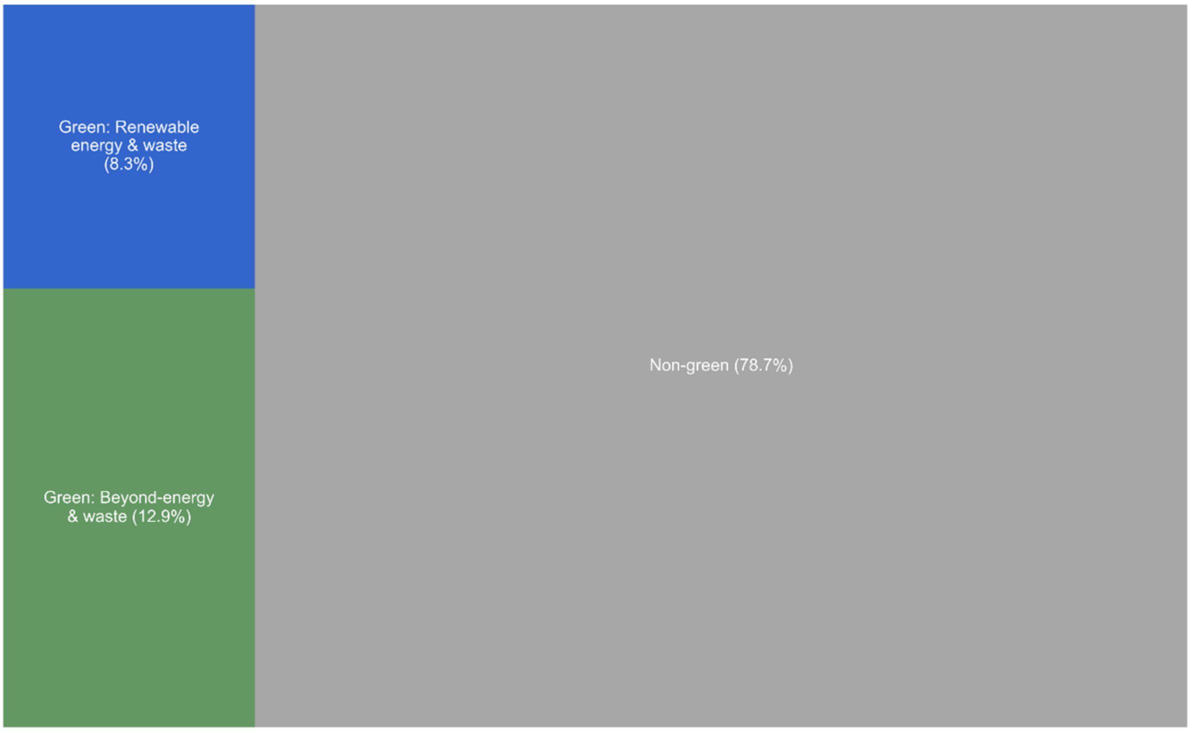
Figure S19: The regional distribution of green FDI (Num. projects) in EU27+UK: Renewable Energy & Waste FDI (top panel) and beyond-energy green FDI (bottom panel).



Source: Own elaboration using data from Orbis-Crossborder / Bureau van Dijk (2025).

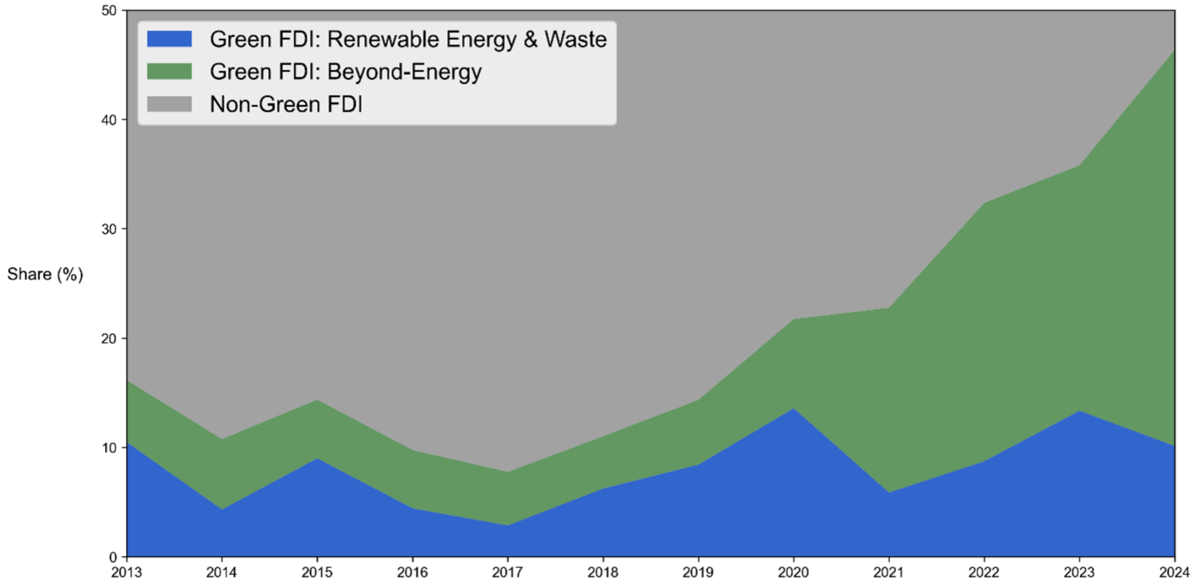
3.2 Green FDI descriptives: 'upper bound' approach

Figure S20: Share of 'green' inward FDI into EU27+UK – upper bound (FDI capital value, 2013–2024)



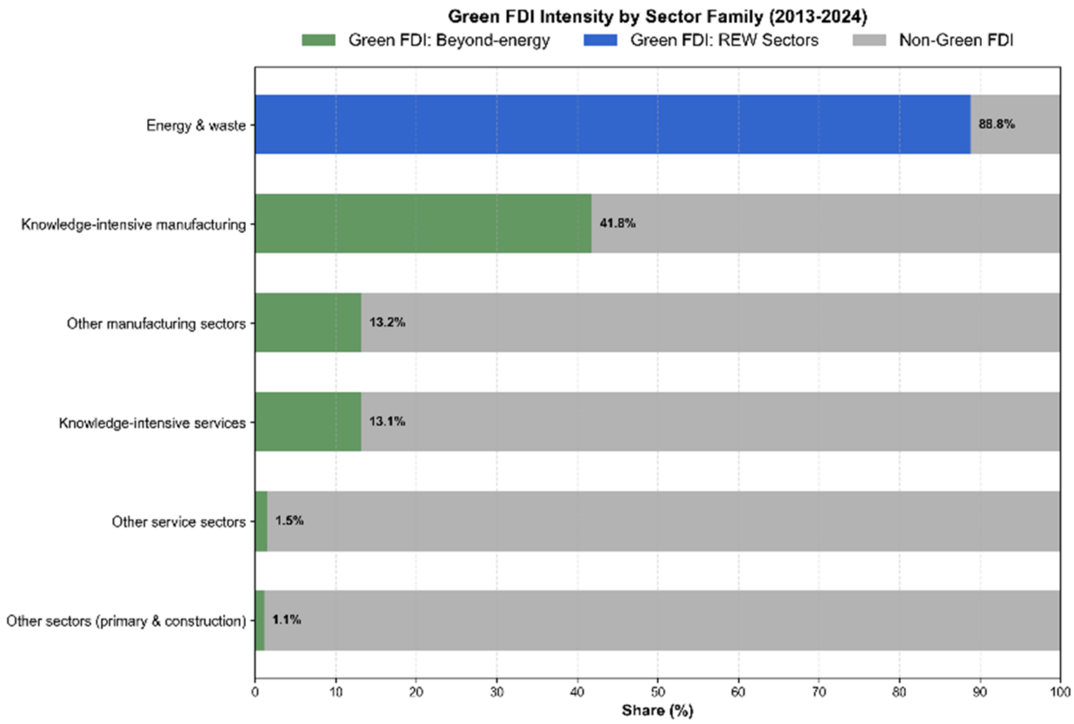
Source: Own elaboration using data from Orbis-crossborder / Bureau van Dijk (2025)

Figure S21: Evolution in the share of 'green' inward FDI into EU27+UK – upper bound (FDI capital value, 2013–2024)



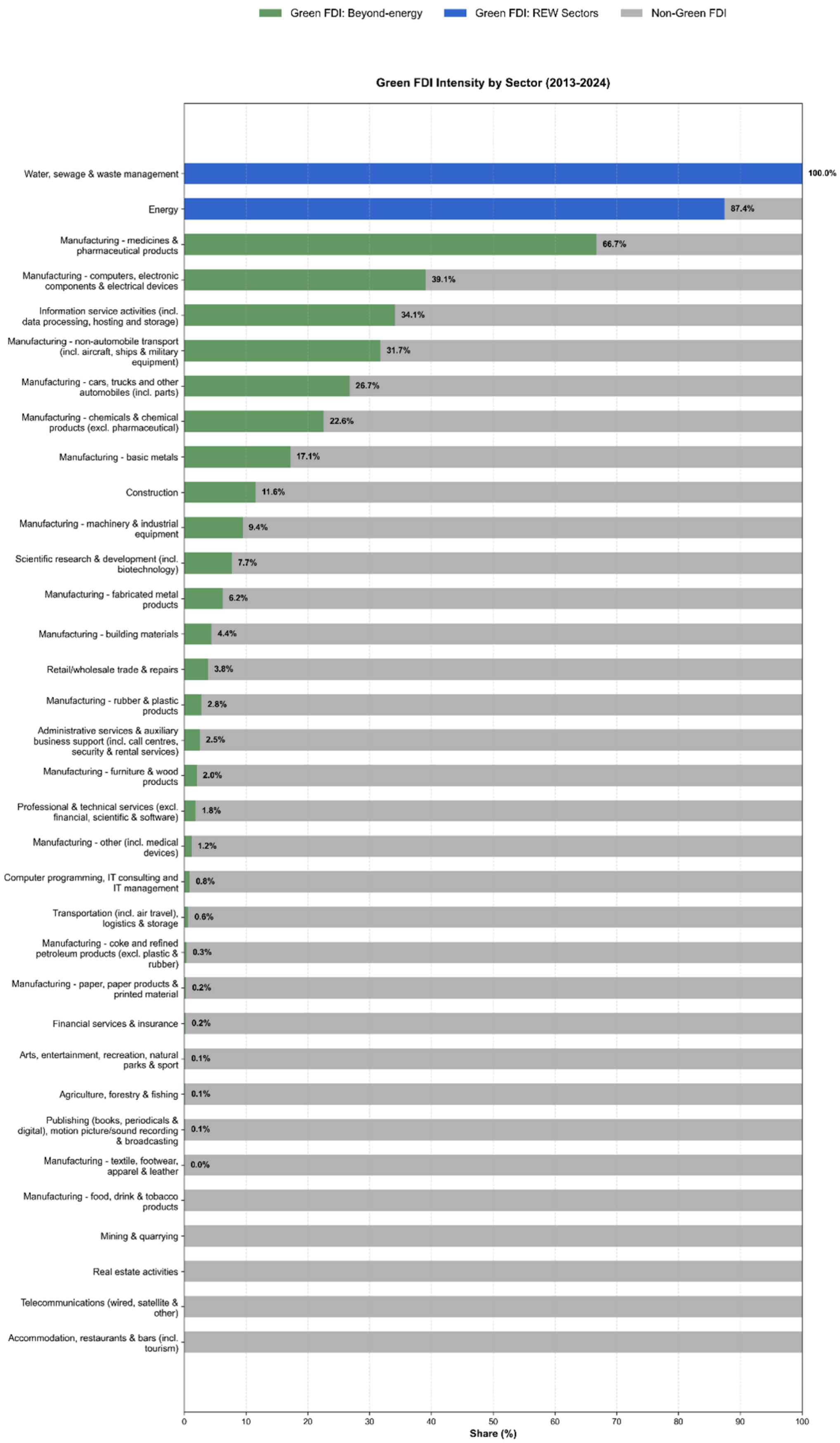
Source: Own elaboration using data from Orbis-crossborder / Bureau van Dijk (2025)

Figure S22: Share of inward FDI into EU27+UK that is 'green' (FDI capital value, 2013–2024) – upper bound, by sector family



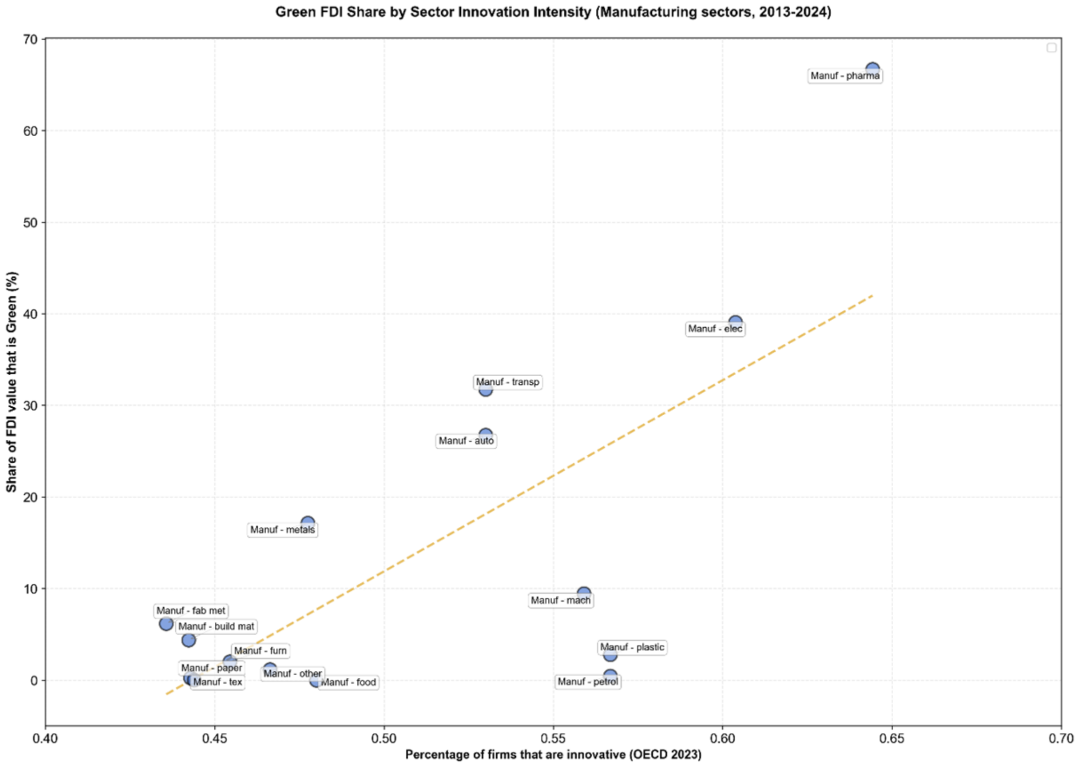
Source: Own elaboration with data from Orbis-crossborder / Bureau van Dijk (2025)

Figure S23: Share of inward FDI into EU27+UK that is 'green' (FDI capital value, 2013–2024) – upper bound, by sector



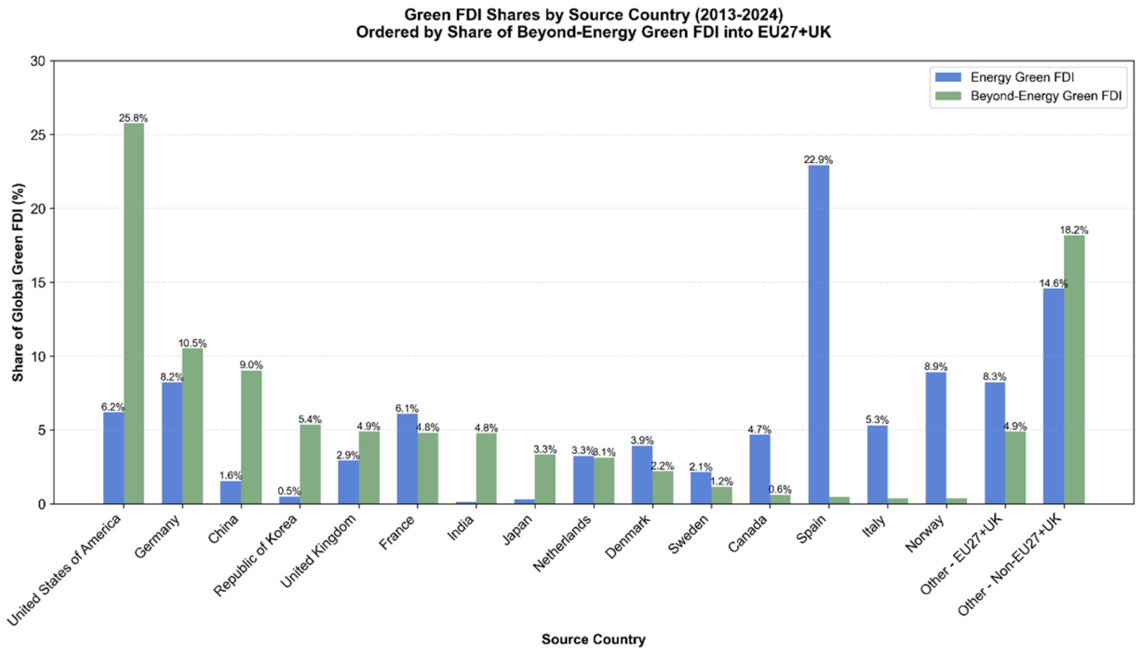
Source: Own elaboration with data from Orbis-crossborder / Bureau van Dijk (2025)

Figure S24: Share of inward FDI into EU27+UK that is 'green' (FDI capital value, 2013–2024) in each manufacturing sector – upper bound – vs. share of innovative firms in sector (2023)



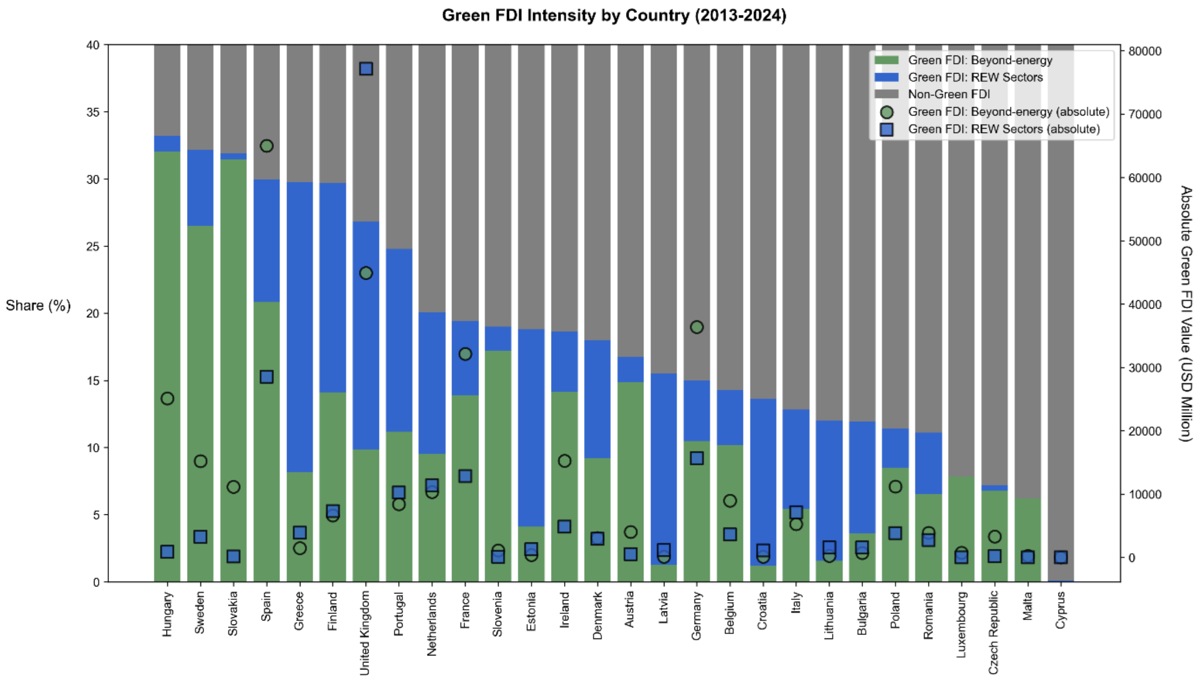
Source: Own elaboration with data from Orbis-crossborder / Bureau van Dijk (2025); OECD, 2024.

Figure S25: Share of inward FDI into EU27+UK that is 'green' (FDI capital value, 2013–2024) by country of origin – upper bound, for both energy and beyond-energy green FDI, top 15 source countries



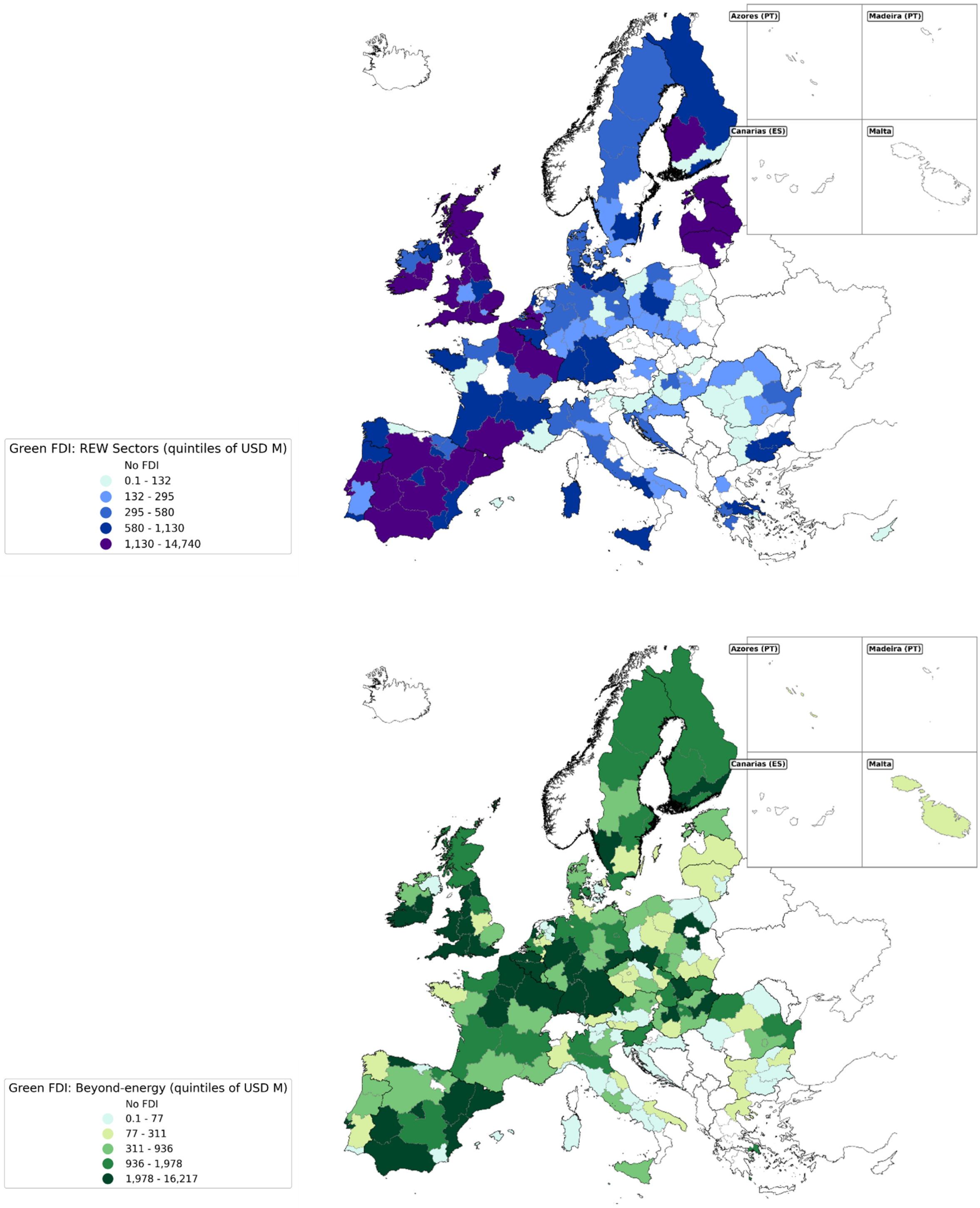
Source: Own elaboration with data from Orbis-crossborder / Bureau van Dijk (2025)

Figure S26: Value and share of inward FDI into EU27+UK that is 'green' (FDI capital value, 2013–2024) – upper bound, by country



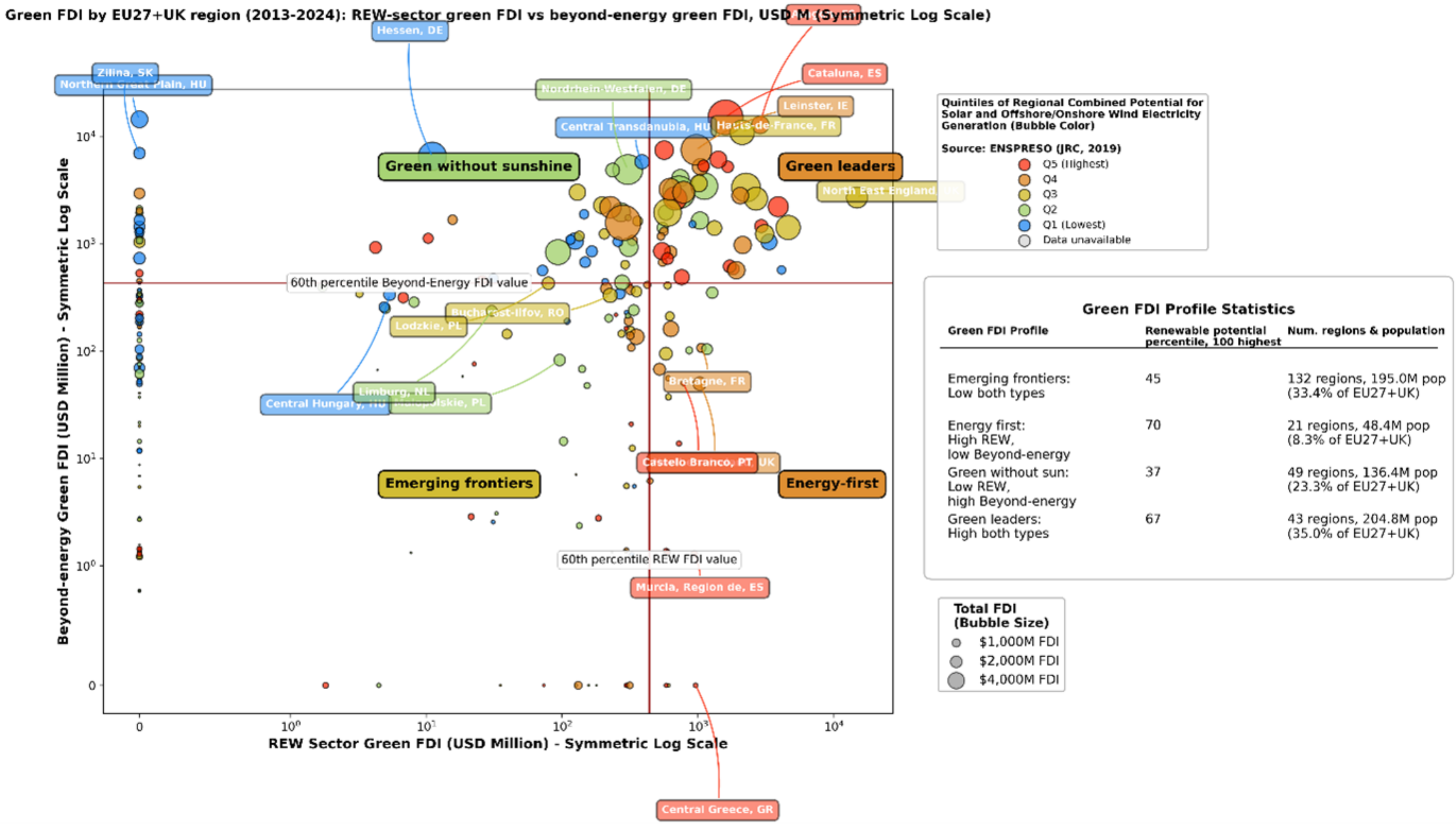
Source: Own elaboration with data from Orbis-crossborder / Bureau van Dijk (2025)

Figure S27: The regional distribution of green FDI (USD value) in EU27+UK – upper bound: Renewable Energy & Waste FDI (top panel) and beyond-energy green FDI (bottom panel)



Source: Own elaboration using data from Orbis-Crossborder / Bureau van Dijk (2025).

Figure S28: Green FDI by region in EU27+UK (2013–2024) – upper bound: Share of REW Sector green FDI vs Share of Beyond-Energy Green FDI (FDI USD value), symmetric log scale



Source: Own elaboration using data from Orbis-Crossborder / Bureau van Dijk (2025).

SI.4. LLM Evaluation

Several methods have been applied to verify the quality of LLM outputs and assess the methodology's overall usability. We focus these tests around the following three axes: the accuracy of the LLM in carrying out our request, as judged by the agreement of its conclusions vis-à-vis those reached by trained human experts; the validity of its rationale, as proxied by its 'faithfulness' to the inputs and instructions given; the reproducibility of the LLM's decision-making, as indicated by the sensitivity of outputs when repeating the request, both via identical prompts and model parameters and via perturbation, where we vary the prompts and parameters and compare results. These axes are deemed the most relevant given the nature of the task at hand, and are in line with the most common rubrics for human evaluation of NLG outputs that have been highlighted by the literature (Chang et al., 2023).

4.1 Accuracy

To query the accuracy of responses, in terms of whether the conclusions reached by the LLM are 'correct', we undertake human evaluation checks with postgraduate students who have advanced knowledge of the domain. While it is possible to carry out a series of automated checks and/or comparison to 'benchmarks' for specific kinds of NLP and NLG tasks (and in fact we compute some of these automated metrics too), these often fall short of replicating human decisions (Krahmer and Theune, 2010; Reiter, 2018; Reiter and Belz, 2009), so human expert evaluation checks are considered the 'gold standard' (Celikyilmaz et al., 2021; Clark et al., 2021).

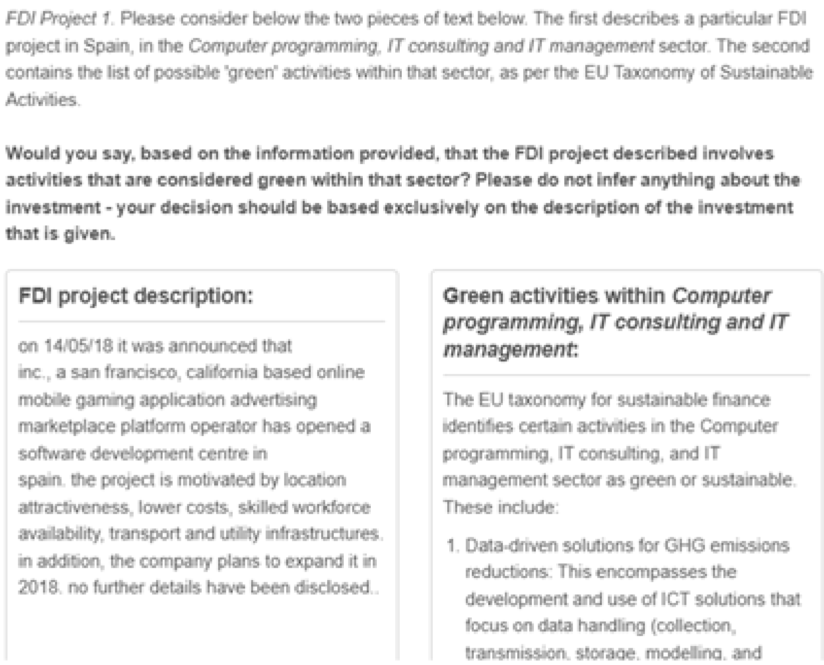
In our case, we recruit students to carry out an intrinsic evaluation of the GPT-4 output itself, rather than an 'extrinsic' evaluation of how well the output enables the success of a downstream task (Celikyilmaz et al., 2021; van der Lee et al., 2021). We opt for human 'experts' (as opposed to non-experts[19]) in reflection of the audience for which this was developed (van der Lee et al., 2021). Someone carrying out this task manually would require some domain expertise/context beyond a simple understanding of the English language, for example in terms of synonymous terminology for the green activities/technologies which may be referenced in both the EUTSA and the FDI project descriptions. As such, we opt for postgraduate students in the fields of economic geography, environmental economics and international development at the London School of Economics, all of whom are directly exposed in their studies to concepts around the green transition, sustainable technologies and FDI.[20]

[19]Depending on the nature of the NLG task, evaluation may well be undertaken by any human that is able to understand the given language, for example via platforms like Amazon Mechanical Turk.

[20]In this respect, the selection of postgraduate students with *some* relevant knowledge and cognition could in fact be more suitable than senior experts in EU sustainable finance (a topic which is known to be contentious and somewhat politicised), who may in fact draw on their own insights and/or notions of what activities are green in making this decision. However, it is important to remember that we are using the LLM to identify green FDI

In total, we ask 34 students to review the LLM outputs for 10 FDI projects each (giving a total of 340). These were allocated from a random selection of 500 FDI projects which the LLM considers to be green, and 500 which the LLM concludes are not green, after the 'first stage' of processing.[21] Our human verification stage compares favourably and is more transparent than is common in NLG studies. A meta-analysis of INLG and ACL papers (van der Lee et al., 2021) reports a median of just 3 human evaluators per study, with 39% using 5 or fewer. Our panel of 34 trained postgraduates also falls within the top 15% of studies that recruit at least 10 human raters. Similarly, whereas the median evaluation in their analysis is 100 texts, we assess 340 FDI descriptions. We also provide demographic details and operate a blind, randomised protocol, practices that van der Lee et al. (2021) show are missing from much published work in this space. Moreover, in reviewing 5% of all FDI projects that are identified as green after the first stage (out of a total of 3,414), this exercise is carried out over a substantial portion of our full sample.

Figure S29: Screenshot of Qualtrics evaluation interface shown to human evaluators



Source: Own elaboration.

projects based on the strict application of an existing classification logic (in the form of the EUTSA), and it is this that we wish our human experts to replicate.

[21]We opt to conduct our human expert evaluation exercise based on the LLM outputs following the first stage of processing, where the prompt includes more general (and often vague) detail describing the green activities for the corresponding sector. This decision was taken in order to limit the technical detail which the human evaluators had to digest in order to evaluate the green-ness of each FDI project, and enable them to process more cases. If, at this stage where the EUTSA criteria can be generic and vague, the LLM already demonstrates a high-level of accuracy vis-à-vis the human expert judgments, we expect that when the prompt is strengthened by adding more technical detail and the decisions are more 'clear-cut', the agreement between the LLM and human experts would likely only increase. In this way, our methodology for evaluating accuracy through human checks is likely providing a conservative (or 'lower-bound') approximation of our LLM methodology's ability to accurately identify which FDI projects are green in each sector.

As shown in Figure S29, students were presented with the task exactly as it was to the LLM, although via survey software Qualtrics in order to facilitate readability. In line with best-practice for these kinds of evaluations, student evaluators carry out their assessment 'blind', in that they do not get to view the response given by the LLM until after giving their own verdict. We gather the responses given by the student evaluators, compare these to the LLM's decisions for each FDI project, and compute an F1 score. This is a commonly used metric for evaluating the performance of LLMs in conducting binary classification tasks (Chang et al., 2023). It combines a model's 'precision' (the share of false positives) with its 'recall' (the share of false negatives), effectively computing a single measure for how 'correct' or accurate the model is. As shown below in Equation 2, the F1 score is effectively the harmonic mean of these two scores:

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{2}$$

Table S9: F1, precision and recall scores from human evaluation checks

Agreement with GPT-4 when...	F1 score	Precision	Recall
Judgment made 'blind'	0.844	0.844	0.837
Respondent shown GPT-4 decision and justification	0.932	0.943	0.921

Source: Own elaboration.

The F1 scores shown in Table S9 denote the accuracy of GPT-4 with respect to two sets of human-evaluator conclusions: firstly, the conclusion when the evaluator is presented with the task 'blind', in the sense of not viewing the LLM's own conclusion; secondly, the conclusion after, in cases where the evaluator's judgment differs from that of the LLM, the evaluator is shown GPT-4's own decision and justification.[22] In the first instance, the LLM decision as to whether an investment can be considered green (or not) has an F1 score of 0.844. Moreover, the precision of the LLM is virtually equal to its recall. This indicates that, among any errors, the frequency of 'false positives' (non-green FDI projects that the LLM wrongly identifies as green) is basically the same as the frequency of 'false negatives' (green FDI projects that the LLM wrongly identifies as non-green). If we switch the baseline to be the human evaluator's judgment after being able to view the LLM's own assessment, the F1 score increases further to 0.932. Again, the precision and recall scores which make up the F1 metric are well-balanced. These scores are firmly within what the literature considers robust (typically above 0.7) for binary classification tasks[23], and enables us to conclude that our use of the LLM produces accurate conclusions on whether an FDI project can be considered green.

[22] In such cases, we identify which evaluators 'changed their mind' toward agreement with GPT-4 by asking them to indicate whether they agree (or not) with GPT-4's own conclusion. Of 5 options (completely agree, somewhat agree, neither, somewhat disagree, completely disagree), we consider 'opinion changes' those cases where respondents select either of the first two options. The new set of human-evaluator conclusions that this process generates then becomes the baseline for comparison, enabling the calculation of the second F1 accuracy score shown in Table S9.

[23] See, for example: https://serokell.io/blog/a-guide-to-f1-score

4.2 Validity

To explore the validity of the outputs, we focus on the justifications that we ask the LLM to provide in relation to every FDI project. The intention is to determine whether, beyond the accuracy of its final judgment, the agent applies logic that is valid in the sense of following the instructions given. For our particular case, this would imply verifying that the LLM is determining whether an FDI project is green based on the specific EUTSA criteria given, rather than on its own 'intuitions' or parametrised knowledge. Even if an FDI project is correctly identified as green (or not green), it may well be that this was for the 'wrong' reason, for example because it draws on alternative, erroneous set of criteria beyond the EUTSA. While our prompt to the LLM was worded to avoid this, the potential for LLMs to override instructions and/or 'hallucinate' is a well-known phenomenon that NLG researchers and users have to contend with.

We address this via a combination of automated checks. Firstly, we conduct simple word-searches within the justifications given, in order to verify that the term "EU Taxonomy" is mentioned. This would provide an initial indication that the logic and/or criteria applied are in fact based on the EUTSA, and therefore that the LLM is likely faithfully carrying out its task. Word-searches across all 109,084 justifications show that 93.7% of green classifications and 97.2% of non-green classifications explicitly mention "EU Taxonomy," "EUTSA," or "Taxonomy criteria". The small number of non-mentions were manually reviewed and found to implicitly reference EUTSA concepts without using the exact terminology.

To bolster this, we also check whether the content of the justification is relevant, in the sense of aligning with the content of the corresponding EUTSA section. We do this by calculating BLEU similarity scores (Papineni et al., 2002) between each justification and the EUTSA criteria that is extracted for every sector. This measures the similarity between texts by comparing *n*-grams (sequences of *n* words) in a particular text with a 'reference' text, and is defined mathematically as shown in Equation 3, below:

$$\text{BLEU} = BP \times \exp\left(\sum_{n=1}^{N} w_n \cdot \log(p_n)\right) \tag{3}$$

where w_n are weights for each *n*-gram, p_n is the precision of an *n*-gram, and N is the maximum order of *n*-grams used (from one to four). The term BP denotes a 'brevity penalty' which captures the difference in text length, and which we set to uniform for our purposes. A BLEU score ranges between 0 and 1, with 1 representing 'perfect' *n*-gram similarity. We calculate the BLEU similarity score of the LLM output with respect to the relevant Taxonomy text. This should be relatively high if the LLM output is indeed quoting from or referencing this section of the Taxonomy. We then calculate the BLEU similarity scores vis-à-vis every other section of the Taxonomy – i.e., those corresponding to the other 41 composite NACE sectors, which the LLM should *not* be considering. We calculated these 41 scores for 1,000 randomly selected FDI projects, and compared these to the BLEU similarity score corresponding to the 'correct' EUTSA text for each FDI project. We find that, for 88.1% of cases, the 'correct' EUTSA text yielded the highest BLEU score of all, while for 92.5% of cases it was above the 90$^{\text{th}}$ percentile. Table S10, below, shows the results of this exercise as average BLEU scores for each sector represented, with the original or 'correct' score in bold, plus the average and 90$^{\text{th}}$ percentile of the alternative BLEU scores on the right. This shows clearly how, for all but one sector in our random sample of 1,000 FDI projects, the highest BLEU similarity score corresponds to the correct section of the Taxonomy.[24] The mean and 90$^{\text{th}}$ percentile of scores that correspond to alternative sections of the Taxonomy are substantially lower for every sector. Taken together, this provides evidence that the LLM is referencing the correct section of the Taxonomy (that which is provided in the prompt), and therefore making its judgment in-line with our instructions.

As a final check for validity, we re-process a random sample of 1,000 FDI projects, evenly split into 500 identified as green and 500 as non-green after our full 2-stage methodology, through the LLM, this time prompting the agent to reflect on the decision and justification made. This draws on recent empirical literature showing that self-reflection by multiple additional or alternative LLM prompts can enhance its performance in certain problem-solving tasks (Renze and Guven, 2024*b*). For each of these 1,000 FDI projects, we present the agent with the original EUTSA criteria and FDI project description, as well as the initial conclusion, and ask it to evaluate its initial judgment. We do this through the prompt shown in Table S11, where the LLM is explicitly asked to reflect on whether the 'correct' EU Taxonomy text (the one provided) is referenced. Due to cost constraints, this exercise was done using the substantially cheaper GPT-4o LLM. This is an offshoot of the GPT-4 Turbo LLM used in our full methodology, which (as demonstrated in the reproducibility section ahead) produces outputs that are highly aligned to those of GPT-4 Turbo when considering the same FDI project and green activity information.

As shown in Figure S30, the level of agreement with the original classification and justification is extremely high – almost 100%.

[24] The only sector where this does not hold is Construction. We have checked the corresponding FDI projects manually and are confident that, despite the low BLEU scores reported with respect to the 'correct' section of the Taxonomy, the LLM is faithfully carrying out our request. We believe that the reason such scores are so low is because the EU Taxonomy description of green activities within the Construction sector uniquely references activities and concepts that are mentioned in the text of many other sectors (for example, the construction of energy and utility infrastructure, an activity which is also present in renewable energy and/or water sectors), so that the highest BLEU scores are found there.

Table S10: Average BLEU scores denoting similarity of LLM output vs. 'correct' EUTSA text, versus against every other 'incorrect' sector (as average & 90th percentile of 41 alternatives), averaged for every sector represented

Sector	Average BLEU score – correct	Average BLEU score mean of all 'incorrect' alternatives	Average BLEU score 90th pct of all 'incorrect' alternatives
Administrative Services & Auxiliary Business Support	**0.051**	0.007	0.019
Agriculture, forestry & fishing	**0.094**	0.010	0.034
Arts, Entertainment, Recreation, Natural Parks & Sport	**0.115**	0.010	0.029
Computer Programming, IT Consulting, and IT Management	**0.181**	0.026	0.048
Construction	**0.010**	0.010	0.030
Financial Services & Insurance	**0.126**	0.012	0.036
Information Service Activities	**0.230**	0.021	0.047
Manufacturing – Basic Metals	**0.181**	0.017	0.039
Manufacturing – Building Materials	**0.043**	0.013	0.032
Manufacturing – Cars, Trucks, and Other Automobiles (incl. parts)	**0.107**	0.015	0.040
Manufacturing – Chemicals & Chemical Products (Excl. Pharmaceutical)	**0.108**	0.016	0.038
Manufacturing – Computers, Electronic Components & Electrical Devices	**0.029**	0.011	0.030
Manufacturing – Fabricated Metal Products	**0.083**	0.023	0.048
Manufacturing – Furniture & Wood Products	**0.218**	0.044	0.118
Manufacturing – Machinery & Industrial Equipment	**0.053**	0.012	0.028
Manufacturing – Medicines & Pharmaceutical Products	**0.191**	0.005	0.017
Manufacturing – Non-Automobile Transport	**0.163**	0.015	0.036
Manufacturing – Paper, Paper Products & Printed Material	**0.163**	0.034	0.074
Manufacturing – Rubber & Plastic Products	**0.083**	0.014	0.036
Professional & Technical Services (Excl. Financial, Scientific & Software)	**0.126**	0.019	0.045
Publishing, Picture/Sound Recording, Broadcasting, and IT Solutions	**0.117**	0.012	0.033
Real Estate	**0.184**	0.026	0.054
Retail/Wholesale Trade & Repairs	**0.245**	0.018	0.047
Scientific Research & Development (incl. Biotechnology)	**0.087**	0.011	0.031
Second-Hand Goods Marketplaces and Classifieds	**0.124**	0.012	0.034
Transportation, Logistics & Storage	**0.094**	0.018	0.041
Total	**0.123**	**0.017**	**0.041**

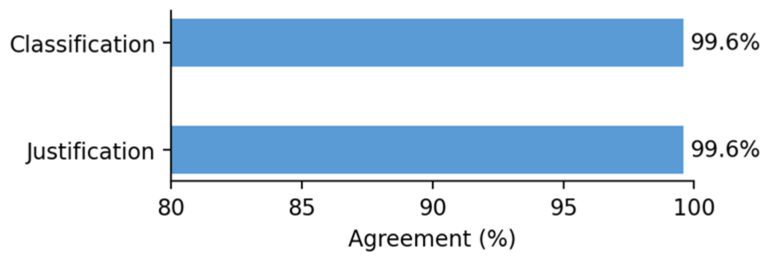
Source: Own elaboration.

Table S11: Specific prompt and temperature parameter settings for self-reflection exercises

Persona/prompt	Temperature parameters tested	LLMs
"Evaluate this existing justification for why the investment was classified as {original_classification}: {justification}. Please provide: 1. Classification: Respond with "Yes" if the investment is green, "No" if it is not. 2. Agreement: Respond with "Agree" or "Disagree" regarding the justification. 3. Agreement Rationale: Provide a 1–2 sentence explanation for your agreement/disagreement"	0.2	GPT-4o

Source: Own elaboration.

Figure S30: Average agreement with classification and qualitative rationale for the same prompt & temperature as GPT-4 Turbo



Source: Own elaboration.

4.3 Reproducibility

Finally, to ensure the reliability of our methodology, a series of robustness checks were conducted to assess the consistency of our results when varying the prompts and temperature settings, as well as switching to other LLMs (both by OpenAI and other providers). The primary aim is to evaluate whether our approach results in reproducible classifications for other researchers performing similar classification tasks, and the extent to which 'prompt engineering', output temperature or inherent model capabilities might negatively impact that reproducibility.

We conduct our reproducibility checks for the same randomly-selected subset of 1,000 FDI projects as for our validity checks, 500 of which are classified as green following our full 2-stage methodology, and 500 which are classified as non-green. For this subset of FDI projects, we first re-process the project descriptions against the EUTSA using three alternative prompts (shown in Table S12, below) and two alternative temperature parameter settings, in all possible combinations. This gives a total of 11 variations on our original approach. We carry out these checks using the GPT-4 Turbo LLM – the same as was used for the full sample of EU27+UK FDI projects – so we are able to isolate the effect of adjusting the model temperature and/or prompt. To evaluate how reproducibility is affected by model changes, we then also try an alternative OpenAI model (GPT-4o) and two models built by other companies: Anthropic's *Sonnet 4.5* and Google's *Gemini Flash 2.5*.

The alignment in conclusions reached, expressed as the share of FDI projects which were returned with the same green/non-green label as in our full sample, is shown in Figure S31 for each of the prompt-temperature combinations tried. By way of summary, the results for each possible prompt show alignment percentages in the following range:

- Prompt 1 (detailed and objective – the prompt used for the full sample processing, constitutes perfect reproduction): $\sim$95% alignment
- Prompt 2 (concise and objective): $\sim$85% alignment
- Prompt 3 (subjective negative impact consideration): $\sim$90% alignment
- Prompt 4 (subjective persona): $\sim$87% alignment

While there is some variation across prompts – particularly prompt 2 which yields the lowest alignment with the original classification – overall agreement rates are high and relatively stable within them. In particular, it is worth noting how alignment is particularly high when the project was initially classified as green (denoted by the green bars): in such cases, re-running again finds these as green close to 100% of the time, even when varying the model prompt (to number 2 and 3) and temperature parameters. This suggests that our main approach is relatively "cautious" and careful in its designation of green status, being more likely to under-report (be overly strict) than over-report (and fall foul of greenwashing), as only a tiny percentage of green projects are

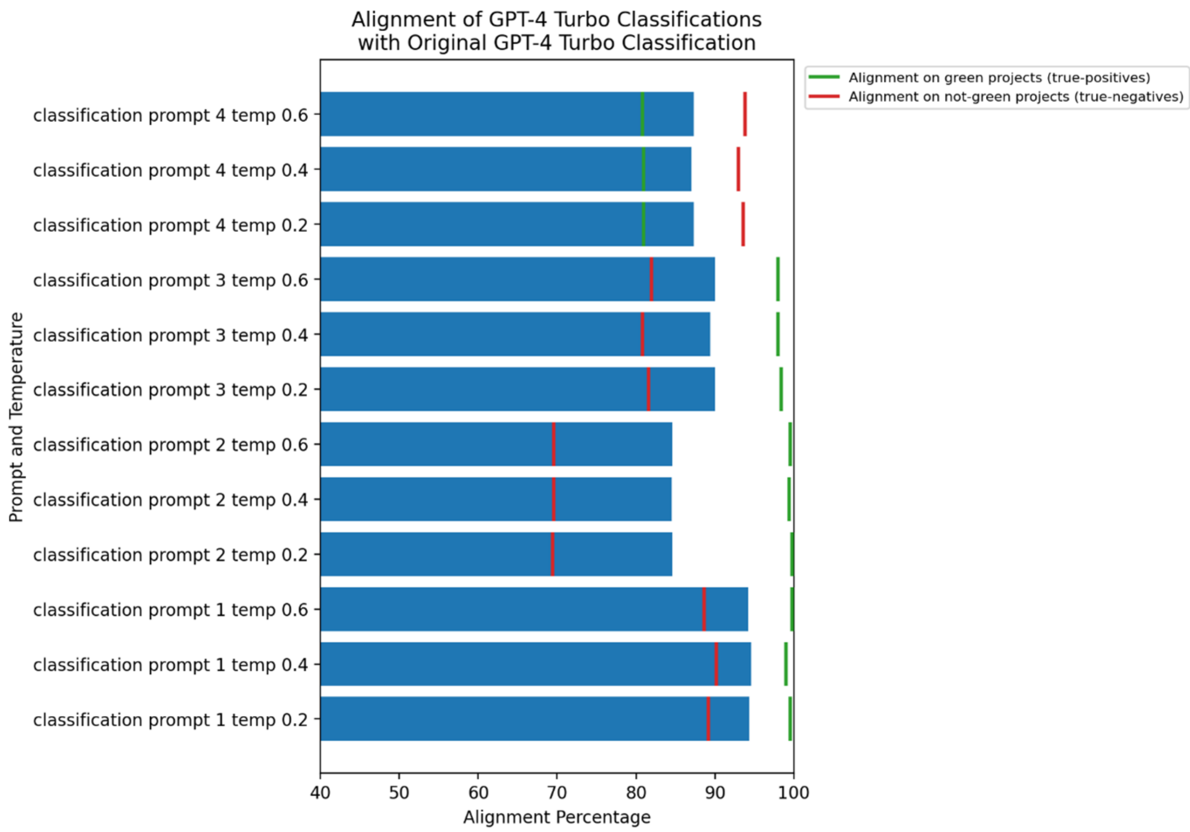
Table S12: Within-model variations for reproducibility checks

Element varied	Setting in original approach	Variations tried
Model prompts	The original prompt for the full classification task: "You are an expert, systematic classifier of whether foreign investments made by companies can be considered 'green' investments. To do this, I will give you two pieces of information. Firstly, ahead is some information describing a particular foreign investment made by a company: {invest_desc}. Secondly, the following describes the kinds of activities by companies operating in the same sector that are considered 'green' by the EU Taxonomy on sustainable finance: {eutsa_green_criteria}. Based strictly on the second piece of information – the criteria for green activities in that particular sector – would you say that the investment described can be considered 'green'? Please do not infer anything about the investment. Base your decision exclusively on the description of the investment that is given. It is essential that you provide your response in the following format: Respond only with 'Yes' or 'No'.'	**Prompt 2**: A much more concise prompt asking for straightforward classification: "Classify the following investment as 'green' or 'not green' based on EU sustainable finance standards: {invest_desc}. Here are the green criteria: {eutsa_green_criteria}. Respond only with 'Yes' or 'No'.' **Prompt 3**: A similarly concise prompt which introduced normative reflection around potential negative environmental impacts: "Analyse the following investment: {invest_desc}. Determine if it qualifies as 'green' under EU Taxonomy, considering both potential positive environmental impacts and any possible negative effects. Here are the green criteria: {eutsa_green_criteria}. Respond only with 'Yes' or 'No'.' **Prompt 4**: A persona-based prompt from the perspective of a sceptical investor concerned about greenwashing: "As a sceptical investor concerned about greenwashing, evaluate this investment: {invest_desc}. Does it genuinely qualify as 'green' according to strict EU Taxonomy standards? Here are the green criteria: {eutsa_green_criteria}. Respond only with 'Yes' or 'No'.'
Temperature parameter	0.2	0.4, 0.6

Source: Own elaboration.

overruled by subsequent reruns. Instead, upon re-processing through the first three prompts, it is more likely that more green FDI projects are found which were initially labelled non-green. This should not necessarily be a surprise given that some FDI projects are very much 'edge-cases' which are not clear cut, especially given the observed ambiguity of the Taxonomy with respect to certain activities and/or sectors, plus the brevity of certain FDI descriptions. Interestingly, for prompt 4 this pattern in fact flips and the overall number of green projects falls slightly (as we might expect), yet the true-positive rate remains decently high, at around 83%. This suggests that, even when prompted to be specifically cautious of greenwashing, our approach is likely to produce very similar results.

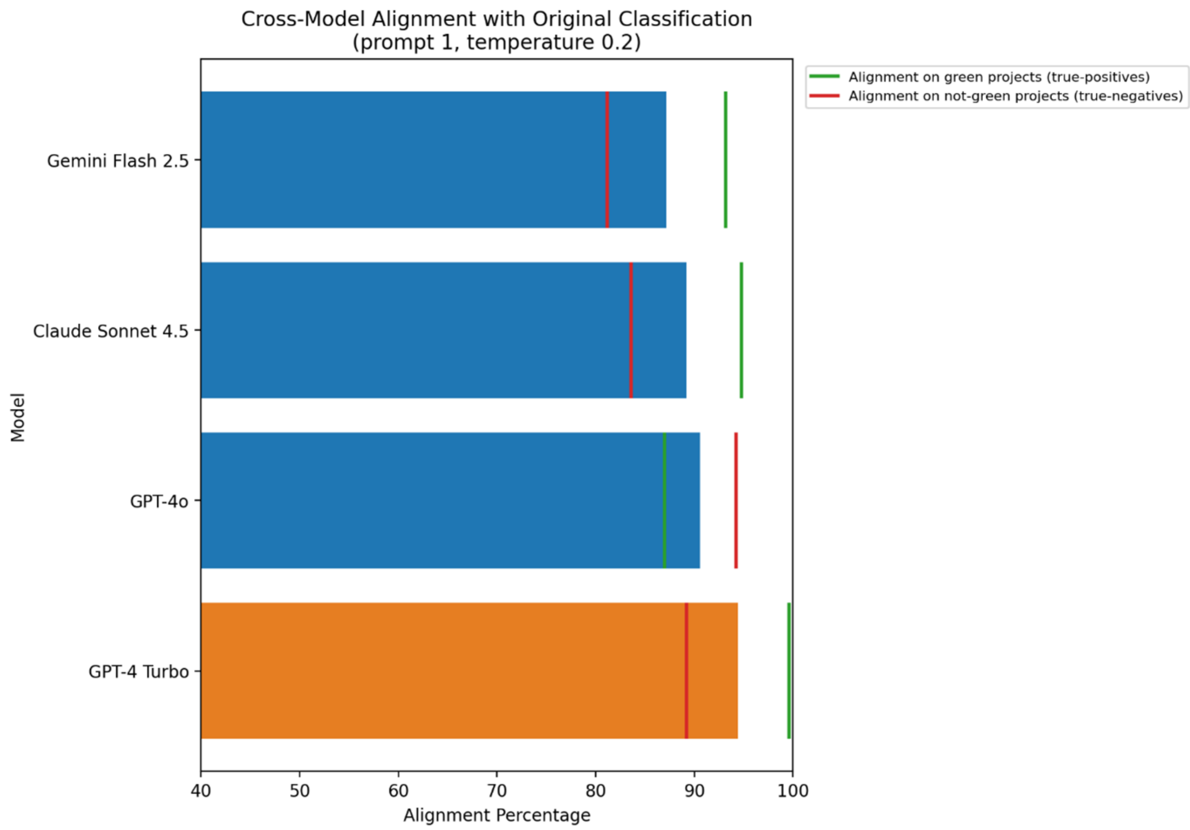
Figure S31: Alignment percentages when re-processing subsample of 1,000 FDI projects through GPT-4o, for 12 prompt-temperature combinations



Source: own elaboration.
Note: Bottom bar = alignment when GPT-4 Turbo was re-tested with the same prompt and temperature as the original.

Lastly, we process the 1,000-strong sample of FDI projects through 3 additional LLMs (Open AI's GPT-4o; Google's Gemini Flash 2.5 and Anthropic's Sonnet 4.5), whilst holding prompt and temperature parameter constant. The results, shown below in Figure S32, demonstrate that even when changing the model, in around 90% of cases the outcome is the same. Again, and reassuringly, for Gemini Flash 2.5 and Sonnet 4.5, the true-positive alignment is particularly high, at around 94%.

Figure S32: Alignment percentages when re-processing subsample of 1,000 FDI projects through 3 additional LLMs, whilst holding prompt and temperature parameter constant



Source: own elaboration.
Note: Orange bar = alignment when GPT-4 Turbo was re-tested with the same prompt and temperature as the original.